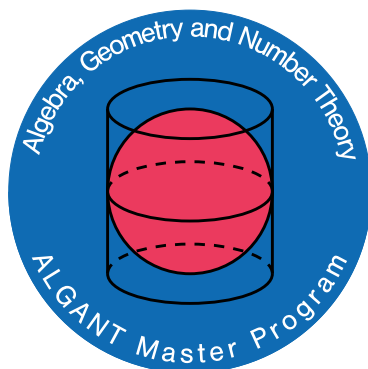




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FAKULTÄT FÜR MATHEMATIK

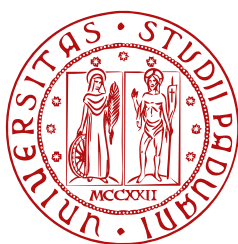
Implementing the Sarkar-Wang Nicefication Algorithm for Heegaard Diagrams

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"And now, Harry, let us step into the night
and pursue that flighty temptress, adventure."

Albus P. W. B. Dumbledore
"Harry Potter and the Half-Blood Prince"

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Introduction

The main goal of this work is to give an introduction to (tangle) Heegaard Floer homology computed via *nice Heegaard diagrams* and to provide a software able to compute such invariant.

Heegaard Floer homology was first developed by Peter Ozsváth and Zoltán Szabó as a variant of Floer homology. They were able to compute this new invariant, later proved to be isomorphic to Seiberg–Witten Floer homology and embedded contact homology, with the use of *Heegaard diagrams*. Heegaard diagrams are a powerful tool of 3-dimensional topology: they are a way to represent a 3-manifold with a genus g surface along with the data of some curves embedded on the surface. The main technical issue is the computation of the proper boundary maps of $\widehat{CF}$, the chain complex built from a Heegaard diagram on which the hat version of Heegaard Floer homology $\widehat{HF}$ is computed. In 2006, Sarkar and Wang were able to find a way around this problem, publishing "An algorithm for computing some Heegaard Floer homologies" ([SW06]). In this article, they proved that there exists a family of diagrams for which the computations of Heegaard Floer homology can be done in a completely combinatorial way: it is sufficient to count the bigons and rectangles of such special diagrams. These were called *nice Heegaard diagram*, and it was further proved that any 3-dimensional manifold admits such a diagram. The construction of a nice Heegaard diagram is possible through an algorithm called *Nicefication algorithm*; this takes any Heegaard diagram and makes it nice using the Heegaard moves, some operations on the embedded curves which do not change the manifold represented by the diagram. This algorithm was implemented in the course of this Master thesis project, and can be found in [nicepy]. Some changes were done in order to optimise the final nice Heegaard diagram obtained with it; these are going to be discussed in Section 4.5.

The focus on the work then shifted on the study of tangles: roughly, a tangle is obtained from a knot by cutting it in n points and pulling apart the endpoints of the n strands created. For tangles, another version of Heegaard Floer homology was developed: tangle Floer homology, denoted by HFT. Tangle Floer homology is also called *multicurve invariant*, as the topological invariant yielded by it is a collection of loop on the four punctured sphere $\mathcal{S}_4^2$. See Figure 1 for an example.

To be able to nimbly compute nice Heegaard diagrams for general 4-ended tangles, two pieces of software were developed and can be found inside the [nicepy] Package: a first one capable of generating a Heegaard diagram for any rational tangle $Q_{p/q}$

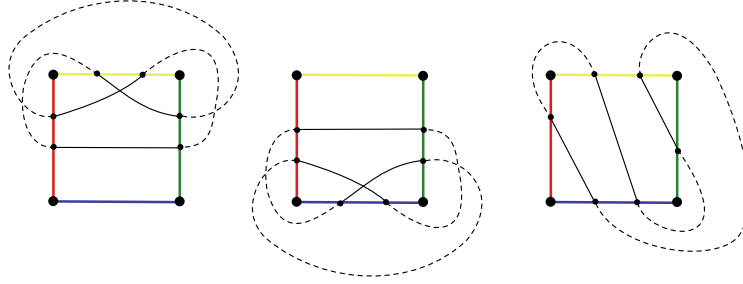


Figure 1: Examples of loops on the four punctured sphere $\mathcal{S}_4^2$: from the left $\mathfrak{s}_2(0; 1, 4)$, $\mathfrak{s}_2(0; 2, 3)$ and $\mathfrak{r}(-2)$.

for $p, q \in \mathbb{Z}$ coprime and a second one able to glue together such diagrams. In this setting, it is possible to take any 4-ended tangle, cut it in a sum of n rational tangles and quite easily generate a Heegaard diagram of genus n for the initial tangle: in fact, it is sufficient to give as input the values of p and q for every tangle, the data of how they are glued together and the Alexander grading of the whole tangle. Of course, as the number of rational tangles needed to obtain the initial tangle grows, the number of generators of the nice Heegaard diagram that we obtain will grow as well; this is something important to keep in mind when trying to compute HFT as a higher number of generators will require more time to do the computations.

With the [nicepy] Package and the [PQM.m] Mathematica Software developed by Claudius Zibrowius, it was possible to compute several invariants for some families of tangles, obtaining some promising results in the search for some general patterns.

The families of tangles that were mostly investigated are:

- the "natural" gluing of a rational tangle and its mirror image, i.e. $Q_{p/q} + Q_{-p/q}$ for q odd;
- the family of tangles of the form $Q_{p/q} + Q_{1/(2n)}$ for q odd.

See Figure 2 to see two examples of elements of the families described.

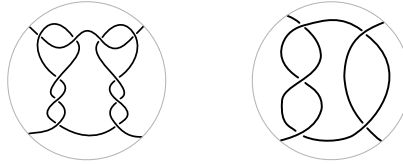


Figure 2: The tangle $Q_{3/7}$ glued with its mirror image $Q_{-3/7}$ (on the left) and the tangle $Q_{1/3}$ glued with $Q_{1/2}$ (on the right).

In Chapter 5, it is possible to see the results obtained and the possible patterns that arise from these initial computations. Among these possible formulas developed, one stands out for its generality and for being in agreement with the known multicurve invariants of Pretzel tangles; we report it here.

Conjecture 0.1 (Conjecture 5.11). *Let $S_{p/q}(2n)$ be the 4-ended tangle obtained as $S_{p/q}(2n) = Q_{p/q} + Q_{1/2n}$, where $p/q \in \mathbb{QP}^1$, q is odd and $0 < 2n \cdot p < q$. Then,*

$$\text{HFT}(S_{p/q}(2n)) = (q - 2n \cdot p) \cdot \mathfrak{r}(1/(2n)) + p \cdot \left(\sum_{i=0}^{n-1} \mathfrak{s}_4(1/(2i+1)) \right).$$

Outline of the thesis

We present here the structure of the thesis.

Chapter 1 is devoted to the introductions of all the topological objects that we will need through the developing of this thesis. It starts with a brief discussion about n -manifolds and their basic properties that we will request in later definitions. We then explore handle decompositions and Heegaard splittings, which are the basic idea on which we will then develop Heegaard diagrams and Heegaard Floer homology. We then shift our interest to knots and tangles, focusing in particular on 4-ended tangles. After the basic definitions, we define a covering space for the 4-punctured sphere (the ambient in which we embed 4-ended tangles) with which are then able to define the very special and useful class of rational tangles. We moreover give the foundations of an arithmetic for 4-ended tangles. We then conclude the first chapter by briefly looking into the theory of sutured, bordered and sutured bordered manifolds, generalisations of the notion of manifolds for which it also exists a Heegaard Floer homology.

In Chapter 2 we construct Heegaard diagrams for all the objects presented in Chapter 1. We first construct them for closed 3-manifolds, introducing Heegaard moves and pointed Heegaard diagrams. Moreover, we explore the link between Morse theory and closed Heegaard diagrams, understanding how a Morse function on a 3-manifold can define a diagram. We then construct Heegaard diagrams for sutured, bordered and sutured bordered manifolds, highlighting similarities and differences from the closed case. We then present Heegaard diagrams for knots, explicitly constructing them with an easy algorithm, and showing how they can be regarded as a special case of the sutured ones. Lastly, we construct Heegaard diagrams for tangles and we focus in particular on the ones for rational tangles; also here we see how we can interpret these diagrams as a special bordered sutured case.

In Chapter 3, we see how we can construct different flavours of Heegaard Floer homology, following the structure already used for Chapter 2. We start by developing in details the hat version $\widehat{\text{HF}}$ for closed 3-manifolds, building up all the machinery needed in order to compute the chain complex $\widehat{\text{CF}}$. We compute also a concrete example: $\widehat{\text{HF}}(L(p, q))$. We then shift our attention to sutured Floer homology SHF and knot Floer homology HFK , seeing how we can regard the latter as a special case of the former. Lastly, we construct tangle Heegaard Floer homology, the homology

that we are most interested into, seeing in details the invariant that this theory yields.

In Chapter 4, we introduce the central notion of nice Heegaard diagram, understanding how Heegaard Floer homology for closed 3-manifolds can be computed more easily when working with diagrams of this special type. We then explore in details the Nicefication algorithm for closed Heegaard diagrams, which allows us to construct a nice Heegaard diagram starting from a generic one; moreover we prove that it does not affect the admissibility of a diagram. Subsequently, we generalise the algorithm to the other settings of our interest: we obtain an algorithm to nicefy bordered diagrams, sutured diagrams, bordered sutured diagrams and peculiar diagrams. At last, we discuss some changes made to the original algorithm while implementing `[nicepy]` in order to optimise the final result.

In Chapter 5 we discuss the results obtained applying HFT to the families of tangles $Q_{p/q} + Q_{-p/q}$ for q odd and $Q_{p/q} + Q_{1/(2n)}$ for q odd. These result were obtained with the use of the Python Software `[nicepy]` and the `[PQM.m]` Mathematica Package: the first computed a nice Heegaard diagram for the tangles investigated, which was then passed to the latter software, obtaining its peculiar module CFT^∂ . These were then expressed as multicurves and collected in tables, presented in this chapter. We discuss then some possible patterns that HFT seems to follow for some sub-families of tangles, conjecturing some more general formulas.

Appendix A is a manual for the `[nicepy]` Python Package. It contains the instruction for setting the input (for closed, bordered sutured and tangle Heegaard diagrams) and the addition parameters that it is possible to set for a run and some examples for these. It is also possible to find examples of outputs and a description of the different stages of a run.

Chapter 1

Topologist's toolbox

This first chapter is meant to be a brief survey of all the topological objects that we will need in the course of the thesis. In each section, one or more references are pointed out for further reading.

1.1 Basics on 3-manifolds

We start by giving the basic definitions about 3-manifolds and their properties. Roughly, an n -manifold is something that locally "resembles" the Euclidean space $\mathbb{R}^n$. To formalise this idea, we need to talk about *charts*, homeomorphisms from the topological space to $\mathbb{R}^n$ for some n that "maps" locally the topological space into the Euclidean space. We now explore this idea formally.

The main reference used for this section is [Bar22].

Definition 1.1 (n -dimensional chart). Let Y be a topological space. An n -dimensional *chart* for Y at a point $y \in Y$ is a homeomorphism

$$\begin{array}{ccc} Y & & \mathbb{R}^n \\ \cup & & \cup \\ U & \xrightarrow{\Phi} & V, \end{array}$$

where U is an open neighbourhood of y and one of the following holds:

- (i) V is an open subset of $\mathbb{R}^n$, or
- (ii) V is an open subset of the upper half-space $\mathbb{H}^n = \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_n \geq 0\}$.

We distinguish between charts of type (i), when case (i) occurs, and charts of type (ii), when case (ii) occurs.

Definition 1.2 (n -dimensional atlas, transition maps). An n -dimensional *atlas* $\mathcal{A}$ for a topological space Y is a family of n -dimensional charts $\mathcal{A} = \{\Phi_i : U_i \rightarrow V_i\}_{i \in I}$ such that "they chart all Y ", i.e. $\cup_{i \in I} U_i = Y$.

The *transition maps* of $\mathcal{A}$ are the maps that allow us to go from one chart to another, i.e. the maps of the form

$$\begin{array}{ccc} \mathbb{R}^n & & \mathbb{R}^n \\ \cup & & \cup \\ \phi_i(U_i \cap U_j) & \xrightarrow{\Phi_j \circ \Phi_i^{-1}} & \phi_j(U_i \cap U_j) \end{array}$$

for $i, j \in I$.

Definition 1.3 (*n*-manifold). A topological space Y is said to be an *n-dimensional topological manifold* (or *n-manifold*), if

1. it is second-countable (i.e. its topology has a countable basis),
2. it is Hausdorff (i.e. open subsets separate points),
3. for every $y \in Y$ there exists an *n*-dimensional chart of type (i) at y .

Moreover, we introduce the notion of *closedness* for a manifold, which we will require for the construction of the hat version of Heegaard Floer homology.

Definition 1.4 (Closed *n*-manifold). An *n*-manifold Y is said to be *closed* if it is compact.

We now generalise to the case in which we admit also charts of type (ii), obtaining *n-manifolds with boundary*.

Definition 1.5 (*n*-manifold with boundary). A topological space Y is said to be an *n-dimensional topological manifold with boundary* (or *n-manifold with boundary*), if

1. it is second-countable (i.e. its topology has a countable basis),
2. it is Hausdorff (i.e. open subsets separate points),
3. for every $y \in Y$ there exists an *n*-dimensional chart of type (ii) at y .

Remark 1.6. Let Y be an *n*-manifold with boundary. The *boundary* of Y , denoted by ∂Y , is the set of all boundary points of Y , described in charts as

$$\partial Y = \bigcup_{i \in I} \Phi_i^{-1}(\{x_n = 0\} \cap \text{Im} \Phi_i).$$

Roughly, a point $y \in Y$ is a *boundary point* if it does not admit a chart of type (i).

Another property of manifolds that we will request for Heegaard Floer homologies, is *orientability*. In order to define this new property, we need to restrict ourselves to *smooth manifolds*.

Definition 1.7 (Smooth atlas, smooth *n*-manifold (with boundary)). An atlas is *smooth* if all the transition maps are smooth (i.e., $\mathcal{C}^\infty$).

A *smooth n-manifold (with boundary)* is an *n*-manifold (with boundary) together with a smooth atlas. This is also sometime called a *differentiable manifold (with boundary)* of class $\mathcal{C}^\infty$.

We also need to define a relations between atlases.

Definition 1.8 (Compatible atlases, maximal atlas). Two smooth atlases $\mathcal{A} = \{\Phi_i\}_{i \in I}$ and $\mathcal{A}' = \{\Phi'_j\}_{j \in J}$ are said *compatible* if every change of chart is smooth, i.e., if the union $\mathcal{A} \cup \mathcal{A}'$ is still a smooth atlas. A smooth atlas is called *maximal* if it is not contained in any strictly larger compatible smooth atlas.

Furthermore it is possible to see that the following proposition holds.

Proposition 1.9 ([Bar22, Proposition 1.9]). *Let Y be a smooth manifold. Then every smooth atlas is contained in a unique maximal smooth atlas and two atlases are contained in the same maximal atlas if and only if they are compatible.*

In what follows, we then assume that each atlas describing a smooth manifold is maximal.

Example 1.10. We can see some basic examples of smooth n -manifolds and n -manifolds with boundary.

1. $\mathbb{R}^n$ is a smooth n -manifold. As charts, we can take identity maps on some covering; therefore it is also smooth. Notice that it is not closed: it has no boundary, but it is not compact.
2. Analogously, an open set $U \subset \mathbb{R}^n$ is an n -manifold, not closed for the same arguments as above.
3. A closed set $C \subset \mathbb{R}^n$ is a smooth n -manifold with boundary.
4. The sphere $\mathcal{S}^n$ is a smooth, closed n -manifold.

Consider the sphere to be the following subset of $\mathbb{R}^{n+1}$, with the topology induced by $\mathbb{R}^{n+1}$

$$\mathcal{S}^n = \{\underline{x} = (x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1} \mid |\underline{x}|_{\mathbb{R}^n} = 1\} \subset \mathbb{R}^{n+1}.$$

Then consider the two open subsets $U_N = \mathcal{S}^n \setminus \{N\}$ and $U_S = \mathcal{S}^n \setminus \{S\}$, where $N = (0, \dots, 0, 1)$ is the north pole and $S = (0, \dots, 0, -1)$ is the south pole. We want to construct an atlas with only two charts, defined on these open subsets that cover the sphere. Notice that this is the minimum number of charts that we can use for an atlas of $\mathcal{S}^n$: if we could construct an atlas with only one chart, then it would mean that $\mathcal{S}^n$ is homeomorphic to an open subset of $\mathbb{R}^n$, which is false since $\mathcal{S}^n$ is compact.

We define the charts $\Phi_N : U_N \rightarrow \mathbb{R}^n$ and $\Phi_S : U_S \rightarrow \mathbb{R}^n$ to be the stereographic projections (from N and S respectively); in coordinates this is expressed as

$$\begin{array}{ccc} U_N & \xrightarrow{\Phi_N} & \mathbb{R}^n \\ \underline{x} & \longmapsto & \frac{1}{1-x_{n+1}}(x_1, \dots, x_n) \end{array} \qquad \begin{array}{ccc} U_S & \xrightarrow{\Phi_S} & \mathbb{R}^n \\ \underline{x} & \longmapsto & \frac{1}{1+x_{n+1}}(x_1, \dots, x_n). \end{array}$$

The transition map $\Phi_S \circ \Phi_N^{-1}$ is described as

$$\mathbb{R}^n \setminus \{0\} \xrightarrow{\Phi_S \circ \Phi_N^{-1}} \mathbb{R}^n \setminus \{0\} \quad \underline{x} \longmapsto \frac{\underline{x}}{|\underline{x}|_{\mathbb{R}^n}^2}.$$

We want to construct the orientation of Y using some the charts. This is done using a smooth atlas and the Jacobian matrices¹ of the transition maps.

Definition 1.11 (Equioriented charts, oriented atlas). Two smooth charts $\Phi_1 : U_1 \rightarrow V_1$ and $\Phi_2 : U_2 \rightarrow V_2$ of some manifold Y are said to be *equioriented* if $\det J_f(x_0)(\Phi_2 \circ \Phi_1^{-1}) > 0$ for any $x_0 \in \Phi_1(U_1 \cap U_2)$, where $J(f)$ is the Jacobian matrix of f .

An atlas is *oriented* if every pair of charts are equioriented.

We can now define oriented manifolds.

Definition 1.12 (Oriented n -manifold (with boundary)). A smooth n -manifold Y is said to be *orientable* if it admits an oriented atlas. An *orientation* of Y is a choice of an oriented atlas of Y .

Example 1.13 ($\mathcal{S}^n$ is orientable for any $n \geq 1$). We can easily prove that $\mathcal{S}^n$ is orientable for any $n \geq 1$.

The case $n = 1$ is done by hand. To prove the case $n \geq 2$, one uses the fact that local diffeomorphisms do not change sign on connected sets and have sign different than zero. Therefore, the signs of the Jacobians (which are local diffeomorphisms) of the only transition map are different than zero and they all agree because the intersection $U_N \cap U_S$ is connected. If the sign is positive, then we are done. If it is negative, we just to post-compose the second chart to a map that swaps the last two coordinates, obtaining the following new chart

$$U_S \xrightarrow{\Phi'_S} \mathbb{R}^n$$

$$\underline{x} \longmapsto \frac{1}{1+x_{n+1}}(x_1, \dots, x_{n-2}, x_n, x_{n-1}).$$

¹Let f be a map $f : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$, defined by $\underline{x} = (x_1, \dots, x_n) \mapsto (f_1(\underline{x}), \dots, f_m(\underline{x}))$. Recall that the Jacobian of f in $x_0 \in U$ is defined to be

$$J_f(x_0) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(x_0) & \cdots & \frac{\partial f_1}{\partial x_n}(x_0) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(x_0) & \cdots & \frac{\partial f_m}{\partial x_n}(x_0) \end{pmatrix}.$$

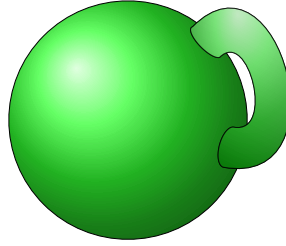


Figure 1.1: Handlebody of genus 1.

1.2 Handle decomposition and Heegaard splittings

The underlying idea of Heegaard splittings (also called Heegaard decompositions) is to decompose a 3-manifold in simpler pieces, called *handlebodies*. Intuitively, we will carve some holes in the 3-manifold, g holes for examples, obtaining a handlebody of genus g . Then, we consider the complement of this handlebody in our 3-manifold, obtaining another handlebody of genus g which has the border in common with the previous handlebody: this is then a Heegaard splitting.

The main references for this section are [OS06] and [Hom19].

We are going to assume always that 3-manifolds are closed and oriented and that homeomorphisms are orientation-preserving (unless otherwise stated).

Definition 1.14 (n -dimensional k -handle). Consider $0 \leq k \leq n$. We call a n -dimensional k -handle a copy of $B^k \times B^{n-k}$, attached to the boundary of a smooth n -manifold Y along $(\partial B^k) \times B^{n-k}$ by an embedding $\phi : (\partial B^k) \times B^{n-k} \hookrightarrow \partial Y$. As there is a canonical way to smooth corners, we think of Y with the attached handle as a smooth n -manifold.

Often, we think about an n -dimensional k -handle as a k -cell that we thicken-up until it become n -dimensional.

Definition 1.15 (Handlebody of genus g). We define an *handlebody of genus g* to be a 3-dimensional ball to which we attach g 3-dimensional 1-handles to it.

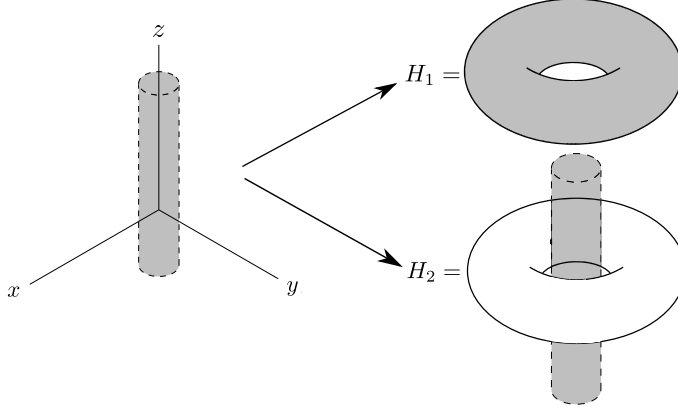
We can clearly see that the surface of a handlebody of genus g is a surface of genus g .

Definition 1.16 (Heegaard splitting of genus g). Let Y be a closed and oriented 3-manifold. An *Heegaard splitting of Y of genus g* (also called *Heegaard decomposition*) is a decomposition

$$Y = H_1 \cup_{\Sigma} H_2$$

in which H_1 and H_2 are handlebodies of genus g such that $\partial H_1 \simeq \partial H_2$ (they are homeomorphic via an orientation-reversing homeomorphism).

Example 1.17 (First examples). We see now some examples that can help to visualise the above definition.

Figure 1.2: Heegaard splitting for $\mathcal{S}^3 \simeq \mathbb{R}^3 \cup \{\infty\}$.

1. Heegaard splitting of genus 0 of $\mathcal{S}^3$

Consider the sphere $\mathcal{S}^3$, cut it in half in two hemispheres H_1 and H_2 , namely $H_1 = \{(x, y, z, w) \in \mathcal{S}^3 \subset \mathbb{R}^4 \mid w \geq 0\}$ and $H_2 = \{(x, y, z, w) \in \mathcal{S}^3 \subset \mathbb{R}^4 \mid w \leq 0\}$. We have that $H_i \simeq B^3$, i.e. they are diffeomorphic the 3-dimensional ball and that $\partial H_1 = \partial H_2 \simeq \mathcal{S}^2$. Therefore this is a genus 0 Heegaard decomposition of $\mathcal{S}^3$.

2. Heegaard splitting of genus 1 of $\mathcal{S}^3$

Regard $\mathcal{S}^3$ as the compactification of $\mathbb{R}^3$ with a point to the infinite. Consider a closed tubular neighbourhood of the circle given by the z -axis and the point to the infinite, this is the first genus-1 handlebody H_1 . Let H_2 be the closure of the complement of H_1 ; this is another handlebody of genus 1. Therefore we have constructed a genus 1 Heegaard splitting of $\mathcal{S}^3$.

3. Heegaard splitting of genus g of $\mathcal{S}^3$

Starting from the 0-genus splitting, we can obtain a g -genus for any g in the following way. Take the two 3-balls H_1, H_2 , in H_1 "drill" g unknotted holes (they have to be disjoint) and in H_2 attach g disjoint handles on the outsides; call H'_1, H'_2 the two new handlebodies. Now we should identify each handle of H'_2 with a hole in H'_1 . Therefore here the surface Σ' of the handlebodies is of genus g and we have that $\mathcal{S}^3 = H'_1 \cup_{\Sigma'} H'_2$.

4. Heegaard splitting of genus 1 of lens spaces $L(p, q)$

We give first the definition of such space. Let p, q be coprime integers and consider the 4-dimensional sphere embedded in $\mathbb{C}^2$

$$\mathcal{S}^3 = \left\{ (z, w) \in \mathbb{C}^2 \mid |z|^2 + |w|^2 = 1 \right\} \subset \mathbb{C}^2.$$

Then we consider the following $\mathbb{Z}/p\mathbb{Z}$ action on $\mathcal{S}^3$

$$\begin{aligned} \frac{\mathbb{Z}}{p\mathbb{Z}} \times \mathcal{S}^3 &\longrightarrow \mathcal{S}^3 \\ (k + p\mathbb{Z}, (z, w)) &\longmapsto \left(e^{\frac{2\pi i k}{p}} z, e^{\frac{2\pi i k q}{p}} w \right) \end{aligned}$$

and define $L(p, q)$ as the quotient of this action. We can also see that $L(p, q)$ is a closed, 3-dimensional, oriented, smooth manifold, whose orientation is given by the initial orientation of $\mathcal{S}^3$. In fact, since the $\mathbb{Z}/p\mathbb{Z}$ action acts freely, properly and continuously, we have that $L(p, q)$ is a closed 3-manifold and that the projection $p : \mathcal{S}^3 \rightarrow L(p, q)$ is a local homeomorphism; moreover, since the action is smooth (i.e., for every $\bar{a} \in \mathbb{Z}/p\mathbb{Z}$ the map $\mathcal{S}^3 \rightarrow \mathcal{S}^3$ defined by $x \mapsto \bar{a} \cdot x$ is smooth), we also have that $L(p, q)$ has a unique smooth structure that makes the projection p a local diffeomorphism.

We want to construct now a genus 1 Heegaard splitting. Consider the following two subsets of $\mathcal{S}^3$

$$\begin{aligned} X &= \left\{ (z, w) \in \mathbb{C}^2 \mid |z|^2 + |w|^2 = 1 \text{ and } |z|^2 \geq \frac{1}{2} \right\}; \\ Y &= \left\{ (z, w) \in \mathbb{C}^2 \mid |z|^2 + |w|^2 = 1 \text{ and } |w|^2 \geq \frac{1}{2} \right\}. \end{aligned}$$

These are both diffeomorphic to the solid 1-torus, $\mathcal{S}^1 \times B^2$ and their intersection is exactly the surface of a torus $\mathcal{S}^1 \times \mathcal{S}^2$. We can now notice that the action of $\mathbb{Z}/p\mathbb{Z}$ on $\mathcal{S}^3 = X \cup Y$ preserves these two tori set-wise. More precisely, the quotients $X/\mathbb{Z}_p$ and $Y/\mathbb{Z}_p$ are again diffeomorphic to the solid torus. To see this, we pick $r \in \mathbb{N}$ with $r \cdot q \equiv -1 \pmod{p}$; it is straightforward to see that the maps

$$\begin{aligned} \frac{X}{\mathbb{Z}/p\mathbb{Z}} &\xrightarrow{\Phi} S^1 \times B^2 & \frac{Y}{\mathbb{Z}/p\mathbb{Z}} &\xrightarrow{\Psi} S^1 \times B^2 \\ [(z, w)] &\longmapsto \left(\frac{z^p}{|z|^p}, \frac{z^{-q} w}{|z^{1-q}|} \right) & [(z, w)] &\longmapsto \left(\frac{w^p}{|w|^p}, \frac{z w^r}{|w^{1+r}|} \right) \end{aligned}$$

are well-defined diffeomorphisms and that Φ is orientation-preserving (to see the latter, it suffices to show that Φ is orientation-preserving at a single point. Using that $p > 0$ one can calculate explicitly that Φ is orientation preserving at the point $[(1, 0)] \in X/\mathbb{Z}/p\mathbb{Z}$). This shows that

$$L(p, q) = X/\mathbb{Z}_p \cup Y/\mathbb{Z}_p$$

is a genus 1 Heegaard decomposition of the lens space $L(p, q)$.

At first glance, it may seem that admitting a Heegaard splitting is a quite special property. This is in reality not true, as any closed 3-manifold admits a Heegaard splitting; a proof of this fact was given by Singer in [Sin33].

Theorem 1.18 (Existence of Heegaard splitting). *Let Y be a closed 3-manifold. Then it admits a Heegaard splitting.*

Proof ([Hom19]). Cairns and Whitehead proved that to every smooth manifold can be given a simplicial structure, i.e. a triangulation.

Consider now the 1-skeleton of a triangulation of Y (vertices and edges); this is a graph embedded in Y . Let H_1 be a small tubular closed neighbourhood of the 1-skeleton: to visualise this think of replacing each vertex with a closed ball and each edge with a solid cylinder between the two balls at the extremities. It is clear that H_1 is a handlebody of some genus g . If we now call H_2 the closure of $Y \setminus H_1$, we see that this is another handlebody of genus g given by a regular neighbourhood of a graph which vertices stay in the centre of the triangles and tetrahedrons of the 1-skeleton above.

As $\partial H_1 = \partial H_2$, we have a Heegaard splitting of Y . □

Another way to construct a Heegaard splitting for a generic 3-manifold is to use a self-indexing, Morse function on it. We will explore this particular construction briefly in Section 2.3.

As we have seen in the examples above, there could be different choices of Heegaard splittings for the same 3-manifold. We are now going to see some definitions which will give us a clearer idea of how these choices are related one to another.

Definition 1.19 (Homeomorphic Heegaard splittings). Given two Heegaard splittings of Y , namely

$$Y = H_1 \cup_{\Sigma} H_2 = H'_1 \cup_{\Sigma'} H'_2,$$

we say that they are *homeomorphic* if there exists a homeomorphism $Y \xrightarrow{\phi} Y$ such that $\phi(H_i) = H'_i$ for $i = 1, 2$.

We also have a stricter relation between Heegaard splittings that we can consider.

Definition 1.20 (Isotopic Heegaard splittings). Given two Heegaard splittings of Y , namely

$$Y = H_1 \cup_{\Sigma} H_2 = H'_1 \cup_{\Sigma'} H'_2,$$

we say that they are *isotopic* if there exists a map $Y \times [0, 1] \xrightarrow{\psi} Y$ such that

1. $\psi|_{Y \times \{0\}} = id_Y$;
2. $\psi|_{Y \times \{t\}}$ is a homeomorphism for any t ;
3. $\psi|_{Y \times \{1\}}$ sends H_i to H'_i for $i = 1, 2$.

This definition is clearly stricter than the one above since we can use $\psi|_{Y \times \{1\}}$ as homeomorphism.

Definition 1.21 (Stabilisation). Let $Y = H_1 \cup_{\Sigma} H_2$ be a Heegaard splitting of genus g . A *stabilisation* (or *1-fold stabilisation*) of $H_1 \cup_{\Sigma} H_2$ is a genus $g + 1$ Heegaard splitting of Y obtained by adding one more handle to the existing splitting. More precisely, it is constructed in the following way. Choose two point on Σ and connect them with two arcs γ_1 and γ_2 with the following properties:

- γ_1 is embedded in Σ ;
- γ_2 is embedded in H_1 ;
- the closed curve obtained by $\gamma_1 \cup \gamma_2$ bounds an embedded disc $D_\star$ completely in H_1 .

Let N be a small tubular neighbourhood of γ_2 ; define then $H'_1 = H_1 \setminus N$, and H'_2 as the union $H_2 \cup N$. We call an n -fold *stabilisation* a sequence of n stabilisations.

Remark 1.22. Notice that the isotopy type of a stabilisation of a Heegaard splitting does not depend by the choice of γ_1 and γ_2 . In fact, to perform a stabilisation means to choose a disc $D_\star \subset H_1$ as above described; as it is contractible in the handlebody H_1 , the stabilisation does not depend on where it is.

Therefore we can talk of *the* stabilisation of a Heegaard splitting.

Example 1.23. As an example, we can see that in Example 1.17, the genus 1 decomposition of $\mathcal{S}^3$ is the stabilisation of the Heegaard decomposition of genus 0.

We have a theorem of Reidemeister and Singer, which shows us that any two splittings of a 3-manifold can be connected via stabilisations. A proof can be found in [Rei33] and [Sin33].

Proposition 1.24. *Let Y be a 3-manifold with two different Heegaard splittings $Y = H_1 \cup_\Sigma H_2 = H'_1 \cup_{\Sigma'} H'_2$ of genus g and g' respectively. Then, for k large enough, the $(k - g')$ -fold stabilisation of the first decomposition is diffeomorphic to the $(k - g)$ -fold stabilisation of the second decomposition.*

1.3 Knots and tangles

In this section, we introduce the notions of knots and tangles, explaining how we can work with them. As main reference, one can look at Section I.1 of [Zib17]. Another reference which can be very helpful for visualising this notions, is [Ada94].

We first introduce the notion of *knot*.

Definition 1.25 (Knot). A *knot* K is an embedding of $\mathcal{S}^1$ into the closed 3-ball B^3

$$K : \mathcal{S}^1 \hookrightarrow B^3 \subset \mathbb{R}^3,$$

i.e. a simple, closed curve in the 3-dimensional space.

In a similar fashion, we introduce *tangles*.

Definition 1.26 (Tangle, 4-ended tangle). A *tangle* T is an embedding of a disjoint union of intervals and circles into the closed 3-ball B^3

$$T : \left(\left(\bigsqcup_{i=1}^n I \right) \sqcup \left(\bigsqcup_{j=1}^m \mathcal{S}^1 \right), \partial \right) \hookrightarrow (B^3, S \subset \partial B^3)$$

such that the endpoints of the interval lie on a fixed, oriented circle $S \cong \mathcal{S}^1$ contained in the boundary of B^3 , together with a labelling of the arcs of $S \setminus \text{im}(T)$. We call a tangle *2n-ended* if the number of intervals is n . Moreover, the images of the intervals are called *open components*, the images of the circles are called *closed components*; these components are usually labelled by variables $t_1, t_2, \dots$, which we call *colours of T*.

Remark 1.27. We will be mainly interested in 4-ended tangle without any closed components, i.e. tangles of the form

$$T : (I_1 \sqcup I_2, \partial) \hookrightarrow (B^3, S \subset \partial B^3).$$

However, we will talk in general about tangles (and 4-ended tangles, which admits closed components) and we will highlight when we restrict ourselves to this other family.

We will consider knots and tangles up to *ambient isotopy*, defined as follows.

Definition 1.28 (Ambient isotopy). We can visualise a knot as string in the 3-dimensional ball and a $2n$ -ended tangle as the collection of n strings in the 3-dimensional ball, which endpoints stay fixed on the fixed circle $\mathcal{S}$. We define an *ambient isotopy* to be the act of re-arrange these strings in the 3-dimensional space (hence the word *ambient*) and without letting any string to pass through any other string (itself included), admitting also any deformation of the strings (hence the word *isotopy*). Moreover, for a tangle, we do not allow moving the endpoints of the strings and to modify the orientation of the fixed circle $\mathcal{S}$.

When we want to say that two tangles (or knots) A_1 and A_2 are isotopic, we write $A_1 \simeq A_2$.

Notice that in an ambient isotopy, we are not allowed to erase some part of the knot or tangle by shrinking any part of it down to a point.

Remark 1.29 (On the definition of ambient isotopy for tangles). The definition of ambient isotopy given above is something in the middle between allowing all the isotopies and allowing only isotopies that fix the whole boundary sphere. In this way, we construct equivalence classes in a reasonable way, differentiating between the right tangles. For instance, if the classes were obtained by quotienting for all the isotopies, all the rational tangles (for the reader that do not knows what a rational tangle is, we will define them in Section 1.5) would fall in the same class. In the same way, fixing all the boundary would be a too rigid requirement.

To represent knots and tangles on paper, we define their *diagrams*.

Definition 1.30 (Knot diagram). A *knot diagram* E is immersion of $\mathcal{S}^1$ into the closed 2-disc

$$E : \mathcal{S}^1 \looparrowright \mathbb{D}^2 \subset \mathbb{R}^2,$$

where all the self-intersections are transverse.

Moreover, we require that the image is a graph with the following properties:

1. vertices can only have valence 4;
2. every vertex carries the under/over information.

We call *regions* the connected components of $\mathbb{D}^2 \setminus \text{im}(E)$.

Definition 1.31 (Tangle diagram). A *tangle diagram* D is an immersion of intervals and circles into the closed 2-disc,

$$D : \left(\bigsqcup_{i=1}^n I \sqcup \bigsqcup_{j=1}^m \mathcal{S}^1 \right), \partial \looparrowright (\mathbb{D}^2, \partial\mathbb{D}^2),$$

where all the self-intersections are transverse.

Moreover, we require that the image is a graph with the following properties:

1. vertices can only have valence 1 or 4;
2. every 4-valent vertex carries the under/over information;
3. we have a labelling with some index set $\{a, b, c, \dots\}$ of the arcs on the border of $\mathbb{D}^2$ minus the image of D , i.e. $\partial\mathbb{D}^2 \setminus D(\partial(\bigsqcup_{i=1}^n I))$.

We call *regions* the connected components of $\mathbb{D}^2 \setminus \text{im}(D)$; the ones that intersect $\partial\mathbb{D}^2$ are called *open regions*, the others *closed regions*. We say that a diagram is *connected* if for any open regions we have that its intersection with $\partial\mathbb{D}^2$ is connected.

From here, unless specified otherwise, diagrams are assumed to be connected and have at least one crossing.

Remark 1.32 (Knots, tangles and their diagrams). We want to understand how loose we can be with the notation when speaking of a knot or a tangle and one of its diagrams. We claim that it is always possible to pass from one to the other without troubles. We prove that it is possible to do so for tangles and tangles diagrams, the knot case is done analogously.

We start with a tangle diagram D for some tangle T , we want to reconstruct the tangle starting from D . Let us consider the image of D to stay in the plane $\{z = 0\}$, i.e. $\text{Im}(D) \subset \mathbb{D}^2 = B^3 \cap \{z = 0\}$. We can construct T by pushing the components of the singularities of the immersion D into $\{z < 0\}$ and $\{z > 0\}$, deciding which one goes where using the under/over data of the crossing.

We now see how to construct a diagram starting from a tangle. Let T be a tangle, consider an embedded disc $\mathbb{D}^2 \subset B^3$ bounding the fixed circle $S \cong \mathcal{S}^1$. Then we can generate a well-defined tangle diagram D for the tangle by simply projecting B^3 onto this disc $\mathbb{D}^2$.

Therefore, when we will talk about a tangle (respectively, knot) diagram, it will be clear which unique tangle (respectively, knot) is represented by it.

Clearly there are multiple ways to represent the same knot or the same tangle with a diagram, and by moving the perspective to the 3-dimensional context and operating some ambient isotopy it is clear when two diagrams represent the same object. To express ambient isotopies in a diagram, we use *Reidemeister moves*, which help us to move between diagrams representing the same knot or tangle without having to work in the 3-dimensional space. We now define these moves, more details can be found in [Ada94], Section 1.3.

Definition 1.33 (Reidemeister moves). We call *Reidemeister moves* the following three operations that we can do on a knot or a tangle diagram. In each of these move, we are assuming to change the diagram only locally in the section stated, leaving all the rest unchanged.

- The first Reidemeister move is called *twist move*; it allows us to put in or take out a twist in the diagram.

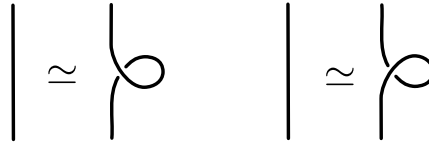


Figure 1.3: The first Reidemeister move.

- The second Reidemeister move is called *poke move*; it allows us to either add two crossings or remove two crossings.

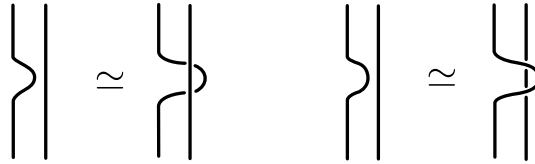


Figure 1.4: The second Reidemeister move.

- the third Reidemeister move is called *slide move*; it allows us to slide a strand of the diagram from one side of a crossing to the other side of the crossing.

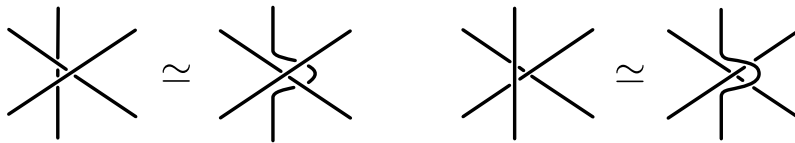


Figure 1.5: The third Reidemeister move.

In 1926, Reidemeister proved the following result.

Theorem 1.34 (Reidemeister, 1926). *If we have two distinct projections of the same knot, we can get from the one to the other via a series of Reidemeister moves and planar isotopies.*

It is possible to generalise such result in the case of tangles, as follows.

Proposition 1.35 ([Zib17]). *Let T_1 and T_2 be two oriented connected tangle diagrams that represent the same tangle. Then it is possible to obtain T_2 operating a sequence of Reidemeister moves T_1 , obtaining connected diagrams as intermediate steps.*

Proof ([Zib17]). By Reidemeister's theorem (Theorem 1.34), it follows that we can always find a sequence of Reidemeister moves connecting the two diagrams.

We want now to ensure that all intermediate diagrams are connected. Let t_1 be one open strand of the tangle that is near the boundary; before starting the sequence of Reidemeister moves, we deform t_1 so that it goes once around the whole diagram, using the second Reidemeister move when we find another strand in our way, "isolating" the border regions of the diagram. We now apply the sequence of Reidemeister moves and we undo the move done on t_1 once we have arrived at T_2 . $\square$

1.4 A covering space for $\mathcal{S}_4^2$

To proceed with our discussion on tangles and to define *rational tangles*, we need to define a covering space for the 4-punctured sphere $\mathcal{S}_4^2$. This will be needed in multiple occasions, such as when we will talk about the invariants yielded HFT.

As main reference for this section, one can see Section 2.2 of [Aub22].

Recall that we defined a 4-ended tangle as an embedding

$$T : \left((I \sqcup I) \sqcup \left(\bigsqcup_{j=1}^m \mathcal{S}^1 \right), \{v_1, v_2, u_1, u_2\} \right) \hookrightarrow (B^3, \mathcal{S} \subset \partial B^3),$$

where we denoted by v_1, v_2, u_1, u_2 the endpoints of the two open strings, which are mapped to $T(v_1), T(v_2), T(u_1), T(u_2) \subset \mathcal{S}$, with $\mathcal{S} \cong \mathcal{S}^1$ is a fixed circle on ∂B^3 .

Definition 1.36 ($\mathcal{S}_4^2$). We define the 4-punctured sphere to be $\mathcal{S}_4^2 = \partial(B^3) \setminus (\partial(\text{Im}T))$. We are going to refer to the 4 punctures as north-west, south-west, south-east and north-east.

Hence, with the images of the strings' ends $T(v_1), T(v_2), T(u_1), T(u_2)$ we are dividing $\mathcal{S}$ in four components, that we are going to call α -arcs sites (or, more simply, *sites*), which are denoted by the letters a, b, c, d starting from the component between the north-west and the south-west punctures and going counter-clock wise. We will use colours to identify the different sites: red for a , blue for b , green for c and yellow for d (see Figure 1.6). Notice that these arcs divide $\mathcal{S}_4^2$ in two components, that we can distinguish thanks to the orientation of $\mathcal{S}$: the *back component* and

the *front component*. The back component is the one whose boundary orientation agrees with the orientation of $\mathcal{S}$ (using the right-hand rule and a normal vector field pointing into B^3), the other one is the front component.

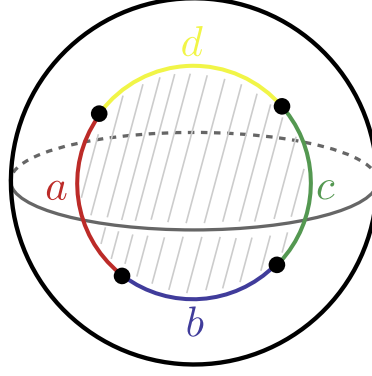


Figure 1.6: The 4-punctured sphere $\mathcal{S}_4^2$ with colours on the sites.

We can now construct a covering space for $\mathcal{S}_4^2$, namely

$$\mathbb{R}^2 \setminus \mathbb{Z}^2 \xrightarrow{\eta} \mathcal{S}_4^2.$$

To prove that this is a covering space, we see how to consider the 4-punctured sphere as a quotient of $\mathbb{R}^2 \setminus \mathbb{Z}^2$. Let G be the subgroup of the isometries of $\mathbb{R}^2 \setminus \mathbb{Z}^2$ generated by s_v and ρ , where s_v is the translation for a vector $v \in (2\mathbb{Z})^2$ (we take "even vectors" so that that translations of the front is mapped to the front and translations of the back to the back) and ρ is the rotation $(x, y) \mapsto (-x, -y)$; we then obtain that $\mathcal{S}_4^2 \cong (\mathbb{R}^2 \setminus \mathbb{Z}^2) / G$. We then have the covering, induced by the projection

$$\mathbb{R}^2 \setminus \mathbb{Z}^2 \xrightarrow{\eta} (\mathbb{R}^2 \setminus \mathbb{Z}^2) / G \cong \mathcal{S}_4^2.$$

See Figure 1.7 for a visualisation of the covering space.

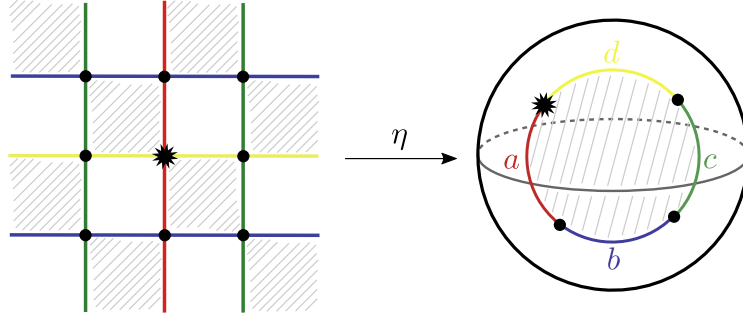


Figure 1.7: Covering space for $\mathcal{S}_4^2$.

Lemma 1.37. *Let G be $\langle s_v, \rho \mid v \in (2\mathbb{Z})^2 \rangle$ described above. Then, $G \cong (\mathbb{Z}/2) \ltimes (2\mathbb{Z})^2$.*

Sketch of proof. In fact, notice that

- $\rho^2 = id$;
- $\rho \circ s_v = s_{-v} \circ \rho$ for any $v \in (2\mathbb{Z})^2$;
- $s_v \circ s_w = s_{v+w}$ for any $v, w \in (2\mathbb{Z})^2$.

We then have the following bijection of sets

$$\begin{aligned} (\mathbb{Z}/2) \ltimes (2\mathbb{Z})^2 &\xrightarrow{\cong} G \\ (r, v) &\longmapsto \rho^r \circ s_v. \end{aligned}$$

At group level, we then obtain $G \cong (\mathbb{Z}/2) \ltimes (2\mathbb{Z})^2$ as we wanted, where the multiplication in the semi-direct product has to be read from right to left (since it is a composition of functions) and it is given by

$$(r_2, v_2) \circ (r_1, v_1) = (r_1 + r_2, v_1 + (-1)^{r_1} v_2).$$

□

We are now going to see how the action of $SL_2(\mathbb{Z})$ in $\mathbb{R}^2 \setminus \mathbb{Z}^2$ induces an action of the 4-punctured sphere, namely the generation of *half-twists*. The group of 2×2 matrices with entries in $\mathbb{Z}$ is generated by τ_1 and τ_2 , where

$$\tau_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \tau_2 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

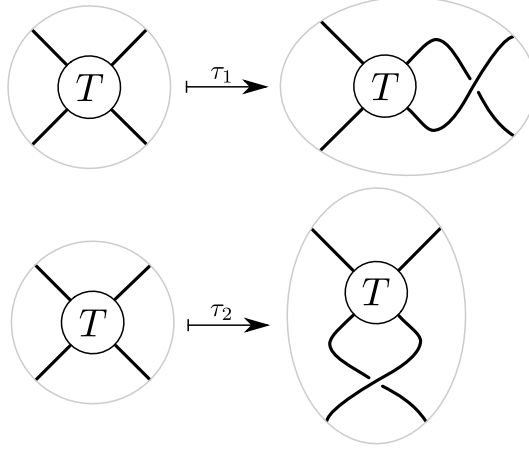
Observe now that for any $\xi \in SL_2(\mathbb{Z})$ we have that

- $\xi \circ s_v = s_{\xi(v)} \circ \xi$ for any $v \in (2\mathbb{Z})^2$;
- $\xi \circ \rho = \rho \circ \xi$.

Therefore, the action of $SL_2(\mathbb{Z})$ on $\mathbb{R}^2 \setminus \mathbb{Z}^2$ induces an action on the quotient $\mathcal{S}_4^2 \cong (\mathbb{R}^2 \setminus \mathbb{Z}^2)/G$; we still denote the generators of this action on the sphere by τ_i for $i = 1, 2$. We can visualise this action on $\mathcal{S}_4^2$ as done in Figure 1.8. It is then clear that τ_1 is a *horizontal half-twist*, i.e. the addition of a crossing on the bottom and that τ_2 is a *vertical half-twist*, i.e. the addition of a crossing on the bottom.

Remark 1.38 (Action of $SL_2(\mathbb{Z})$ on some lines in $\mathbb{R}^2$). It is worth to spend a couple of words about the action of $SL_2(\mathbb{Z})$ on some lines in the plane passing through the origin, as we will need this consideration to talk about HFT. Consider the line of slope p/q , for $p, q \in \mathbb{Z}$ coprime, namely $L = \left\{ t \cdot \begin{pmatrix} q \\ p \end{pmatrix} \mid t \in \mathbb{R} \right\}$. Then, for any $n \in \mathbb{Z} \setminus \{0\}$ we have that

$$\tau_1^n(L) = \left\{ t \cdot \begin{pmatrix} q \\ p + n \cdot q \end{pmatrix} \mid t \in \mathbb{R} \right\}$$

Figure 1.8: Action of $SL_2(\mathbb{Z})$ on $\mathcal{S}_4^2$.

which is the line of slope $(p + nq)/q = n + p/q$ and

$$\tau_2^n(L) = \left\{ t \cdot \begin{pmatrix} n \cdot p + q \\ p \end{pmatrix} \mid t \in \mathbb{R} \right\}$$

which is the line of slope $p/(np + q) = 1/(n + q/p)$.

1.5 Rational tangles

We now specialise in the case of *rational tangles*. This family of tangles is going to be quite useful later, when we will construct Heegaard diagrams for general tangles: in fact, we will often cut a tangle in rational pieces, we will construct diagrams for those and we will glue them back together, obtaining a Heegaard diagram for the initial tangle.

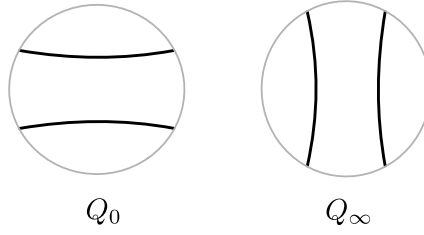
For this section, we mainly follow [Aub22], Sections 2.2 and 2.3.

Rational tangles are very special 4-ended tangles; their most peculiar property is the possibility to fully describe one of them with a fraction of the form p/q for $p, q \in \mathbb{Z}$ coprime and $q > 0$. We will usually write that $p/q \in \mathbb{QP}^1$, where $\mathbb{QP}^1$ is the rational projective line, represented as $\mathbb{QP}^1 \cong \mathbb{Q} \cup \{\infty\}$. One can think of elements of $\mathbb{QP}^1$ as the reduced slopes of lines in $\mathbb{Q}^2$, where $\infty = 1/0$ corresponds to the vertical line.

We start by defining the 0 rational tangle and the ∞ rational tangle.

Definition 1.39 (0 rational tangle and ∞ rational tangle). We call *0 rational tangle* (or *0 tangle*) the unknot tangle with two horizontal strings; it is denote by Q_0 .

We call *∞ rational tangle* (or *∞ tangle*) the unknot tangle with two vertical strings; it is denote by Q_∞ .

Figure 1.9: 0 rational tangle (on the left) and ∞ rational tangle (on the right).

To define all rational tangles, we need the notion of continued fraction.

Definition 1.40 (Continued fraction). Let $n \in \mathbb{Z}_{>0}$, $a_1, \dots, a_n \in \mathbb{Z}$ and $a_i \neq 0$ for $i = 2, \dots, n$. We define the *continued fraction* $[a_1, a_2, \dots, a_n]$ as

$$[a_1, a_2, \dots, a_n] := a_1 + [a_2, \dots, a_n]^{-1},$$

with the convention that $[\] = \infty$.

Remark 1.41 (On the name "continued fraction"). The name "continued fraction" is justified by the behaviour of these objects. In fact, given a continued fraction $A = [a_1, a_2, \dots, a_n]$, it behaves as the fraction

$$[a_1, a_2, \dots, a_n] \approx a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{\ddots + \frac{1}{a_n}}}}.$$

We have therefore a map that sends a continued fraction to an element of $\mathbb{QP}^1$. This map is evidently not injective; for instance, if we consider $A_1 = [2, 3, -2]$ and $A_2 = [3, -2, 3]$, we can see that they have the same image:

$$\begin{aligned} A_1 &\mapsto 2 + \frac{1}{+3 - \frac{1}{2}} = -2 + \frac{2}{5} = \frac{12}{5} \\ A_2 &\mapsto 3 + \frac{1}{-2 + \frac{1}{3}} = 3 - \frac{3}{5} = \frac{12}{5}. \end{aligned}$$

We therefore say that $A_1 \equiv A_2 \in \mathbb{QP}^1$ if two continued fractions are sent to the same element of $\mathbb{QP}^1$ by this map.

We can now link this notion of continued fraction with the construction of some particular tangles. Recall the operators τ_1 and τ_2 , defined in Section 1.4, which add half-twists on a tangle. When we talk about (half-) twists, we distinguish between *positive twists* and *negative twists*: these are defined by the slope of the over-strand: if that is positive, then the twist is positive, otherwise it is a negative twist. Hence, we will write τ_i^n for $n \in \mathbb{Z}$ to indicate the operation of adding n half-twists, positive if $n > 0$ and negative otherwise.

We can define rational tangles.

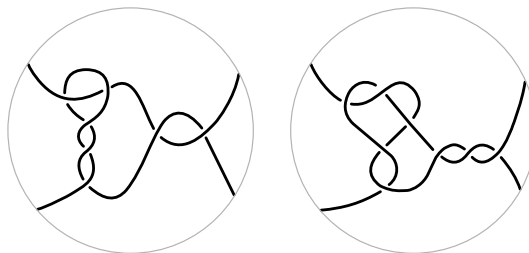


Figure 1.10: Two rational tangles that come from different successions but are equivalent.

Definition 1.42 (Rational tangles). *Rational tangles* are 4-ended tangles without any closed components that are obtained from adding twists to the bottom and to the right of the 0 tangle and the ∞ tangle.

Given a continued fraction $A = [a_1, a_2, \dots, a_n]$, we define the rational tangle Q_A as

$$Q_A = \begin{cases} \tau_1^{a_1} \tau_2^{a_2} \dots \tau_2^{a_n} Q_\infty & \text{if } n \text{ is even;} \\ \tau_1^{a_1} \tau_2^{a_2} \dots \tau_1^{a_n} Q_0 & \text{if } n \text{ is odd.} \end{cases}$$

An intuitive way to describe rational tangles is to give a succession of twist to operate on one of the trivial tangles.

This notation is quite intuitive, but unfortunately is not very effective if we want to have substantially different tangles from different inputs. For instance, the tangles defined by $[2, 3, -2]$ and $[3, -2, 3]$ are actually equivalent (Figure 1.10).

In 1970, John H. Conway proved the following theorem, for which a combinatorial proof was given in [KL03]. Thanks to this result, it is fairly easy to check whether or not two rational tangles are equivalent.

Theorem 1.43 (Conway, [Con70]). *Let A_1 and A_2 be two continued fractions. The rational tangles Q_{A_1} and Q_{A_2} are isotopic if and only if $A_1 \equiv A_2 \in \mathbb{QP}^1$.*

Hence, using continued fractions is possible to see when two rational tangles are actually equivalent. We can try to see it in the previous example with $T_1 = Q_{[2,3,-2]}$ and $T_2 = Q_{[3,-2,3]}$: as seen in Remark 1.41, the continued fractions are sent to the same element of $\mathbb{QP}^1$. Therefore T_1 and T_2 are indeed the same rational tangle and we are going to denote it by $Q_{12/5}$.

In general, given the cleaned continued fraction p/q for $p, q \in \mathbb{Z}$ coprime, we denote the associated rational tangle by $Q_{p/q}$. In general, we will assume that q is always positive and that $p \in \mathbb{Z}$.

1.6 Arithmetic of 4-ended tangles

One characteristic of 4-ended tangles that we will exploit in Chapter 5, is that we can define an arithmetic on them. This arithmetic of tangles will help us to slim the notation and to further understand why the class of rational tangles is so special. The operations that we are now going to define are:

- addition,
- multiplication,
- rotation,
- mirroring,
- inversion,
- flipping.

All of these are well-defined up to isotopy; therefore we can consider them as operations between classes of tangles.

For this section we mainly follow Section 2 of [KL03] and we assume all the tangles to be 4-ended tangles.

We start with the two operations that involve a pair of tangles.

Definition 1.44 (Addition, multiplication). Let T and S be two tangles, which we represent as a circle (inside which the tangle has all its twists) with four strings that come out from the angles, the ends of the 2 strings of it.

We define the *sum* $T + S$ and the *multiplication* $T * S$ of the tangles as the two tangles showed in Figure 1.11.

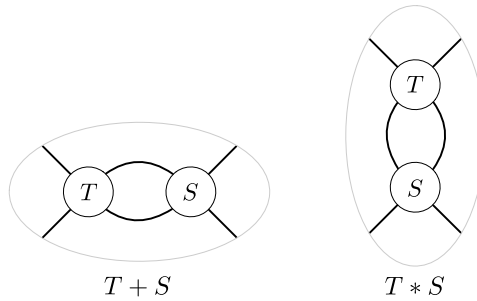


Figure 1.11: Sum and multiplication of two 4-ended tangles.

The remaining operation are defined on a single tangle, as follows.

Definition 1.45 (Rotation). We define the *rotation* of a tangle T_* , denoted by $r(T_*)$, to be a counter-clock wise rotation of the tangle by $\pi/2$.

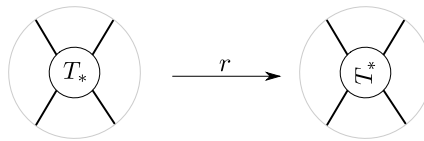


Figure 1.12: Rotation of a tangle T_* .

Definition 1.46 (Mirror image). We define the *mirror image* of a tangle T , denoted by $m(T)$, to be the tangle obtained by swapping all the crossings by T .

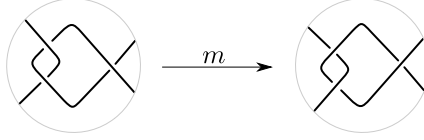


Figure 1.13: Mirror image of a tangle.

Definition 1.47 (Inverse). We define the *inverse* of a tangle T , denoted by $i(T)$, to be the tangle $m(r(T))$.

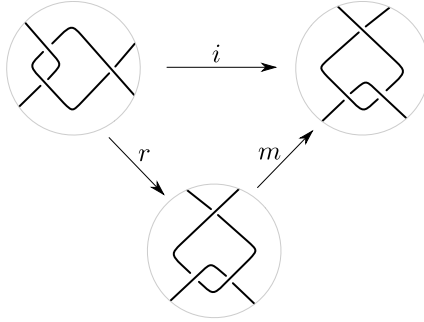


Figure 1.14: Inverse of a tangle.

Remark 1.48. There are several properties that one could prove that explain better of how these operations behave when composed. For instance, one can prove that $m(T + S) = m(T) + m(S)$, $m(T * S) = m(T) * m(S)$ or that for rational tangles holds $m(Q_{p/q}) = Q_{-p/q}$.

Another operation that will be very useful is the *flip* of a tangle.

Definition 1.49 (Flip). Let T_* be a tangle. A *flip* is a rotation in the third dimension by π around some axis. We define the *horizontal flip* of T_* , denoted by T_*^{hflip} , as its rotation around the horizontal axis and the *vertical flip* of T_* , denoted by T_*^{vflip} , as its rotation around the vertical axis.

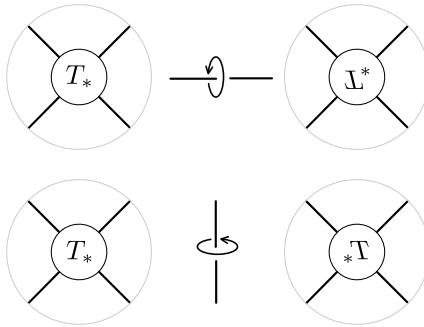


Figure 1.15: Horizontal flip (on top) and vertical flip (on bottom).

Lastly, we define a particular isotopy obtained with a composition of the above operations. This isotopy will be essential to prove an important result of stability under some operations for rational tangles. Consider a tangle T_* and operate the following isotopy: fix the 4 ends of the tangle and flip only the inner box, either horizontally or vertically.

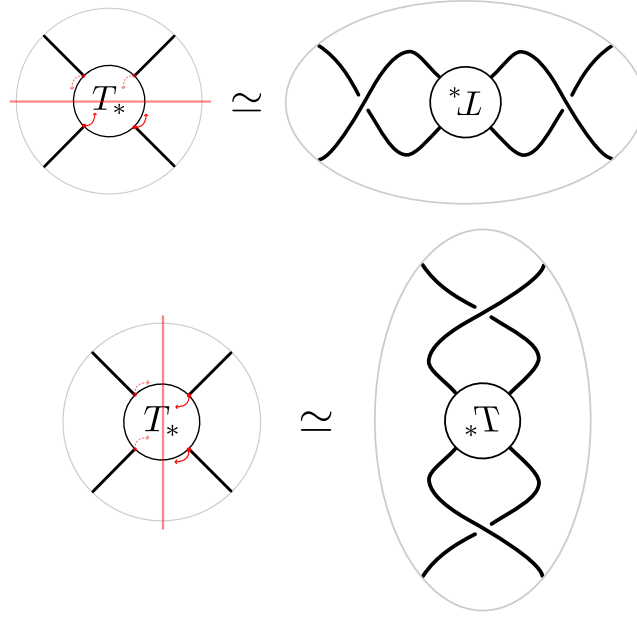


Figure 1.16: Flipping of the inner box with the ends fixed, horizontally (on top) and vertically (on bottom).

The fact that this is an isotopy comes straight from the definition, as we are just re-arranging the inner part of the tangle in the 3-dimensional space without moving the endpoints. using the newly acquired vocabulary, we can say that we proved

$$\begin{aligned} T_* &\simeq Q_1 * T_*^{flip} * Q_{-1} \\ T_* &\simeq Q_1 + T_*^{hflip} + Q_{-1}. \end{aligned}$$

Hence, using the fact that the operations are well defined between classes of isotopy, we can add in both equations Q_{-1} to the left of both sides and obtain

$$\begin{aligned} Q_{-1} * T_* &\simeq T_*^{flip} * Q_{-1} \\ Q_{-1} + T_* &\simeq T_*^{hflip} + Q_{-1}, \end{aligned}$$

since for the second Reidemeister move $Q_{-1} + Q_1 \simeq Q_0 \simeq Q_{-1} * Q_1$. Similarly, it holds also

$$\begin{aligned} Q_1 * T_* &\simeq T_*^{flip} * Q_1 \\ Q_1 + T_* &\simeq T_*^{hflip} + Q_1. \end{aligned}$$

Definition 1.50 (Flype). We call *flypes* the isotopies of the above form. We distinguish between *horizontal flypes* and *vertical flypes* according with the above definition of flips.

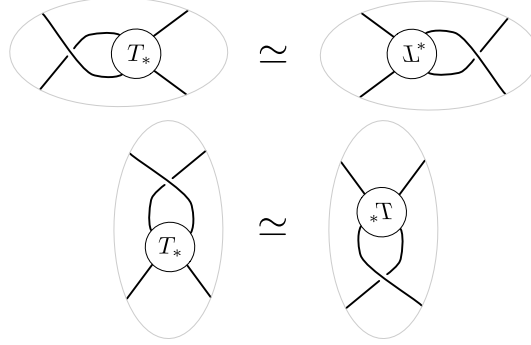


Figure 1.17: Flype isotopies for a tangle T_* : horizontal flype (on top) and vertical flype (on bottom).

We are now able to prove a Lemma which states three distinctive properties of rational tangles, which will be of use in Chapter 5.

Lemma 1.51 (Flipping Lemma, [KL03]). *Let R be a rational tangle. Then*

1. $R \simeq R^{vflip}$;
2. $R \simeq R^{hflip}$;
3. $R \simeq i(i(R)) = r(r(R))$.

Proof ([KL03]). The proof of the first two isotopies is done by induction on the number of crossings. It is clear that the statement holds in the base case, i.e. when R is Q_∞ , Q_1 or Q_{-1} . Suppose it to be true for all the rational tangles with n crossings. Let R be a rational tangle with $n + 1$ crossings; we have then eight possible cases:

- (a) $R = L + Q_{\pm 1}$,
- (b) $R = Q_{\pm 1} + L$,
- (c) $R = L * Q_{\pm 1}$,
- (d) $R = Q_{\pm 1} * L$,

where L is a suitable rational tangle with n crossings. We then prove case (a) as in Figure 1.18 and (d) as in Figure 1.19. The other cases are done analogously.

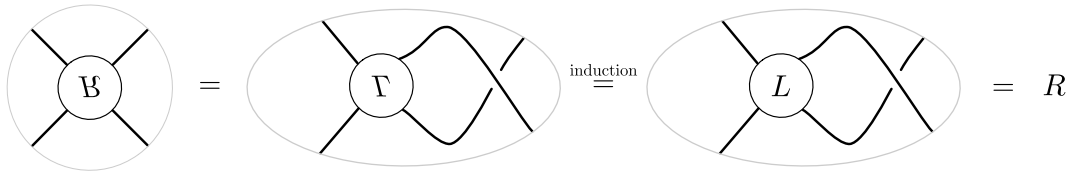


Figure 1.18: Inductive proof of the Flipping Lemma, case (a).

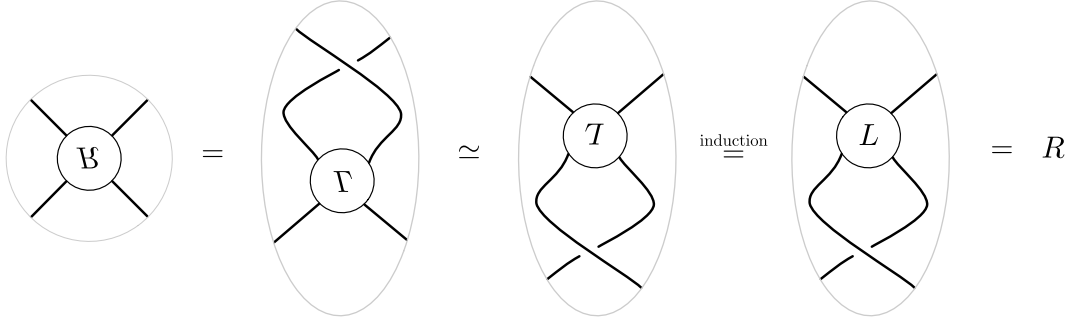


Figure 1.19: Inductive proof of the Flipping Lemma, case (d). The isotopy comes from a vertical flype and the equality comes from applying induction two times.

Lastly, the proof of the third isotopy follows from the first two, as we have that $i(i(R)) = (R^{flip})^{vflip} = r(r(R))$. $\square$

Apply the Flipping Lemma to the definition of flype, we obtain the following corollary.

Corollary 1.52. *For R a rational tangle holds*

$$\begin{aligned} Q_{\pm 1} + R &\simeq R + Q_{\pm 1} \\ Q_{\pm 1} * R &\simeq R * Q_{\pm 1}. \end{aligned}$$

Therefore, more generally, we have that the rational tangles of the form $Q_{\pm n}$ commute with any other rational tangle under addition and that $Q_{\pm 1/n}$ commute with any other rational tangle under multiplication.

Remark 1.53. The above corollary justify the definition of rational tangle gave in Section 1.5: in fact, one could define a rational tangle to be a tangle obtained from Q_0 or Q_∞ by a succession of twists operate on any two adjacent ends of the tangle (not only on the right or on the bottom as we did). But then, with Corollary 1.52, we are able to take a rational tangle constructed with this other definition and "push" all the twists on the right and on the bottom of the tangle, obtaining a rational tangle according to the definition that we gave.

1.7 Bordered and sutured manifolds

The focus of this thesis is on 4-ended tangles and tangle Floer homology HFT. In [Zib17], the author presented it with a close correlation with sutured Floer homology SHF, developed by Andreas Juhász in [Juh06] and bordered sutured Floer homology, established by Rumén Zarev in [Zar11]. Therefore, we briefly re-visit bordered manifolds with the point of view of bordered Heegaard Floer homology and we discuss sutured manifolds and bordered sutured manifolds to better understand the context that we are in.

The interest in the theory of bordered sutured manifolds comes from the possibility to glue such manifolds together and the nice behaviour of this gluing at the level of invariants, which is described with a useful gluing formula for bordered sutured Floer homology ([Zar11]). This gluing property was then adapted to the tangle case, adapting arguments from bordered sutured Floer homology. Further details can be found in Section II.3 of [Zib17].

1.7.1 Bordered manifolds

For this subsection, one can use Sections 3.2 and 4.1 of [LOT18] as reference.

In Section 1.1, we defined 3-manifolds with border as it is done classically. However, we need to briefly re-visit this notion in the bordered Heegaard Floer homology context with the use of *pointed matched circles*.

Definition 1.54 ((Pointed) Matched circle). A *matched circle* is a triple $(Z, \mathbf{a}, M)$, where

1. Z is of an oriented circle,
2. $\mathbf{a} = \{a_1, \dots, a_{4k}\} \subset Z$ is a set of $4k$ distinct points (which we view as 0-spheres),
3. M is a matching (i.e. a 2-to-1 map $\mathbf{a} \rightarrow \{1, \dots, 2k\}$).

Moreover, we require the following property

4. if we perform surgery along the $2k$ pairs of points in Z , we obtain a single circle and not several disjoint circles.

A *pointed matched circle* $\mathcal{Z}$ is a matched circle together with a basepoint $z \in Z \setminus \mathbf{a}$.

Remark 1.55 (Order on $\mathbf{a}$). For a pointed matched circle, there is an order in $\mathbf{a}$ induced by the orientation of Z , with the minimum that is the first point encountered starting from the basepoint.

Remark 1.56. We will represent a pointed matched circle either as a circle or as a segment, whose endpoints are identified and are the basepoint z . See Figure 1.20

Given a pointed matched circle $\mathcal{Z}$, we can construct a unique genus k surface $F(\mathcal{Z})$, called the *surface associated to $\mathcal{Z}$* , in the following way. We take a disc with boundary, whose boundary is Z and we attach oriented 2-dimensional 1-handles along the pairs specified by M . Then, for condition 4 in the definition, we obtain one boundary component homeomorphic to a circle and we fill it with another disc.

Example 1.57. If we take the pointed circle in Figure 1.20, then the surface represented is the torus.

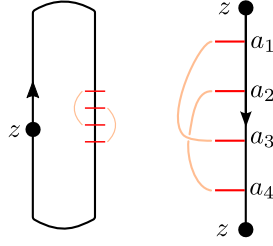


Figure 1.20: Pointed matched circle representing the torus: as a circle (left) and as a segment with identified endpoints (right).

We can now define bordered manifolds in this context, a subclass of 3-manifolds with boundary whose boundary is parametrised by a pointed matched circle.

Definition 1.58 (Bordered 3-manifold). A *bordered 3-manifold* is a triple $(Y, \mathcal{Z}, \phi)$, where

- Y is a compact, oriented 3-manifold with connected boundary ∂Y ,
- $\mathcal{Z}$ is a pointed matched circle,
- $\phi : F(\mathcal{Z}) \rightarrow \partial Y$ is an orientation-preserving homeomorphism.

We say that two bordered 3-manifolds $(Y, \mathcal{Z}, \phi)$ and $(Y', \mathcal{Z}', \phi')$ are *equivalent*, if there exists an orientation-preserving homeomorphism $\psi : Y \rightarrow Y'$ such that $\phi' = \psi \circ \phi$.

Remark 1.59. When we talk about some bordered 3-manifold $(Y, \mathcal{Z}, \phi)$, we will often just denote it by Y and we will be considering the equivalent class of $(Y, \mathcal{Z}, \phi)$.

1.7.2 Sutured manifolds

For this subsection, we mainly follow Sections 2.1 and 3.1 of [Zar09].

We first build our way to *sutured manifolds*; we do this via *divided surfaces*.

Definition 1.60 (Divided surface). A *divided surface* (S, Γ) is the data of

- a closed and oriented surface S ,
- a set of pairwise disjoint, simple, closed, oriented curves $\Gamma = \{\gamma_1, \dots, \gamma_n\}$ on S , which we call *sutures*.

We then define $R(\Gamma) = S \setminus \Gamma$, where every connected component A has the boundary $\partial A \subset \Gamma$, i.e. $\partial A = \cup_{i \in I} \gamma_i$ for some subset $I \subset \{1, \dots, n\}$. Moreover, we impose the following conditions: let A be a connected component of $R(\Gamma)$, then

- A has non-empty boundary,

- consider the orientation on ∂A induced from the orientation of S , then either the orientation on ∂A agrees with the orientation of the suture γ_i on each component of ∂A , or they disagree on every components. If they agree, we say that A is a *positive region*, otherwise we say that it is a *negative region*.

We then divide $R(\Gamma)$ in two pieces: we call $R_+(\Gamma)$ (or R_+) and $R_-(\Gamma)$ (or R_-) the sets

$$R_+(\Gamma) := \overline{\bigcup_{\substack{A \text{ s.t. is} \\ \text{positive region}}} A} \quad R_-(\Gamma) := \overline{\bigcup_{\substack{A \text{ s.t. is} \\ \text{negative region}}} A},$$

where the line above indicate the topological closure of the set below.

Remark 1.61. We do not require S to be connected, but thank to the property of every connected component of $R(\Gamma)$ to have non-trivial boundary, we are requiring each component to have at least one suture.

We then define the notion of balancedness for a divided surface.

Definition 1.62 ((Un-)Balanced divided surface). We say that a divide surface (S, Γ) is *balanced* if $\chi(R_+) = \chi(R_-)$. We say that it is k -unbalanced if $\chi(R_+) = \chi(R_-) + 2k$ for some k non-zero integer.

Remark 1.63. The last definition is well posed, since it is possible to prove that $\chi(R_+) - \chi(R_-)$ is always even. This follows from the facts that S is closed and that $\chi(S) = \chi(R_+) + \chi(R_-)$.

We can define *balanced* and *unbalanced sutured manifolds*.

Definition 1.64 ((Un-)Balanced sutured manifold). We say that (Y, Γ) is a *balanced sutured manifold* if Y is a 3-manifold without closed components such that $(\partial Y, \Gamma)$ is a balanced divided surface.

We say that (Y, Γ) is a k -*unbalanced sutured manifold* if Y is a 3-manifold without closed components such that $(\partial Y, \Gamma)$ is a k -unbalanced divided surface.

Remark 1.65 (Sutured manifold associated to a knot). Following what is done in Section 9 of [Juh06], we see how it is possible to regard knots as a particular case of sutured manifolds.

Consider a knot $K \subset \mathcal{S}^3$. Then, we call $(Y_K, \{s_1, s_2\})$ the *sutured manifold associated to K* , where

- Y_K is the manifold with boundary $Y_K = \mathcal{S}^3 \setminus \nu(K)$, with $\nu(K)$ is a tubular neighbourhood of K ,
- the sutures s_1 and s_2 are two oppositely oriented meridional circles in some part of ∂Y_K .

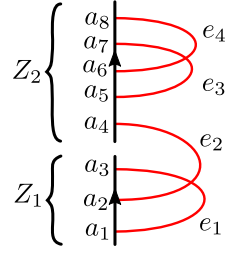


Figure 1.21: Graph of arc diagram.

1.7.3 Bordered sutured manifolds

We want to talk about bordered sutured manifold, a more generalised class of manifolds which contains both bordered manifolds and sutured manifolds. For this, we need the notion of *arc diagram*, a generalisation of pointed matched circle.

As reference, one can look at Section 3.2 of [Zar09].

Definition 1.66 ((Degenerate) Arc diagram). An *arc diagram* $\mathcal{Z} = (Z, \mathbf{a}, M)$ is a triple consisting of

- a collection $Z = \{Z_1, \dots, Z_l\}$ of oriented, open line segments for some $l \geq 0$,
- a collection $\mathbf{a} = \{a_1, \dots, a_{2k}\}$ of distinct points in Z for some $k \geq 0$; we denote by $|Z_i|$ the cardinality of $Z_i \cap \mathbf{a}$,
- a matching M of $\mathbf{a}$, i.e. a 2-to-1 function $\mathbf{a} \xrightarrow{M} \{1, \dots, k\}$,
- it holds the following *non-degeneracy condition*: if we perform surgery on the 1-manifold Z at each $M^{-1}(i)$, which are homeomorphic to $\mathcal{S}^0$, then the resulting 1-manifold cannot have closed components.

If an arc diagram does not satisfy the non-degeneracy condition, we say that it is *degenerate*.

We put on $\mathbf{a}$ the following order: $a_i < a_j$ if $a_i \in Z_{h_i}$, $a_j \in Z_{h_j}$ and $h_i < h_j$; if two points are on the same line segment, we use the order induced by the orientation of the segment.

Given an arc diagram, we can define its graph.

Definition 1.67 (Graph of an arc diagram). consider an arc diagram $\mathcal{Z} = (Z, \mathbf{a}, M)$. We define its *graph*, denoted by $G(\mathcal{Z})$, as the graph obtained by the line segments by adding an edge for each couple of matched points in $\mathbf{a}$ with endpoints on the the two matched points. We denote such edges by $e_1, \dots, e_k$ (where e_i is the edge associated to the matched points $M^{-1}(i)$). See Figure 1.21 for an example.

We can now define *bordered sutured manifolds*, the class of sutured manifolds on which is defined bordered sutured Floer homology.

Definition 1.68 (Bordered sutured manifold). A *bordered sutured manifold* is a tuple $(Y, \Gamma, \mathcal{Z}, \phi)$, where

- (Y, Γ) is a sutured manifold,
- $\mathcal{Z}$ is an arc diagram,
- $\phi : G(\mathcal{Z}) \hookrightarrow \partial Y$ is an orientation-preserving embedding, with
 - $\phi|_{\mathcal{Z}}$ is an orientation preserving embedding into Γ ,
 - $\phi(G(\mathcal{Z}) \setminus \mathcal{Z}) \cap \Gamma = \emptyset$.

Remark 1.69. From the definition of bordered sutured manifold follow that each edge e_i of $G(\mathcal{Z})$ embeds in R_- .

To better set us for the construction of Heegaard diagrams fro bordered sutured manifold, we also define *bordered sutured manifolds with α - and β -arcs*, following what is done in Section 3.1 of [Zib20]. For this, we need *arc diagrams of rank k* .

Definition 1.70 (Arc diagram of rank k). We define an *arc diagram of rank k* to be an arc diagram $\mathcal{Z} = (Z, \mathbf{a}, M)$ such that $\mathbf{a}$ contains $2k$ points and requiring also a type between " α " and " β ". We usually write $\mathcal{Z}_\alpha$ if it is of type α and $\mathcal{Z}_\beta$ if it is of type β .

Definition 1.71 (Bordered sutured manifold with α - and β -arcs). A *bordered sutured manifold with α - and β -arcs* is a tuple $(Y, \Gamma, \mathcal{Z}_\alpha, \phi_\alpha, \mathcal{Z}_\beta, \phi_\beta)$, where

- (Y, Γ) is a sutured manifold,
- $\mathcal{Z}_\alpha = (Z_\alpha, \mathbf{a}_\alpha, M_\alpha)$ and $\mathcal{Z}_\beta = (Z_\beta, \mathbf{a}_\beta, M_\beta)$ are (possibly degenerate) arc diagrams,
- $\phi_\alpha : G(\mathcal{Z}_\alpha) \hookrightarrow \overline{R_-}$ is an embedding such that $\phi_\alpha(Z_\alpha) \subset \Gamma$,
- $\phi_\beta : G(\mathcal{Z}_\beta) \hookrightarrow \overline{R_+}$ is an embedding such that $\phi_\beta(Z_\beta) \subset \Gamma$,
- $\phi_\alpha(Z_\alpha) \cap \phi_\beta(Z_\beta) = \emptyset$,
- the two arc diagrams are *homologically linear independent*, i.e. the map

$$\pi_0(\Gamma \setminus \phi_\alpha(Z_\alpha) \cup \phi_\beta(Z_\beta)) \longrightarrow \pi_0(\partial Y \setminus (\text{Im}\phi_\alpha \cup \text{Im}\phi_\beta))$$

is surjective. This condition assures us that the bordered sutures Heegaard Floer homology is well define. It is possible to prove that if the arc diagrams are non-degenerate, then this property follows from the others (see Proposition 3.6 of [Zar09]).

Remark 1.72 (Tangles as particular bordered sutured manifolds). We see how we can consider a tangle as a bordered sutured manifold. Recall that we defined the α -arc sites of a tangle with n open components to be the connected segments of the fixed circle $\mathcal{S}$ minus the images of the endpoints of the n strings.

Let T be a tangle with n open components and m closed components. We can associate to T a bordered sutured 3-manifold with α -arcs $(X_T, \Gamma, \mathcal{Z}_\alpha, \phi_\alpha)$ in the following way:

- the underlying 3-manifold with boundary is $X_T = B^3 \setminus \nu(T)$, where $\nu(T)$ is a tubular neighbourhood of T in B^3 ;
- we place two oppositely oriented meridional circles around each closed components,
- we place around each tangle end a single suture, where the orientation is chosen such that the boundary of B^3 minus a neighbourhood of the tangle ends lies in R_- ,
- on each one of the four sutures around the tangle ends, we place two "ticks", distant from the end points of the α -arcs and such that there is one tick in each of the connected component of the suture obtained by removing the α -arcs endpoints,
- we define the arc diagram $G(\mathcal{Z}_\alpha)$ via the α -arc sites, defining the line segments Z as the suture opened in correspondence of the ticks.

Chapter 2

Heegaard diagrams

In this chapter, we develop Heegaard diagrams for all the objects that we defined in Chapter 1. We start by defining Heegaard diagrams for closed 3-manifolds, which was also historically the first theory developed. It will be clear from the construction that these diagrams follow quite naturally from Heegaard splittings defined in Section 1.2. However, the construction of the hat version of Heegaard Floer homology $\widehat{HF}$ that we will see in Section 3.1 heavily relies on the notion of *pointed Heegaard diagrams*: we define them in Section 2.2 as diagrams to which we add a *basepoint*. This notion of basepoint seems to appear out of thin air, magically helping us to define a topological invariant for 3-manifolds; it is only in the more general context of sutured manifolds that it is better understood. In fact, we see in Section 2.4 how we are able to consider Heegaard diagrams of closed 3-manifolds as a special case of Heegaard diagrams of sutured manifolds (Remark 2.35).

2.1 Heegaard diagrams for closed 3-manifolds

We can now start to talk about Heegaard diagrams for closed 3-manifolds, which permit us to "forget" the handlebodies of a Heegaard splitting and to deal only with the common surface Σ . The main references for this section are [OS06] and [Hom19].

We first need to define an object which allows us to retrieve in some way the handlebodies structures from their common boundary Σ .

Definition 2.1 (Set of attaching circles). Let H be a genus g handlebody and let Σ_g be its boundary. A *set of attaching circles* $\Gamma = \{\gamma_1, \dots, \gamma_g\}$ for H is a collection of closed embedded curves in Σ_g with the following properties:

1. the curves γ_i are all disjoint from each other;
2. $\Sigma_g \setminus \Gamma$ is connected;
3. the curves γ_i bound disjoint embedded discs in H .

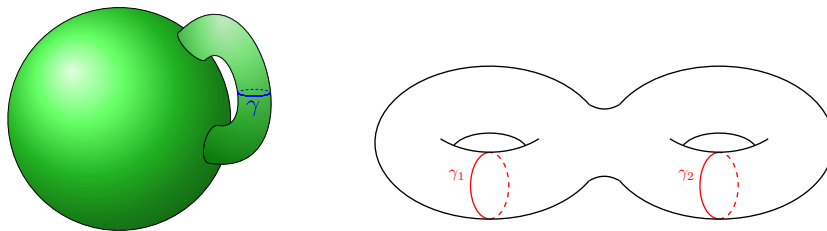


Figure 2.1: Set of attaching circles for a genus 1 (left) and a genus 2 (right) handlebodies.

Remark 2.2. The second property given above is actually equivalent to say that $\{[\gamma_1], \dots, [\gamma_g]\}$ are linearly independent in $H_1(\Sigma, \mathbb{Z})$.

Definition 2.3 (Compatible Heegaard diagram). Consider a genus g Heegaard splitting $Y = H_1 \cup_{\Sigma_g} H_2$ for Y . Then a *compatible Heegaard diagram* is given by $(\Sigma_g, \alpha, \beta)$, where $\alpha = \{\alpha_1, \dots, \alpha_g\}$ is a set of attaching circles for H_1 and $\beta = \{\beta_1, \dots, \beta_g\}$ is a set of attaching circles for H_2 and the intersections between α - and β -circles are transverse. We are going to draw the α -circles in red and the β -circles in blue.

One thing that we can immediately notice is that, given a Heegaard splitting of genus > 1 , it admits a lot of different compatible Heegaard diagrams. On the other way, starting from a Heegaard diagram we have an unique Heegaard splitting for which it is compatible. Therefore from now on we will work predominantly with Heegaard diagrams.

Example 2.4 (Some examples). We now show some Heegaard diagrams starting from the decompositions given in Example 1.17.

1. Heegaard diagram of genus 1 of $\mathcal{S}^3$

The genus 1 Heegaard splitting of $\mathcal{S}^3$ that we saw in Example 1.17 corresponds to a diagram (Σ, α, β) , where the two curves meet transversely at one point x ; see Figure 2.2.

2. Heegaard diagram of genus 1 of $L(p, q)$

The genus 1 Heegaard splitting of $L(p, q)$ corresponds to a genus 1 Heegaard diagram (Σ, α, β) . We denote s and t the generators of $H_1(\Sigma) = \mathbb{Z}^2$, as in Figure 2.3.

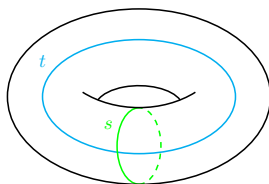


Figure 2.3: Generators of the first homology group of a torus.

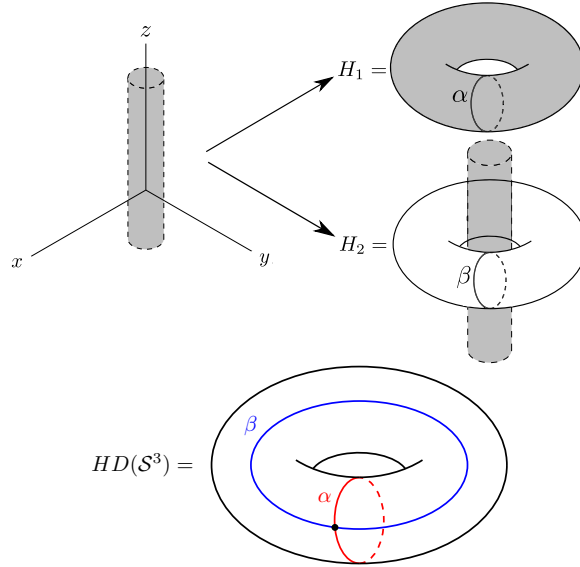


Figure 2.2: Sets of attaching circles for the Heegaard splitting of $\mathcal{S}^3 \simeq \mathbb{R}^3 \cup \{\infty\}$ presented in Example 1.17 (on top) and the respective Heegaard diagram (on bottom).

With this notation, we have that the curves of the diagram are described as $[\alpha] = s$ and $[\beta] = qs + pt$ and they are such that intersect at p points.

3. Heegaard diagram of genus 1 of $L(3, 1)$

Let us make a more explicit example with $L(3, 1)$: we have a diagram (Σ, α, β) where $[\alpha] = s$ and $[\beta] = s + 3t$.

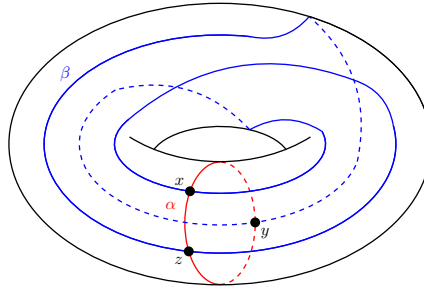


Figure 2.4: Heegaard Diagram of genus 1 for $L(3, 1)$.

Remark 2.5 (Retrieve manifold from its Heegaard diagram). How can we retrieve the original manifold Y from its Heegaard diagram $(\Sigma_g, \alpha, \beta)$? We can construct it in the following way: thicken Σ_g to $\Sigma_g \times [0, 1]$ and glue thickened discs along the $\alpha_i \times \{0\}$ and along the $\beta_j \times \{1\}$; the resulting space has two boundary components, one "on top" and one "on bottom", each homeomorphic to $\mathcal{S}^2$ and we cap each with a three-ball.

$$\begin{aligned}
Y &= H_1 \cup_{\Sigma_g} H_2 \\
&= \left((\Sigma_g \times [0, 1]) \underbrace{\sqcup_{\alpha \times I} (\mathbb{D}^2 \times I)}_{\text{att. circles for } H_1} \underbrace{\sqcup_{\beta \times I} (\mathbb{D}^2 \times I)}_{\text{att. circles for } H_2} \right) \underbrace{\sqcup_0 (B^3)}_{\substack{\text{3-ball} \\ \text{attached} \\ \text{"on bottom"}}} \underbrace{\sqcup_1 (B^3)}_{\substack{\text{3-ball} \\ \text{attached} \\ \text{"on top"}}}. \quad (\boxtimes)
\end{aligned}$$

Lemma 2.6. *Let $(\Sigma_g, \alpha, \beta)$ be a genus g Heegaard diagram for Y . Then we have the following isomorphism of homology groups*

$$H_1(Y) \cong \frac{H_1(\Sigma_g)}{\langle [\alpha_1], \dots, [\alpha_q], [\beta_1], \dots, [\beta_q] \rangle}. \quad (\star)$$

We want to prove $H_1(\tilde{Y}) \cong H_1(\Sigma_g)/\langle [\alpha_1], \dots, [\alpha_g], [\beta_1], \dots, [\beta_g] \rangle$; we use Mayer-Vietoris on the decomposition

$$\begin{aligned}\tilde{Y} &= (\Sigma_g \times [0, 1]) \cup (\sqcup_{\alpha \times I} (\mathbb{D}^2 \times I) \sqcup_{\beta \times I} (\mathbb{D}^2 \times I)) \\ &= (\Sigma_a \times [0, 1]) \cup A.\end{aligned}$$

$$\begin{array}{c} \cdots H_1(\Sigma_g \times [0, 1] \cap A) \xrightarrow{(i_{1*}, i_{2*})} H_1(\Sigma_g \times [0, 1]) \oplus H_1(A) \xrightarrow{g_* - j_*} H_1(\widetilde{Y}) \\ \quad d \\ \hline \hookrightarrow H_0(\Sigma_g \times [0, 1] \cap A) \xrightarrow{(i_{1*}, i_{2*})} H_0(\Sigma_g \times [0, 1]) \oplus H_0(A) \xrightarrow{g_* - j_*} H_0(\widetilde{Y}) \longrightarrow 0, \end{array}$$
$$\begin{array}{ccccc} & & \Sigma_g \times [0, 1] & & \\ & \nearrow i_1 & & \searrow g & \\ \Sigma_g \times [0, 1] \cap A & & & & \tilde{Y} \\ & \searrow i_2 & & \nearrow j & \\ & & A & & \end{array}$$

We start by proving that $d \equiv 0$, i.e. that the map (i_{1*}, i_{2*}) at level zero $H_0(\Sigma_g \times [0, 1] \cap A) \xrightarrow{(i_{1*}, i_{2*})} H_0(\Sigma_g \times [0, 1]) \oplus H_0(A)$, is injective. This is true:

in fact, the connected components of $\Sigma_g \times [0, 1] \cap A$ are exactly the connected components of A , hence are sent injectively by (i_{1*}, i_{2*}) . Hence, we have proven that $d \equiv 0$ and we obtained the exact sequence

$$\cdots \cdots H_1(\Sigma_g \times [0, 1] \cap A) \xrightarrow{(i_{1*}, i_{2*})} H_1(\Sigma_g \times [0, 1]) \oplus H_1(A) \xrightarrow{g_* - j_*} H_1(\tilde{Y}) \longrightarrow 0.$$

We can now prove the isomorphism that we claimed. We need the following observations:

- $g_* - j_*$ is surjective;
- $H_1(\Sigma_g \times [0, 1]) \cong H_1(\Sigma_g)$;
- $H_1(A) = H_1(\sqcup_{\alpha \times I} (\mathbb{D}^2 \times I) \sqcup_{\beta \times I} (\mathbb{D}^2 \times I)) = 0$ (since it is contractible);
- $(\Sigma_g \times [0, 1]) \cap (\sqcup_{\alpha \times I} (\mathbb{D}^2 \times I) \sqcup_{\beta \times I} (\mathbb{D}^2 \times I)) = (\alpha \times [0, 1]) \sqcup (\beta \times [0, 1])$;
- $H_i(\alpha \times [0, 1]) \sqcup (\beta \times [0, 1]) = H_i(\alpha \sqcup \beta)$ for any i ;
- the map $H_1(\Sigma_g \times [0, 1] \cap A) \xrightarrow{(i_{1*}, i_{2*})} H_1(\Sigma_g \times [0, 1]) \oplus H_1(A)$ is the map that sends any circle to its equivalence class in $\Sigma_g \times [0, 1]$.

Therefore, using the first isomorphism theorem, we get the isomorphism

$$H_1(\tilde{Y}) \cong \frac{H_1(\Sigma_g)}{\text{Ker}(g_* - j_*)} \cong \frac{H_1(\Sigma_g)}{\text{Im}(i_{1*}, i_{2*})} \cong \frac{H_1(\Sigma_g)}{\langle [\alpha_1], \dots, [\alpha_g], [\beta_1], \dots, [\beta_g] \rangle},$$

as we wanted. □

2.2 Heegaard moves

We stumble again in the same problem that we had with Heegaard splittings: giving a Heegaard diagram (Σ, α, β) we can construct an unique manifold Y , but the same manifold admits plenty of Heegaard diagrams. As we did with the stabilisation, we are going to define three "moves" that we can do on the sets of attaching circles α, β which do not change the manifold represented by the diagram.

Again, we mainly follow [OS06] and [Hom19] for this section.

Let $\{\gamma_1, \dots, \gamma_g\}$ be a set of attaching circles for a genus g handlebody H , let Σ be the border of H .

Definition 2.7 (Isotopy). An *isotopy* moves $\gamma_1, \dots, \gamma_g$ in a one parameter family in such way that the curves remain disjoint.

Definition 2.8 (Finger move). We will usually use the notion of *finger move* when modifying a Heegaard diagram. This is a particular type of isotopy, constructed as follows: let β_1 be a β -circle and c an arc that has one endpoint on β_1 and that does not intersect any other β -circle; then we push "with our finger" β_1 up c , drawing a small neighbourhood of c that does not touch any β -circle and leaving all the rest of β_1 unaffected. See Figure 2.5 for an example.

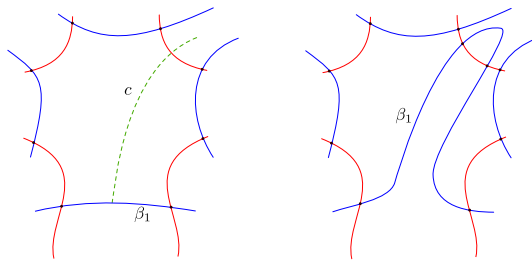


Figure 2.5: Example of a finger move (showed locally).

Definition 2.9 (Handleslide). Choose two of the curves, say γ_1, γ_2 for simplicity. To perform an *handleslide*, we replace the first one with γ'_1 , where this is a simple closed curve disjoint to any γ_i such that γ_1, γ_2 and γ'_1 bound an embedded pair of pants in $\Sigma \setminus \gamma_3 \setminus \cdots \setminus \gamma_g$.

We also have an equivalent definition of handleslide. Suppose that γ_1 and γ_2 can be connected by an arc δ in $\Sigma \setminus \gamma_3 \setminus \cdots \setminus \gamma_g$. Let γ'_1 be the connected sum of γ_1 with a parallel copy of γ_2 , where the connected sum is taken along a neighbourhood of δ . See the figure below for the graphic idea.

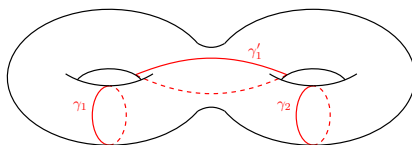


Figure 2.6: Handleslide as embedded pair of pants.

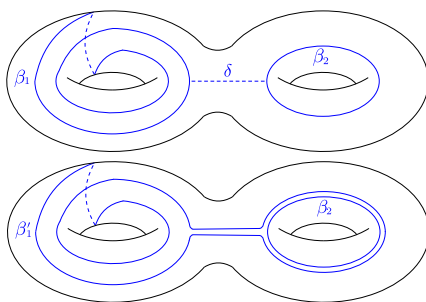


Figure 2.7: Handleslide using the connecting arc.

Lastly we define stabilisation also for Heegaard diagrams. Consider a genus g Heegaard diagram (Σ, α, β) .

Definition 2.10 (Stabilisation). We define the *stabilisation* as follows. Take a genus 1 surface E and consider the connect sum $\Sigma' = \Sigma \# E$, therefore we have a genus $g + 1$ surface; as sets attaching circles take $\alpha' = \alpha \cup \alpha_{g+1}$ and $\beta' = \beta \cup \beta_{g+1}$, where α_{g+1} and β_{g+1} are embedded on E and meet transversely in one point.

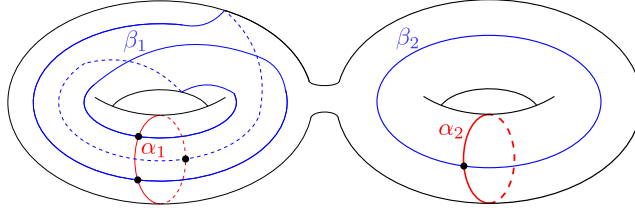


Figure 2.8: Stabilisation of the Heegaard Diagram of $L(3, 1)$.

Notice that, if (Σ, α, β) is compatible with the decomposition $Y = H_1 \cup_{\Sigma} H_2$, then the Heegaard diagram obtained by stabilisation is compatible with the decomposition obtained by stabilisation.

Remark 2.11. In accord to what we requested in Definition 3.7, we ask as well that these moves generate only transverse intersections between α - and β -curves.

As we had Theorem 1.24 for splittings, we also have the following Theorem for diagrams (see [Rei33] and [Sin33]).

Theorem 2.12. *Let Y be a closed and oriented 3-manifold, let $(\Sigma_g, \alpha, \beta)$ be a Heegaard diagram of genus g for Y and $(\Sigma_{g'}, \alpha', \beta')$ be a Heegaard diagram of genus g' for Y . Then, by applying a sequence of stabilisations, isotopies and handleslides, we can change the above diagrams in a way that the new diagrams are diffeomorphic to each other.*

To actually compute $\widehat{HF}$, we will need *pointed Heegaard diagrams*, i.e. Heegaard diagrams with a *basepoint*.

Definition 2.13 (Basepoint, pointed Heegaard diagrams). A *basepoint* on a Heegaard diagram (Σ, α, β) is a point z such that it stays on the complement of the curves, i.e. $z \in \Sigma \setminus (\alpha \cup \beta)$.

we then define a *pointed Heegaard diagram* to be a Heegaard diagram with a basepoint z ; this is denoted by $(\Sigma, \alpha, \beta, z)$.

Since we are going to work with pointed diagrams, we need to define *pointed Heegaard moves*. These are obtained as a generalisation of the Heegaard moves above presented.

Definition 2.14 (Pointed isotopy). A *pointed isotopy*, is an isotopy in which we require that the basepoint is always disjoint from the curves.

Definition 2.15 (Pointed handleslide). A *pointed handleslide*, is a handleslide in which we require that the basepoint is not in the pair of pants region where the handleslide take place.

For the stabilisations there are no differences. As we had Theorem 2.12, we have now the following proposition whose proof can be found in [OS01a].

Proposition 2.16. *Consider two pointed Heegaard diagrams for Σ , $(\Sigma, \alpha, \beta, z_1)$ and $(\Sigma, \alpha, \beta, z_2)$ which differ only for the basepoints. Then the two diagrams can be connected by a sequence of pointed isotopies and pointed handleslides.*

We then have the following corollary of Propositions 2.12 and 2.16.

Corollary 2.17. *Consider two different pointed Heegaard diagrams on Σ , namely $(\Sigma, \alpha, \beta, z)$ and $(\Sigma, \alpha', \beta', z')$. Assume moreover that $z \in \Sigma \setminus (\alpha' \cup \beta')$ and $z' \in \Sigma \setminus (\alpha \cup \beta)$. Then the two diagrams can be connected by a sequence of pointed isotopies and pointed handleslides.*

2.3 Heegaard diagrams rise from Morse functions

In Morse theory, one studies a particular class of smooth functions on n -dimensional manifolds. In this section, we see how we can construct Heegaard diagrams starting from Morse functions; as a reference for Morse theory, one case see [Mil63]. Even though this point of view is not exploited for the most part of this thesis, it is widely used in external literature and hence it is worth to briefly look into it.

For this section, we follow Section 3 of [OS06].

Definition 2.18 (Critical point, non degenerate point). Let f be a smooth function on an n -manifold Y , $f : Y \rightarrow \mathbb{R}$. A point $P \in Y$ is a *critical point of f* if $\partial f / \partial x_i = 0$ for $i = 1, \dots, n$. At a critical point the *Hessian matrix* $H(P)$ is given by the second partial derivatives $H_{ij} = \partial^2 f / \partial x_i \partial x_j$. A critical point P is called *non-degenerate* if $H(P)$ is non-singular.

Definition 2.19 (Morse function). The function $f : Y \rightarrow \mathbb{R}$ is called a Morse function if all the critical points are non-degenerate.

Definition 2.20 (Index of a critical point). Let f be a Morse function and P a critical point of f . Since $H(P)$ is symmetric, it induces an inner product on the tangent space. We call the *index of P* the dimension of a maximal negative definite subspace.

In other words, if we diagonalize $H(P)$ over the reals, and define $\text{index}(P)$ to be the number of negative entries in the diagonal.

Notice that a local minimum of f has index 0 and that a local maximum has index n .

We are only interested in a special class of Morse function, the *self-indexing* Morse functions.

Definition 2.21 (Self-indexing Morse function). A Morse function f is called *self-indexing* if for each critical point P we have $f(P) = \text{index}(P)$.

We have the following result, whose proof can be found in [Mil63].

Proposition 2.22. *Every smooth n -dimensional manifold Y admits a self-indexing Morse function. Furthermore if Y is connected and has no boundary, then we can choose f so that it has unique index 0 and index n critical points.*

By studying how the level sets $f^{-1}((\infty, t])$ change when t goes through a critical value, it is possible to prove the following two facts:

- If $f : Y \rightarrow [0, 3]$ is a self-indexing Morse function on Y with one minimum and one maximum, then f induces a Heegaard splitting with Heegaard surface $\Sigma = f^{-1}(3/2)$, and handlebodies $U_0 = f^{-1}[0, 3/2]$, $U_1 = f^{-1}[3/2, 3]$.
- Show that if Σ has genus g , then f has g index one and g index two critical points. We denote the index 1 critical points of f by $P_1, \dots, P_g$ and the ones of index 2 by $Q_1, \dots, Q_g$.

Moreover, from the Morse function is possible to define a Heegaard diagram on the manifold compatible with the splitting that we found above, as we can see in the next result.

Proposition 2.23. *The Morse function and a Riemannian metric on Y induce a Heegaard diagram for Y .*

Sketch of proof. Consider the gradient vector field ∇f of the Morse function. For each point $x \in \Sigma = f^{-1}(3/2)$ we can look at the gradient trajectory of $\pm \nabla f$ that goes through x . If we consider the critical point P_i , we call α_i the set of points that flow down to it; for the critical point Q_j , we call β_j the set of points that flow up to it.

In [Mil63], is proven the following theorem that studies the local behavior of a Morse function f around a critical point:

Let P be an index i critical point of f . Then there is a diffeomorphism h between a neighbourhood U of $0 \in \mathbb{R}^n$ and a neighbourhood U' of $P \in M^n$ so that

$$h \circ f = - \sum_{j=1}^i x_j^2 + \sum_{j=i+1}^n x_j^2.$$

It follows from the above result and the fact that f is self indexing that α_i, β_i are simple closed curves in Σ . Furthermore, it is also possible to see that $\alpha_1, \dots, \alpha_g$ and $\beta_1, \dots, \beta_g$ are set of attaching circles for U_0 and U_1 respectively; therefore this is a Heegaard diagram of Y compatible with the given Heegaard splitting. $\square$

2.4 Heegaard diagrams for bordered and sutured manifolds

In this section, we develop Heegaard diagrams for bordered, sutured and bordered sutured manifolds. Even though one could simply defined diagrams for bordered sutured manifolds and see the ones for bordered manifolds and sutured manifolds as a particular case, we are going to construct the pieces one by one. This choice is made as in later chapters we are going to work predominantly with bordered diagrams and sutured diagrams.

2.4.1 Heegaard diagrams for bordered manifolds

For this subsection, one can use Section 4.1 of [LOT18] as reference.

Definition 2.24 (Bordered Heegaard diagram). A *bordered Heegaard diagram* is a quadruple $(\Sigma, \alpha = \alpha^a \cup \alpha^c, \beta, z)$, where

- Σ is a genus g , oriented surface with one boundary component,
- $\beta = \{\beta_1, \dots, \beta_g\}$ are g pairwise disjoint circles in the interior of Σ ,
- $\alpha^c = \{\alpha_{g-k}^c, \dots, \alpha_{g-k}^c\}$ are $g - k$ circles in the interior of Σ ,
- $\alpha^a = \{\alpha_1^a, \dots, \alpha_{2k}^a\}$ are $2k$ arcs in Σ with boundary on $\partial\Sigma$ and which are transverse to $\partial\Sigma$,
- all the curves of α are pairwise disjoint,
- z is a point in $\partial\Sigma \setminus (\alpha \cap \partial\Sigma)$.

Moreover, we request as in the closed case that the intersections are transverse and that $\Sigma \setminus \alpha$ and $\Sigma \setminus \beta$ are connected (or that, equivalently, that α is an independent set in $H_1(\Sigma, \partial\Sigma)$ and that β is an independent set in $H_1(\Sigma)$).

We want to see now how we can construct a bordered manifold starting from a bordered Heegaard diagram. For this, we need first to understand how we can give a structure of pointed matched circle to the border of Σ .

Lemma 2.25 ([LOT18, Lemma 4.4]). *Consider a bordered Heegaard diagram $(\Sigma, \alpha, \beta, z)$. Define $Z = \partial\Sigma$ and $\mathbf{a} = \alpha^a \cap \partial\Sigma$. We define a matching M on $\mathbf{a}$ by*

$$M(\alpha_i^a \cap \partial\Sigma) = i.$$

Then, $\mathcal{Z} = (Z, \mathbf{a}, M, z)$ is a pointed matched circle.

Proof ([LOT18]). We only need to check that if we perform surgery along the $2k$ pairs of points in Z , we obtain a single circle. This follows from the connectivity of $\Sigma \setminus \alpha$.

Call Z' the circle obtained from Z after the surgery and Σ' the manifold obtained from Σ after deleting a tubular neighbourhood of each α -circle and gluing two discs to the result. It is clear by the definitions that Z' is homeomorphic to the boundary of a tubular neighbourhood of $\partial\Sigma \cup (\cup_{i=1}^{2k} \alpha_i^a)$ and this is homeomorphic to the boundary of $\Sigma' \setminus \nu(\alpha)$, where $\nu(\alpha)$ is a tubular neighbourhood of all the curves of α . But then, because of the connectivity of $\Sigma \setminus \alpha$, this a circle. $\square$

Definition 2.26 (Boundary of a bordered Heegaard diagram). We are going to call the *boundary of a bordered Heegaard diagram* $\mathcal{H} = (\Sigma, \alpha, \beta, z)$ the pointed matched circle above defined, and we will denote it by $\partial\mathcal{H}$.

Given a bordered bordered Heegaard diagram $\mathcal{H} = (\Sigma, \alpha, \beta, z)$, we can construct an unique bordered 3-manifold $(Y, \mathcal{Z}, \phi)$ in the following way. We thicken the Heegaard surface to $\Sigma \times [0, 1]$, and we attach 3-dimensional 2-handles to each $\alpha_i^c \times \{0\}$ and to each $\beta_j \times \{1\}$. Then, we take $\mathcal{Z} = \partial\mathcal{H}$ and the boundary of the manifold that we get as result of the above construction is then naturally identified with $F(\mathcal{Z})$.

As in the case of closed 3-manifolds, it is possible to prove that any bordered 3-manifolds is represented by some bordered diagram.

Proposition 2.27 ([LOT18, Lemma 4.9]). *Any bordered 3-manifold $(Y, \mathcal{Z}, \phi)$ is represented by some bordered Heegaard diagram.*

A proof of this done via Morse theory can be found in [LOT18].

Moreover, we can still use Heegaard moves to pass from one diagram representing a bordered manifold to another diagram representing the same bordered manifold. There is a difference in the moves that we allow, as it is possible to see in the next result.

Proposition 2.28 ([LOT18, Lemma 4.10]). *Any pair of bordered Heegaard diagrams for the same bordered 3-manifolds can be made diffeomorphic after a sequence of*

- *isotopies of the α -curves and β -circles, not crossing $\partial\Sigma$;*
- *handleslides of β -circles over β -circles,*
- *handleslides of α -curves over α -circles,*
- *stabilisations in the interior of Σ .*

2.4.2 Heegaard diagrams for sutured manifolds

We develop now Heegaard diagrams for sutured manifolds. For this purpose, we mainly follow Section 4.2 of [Alt13].

Traditionally, the approach used to develop sutured Heegaard Floer homology is to define *sutured Heegaard diagrams*, a special class of diagrams with few requirements on the α - and β -circles: we do not ask them to be independent in first homology and we do not ask them to be equal in quantity. Namely, one define this object as follows.

Definition 2.29 (Sutured Heegaard diagram). We define a *sutured Heegaard diagram* to be a tuple (Σ, α, β) , where Σ is a compact and oriented 2-manifold with border and $\alpha = \{\alpha_1, \dots, \alpha_m\}, \beta = \{\beta_1, \dots, \beta_n\}$ are two sets of pairwise disjoint simple closed curves in the interior of Σ .

Then, using some constraints of Heegaard Floer homology constructed via Whitney discs, it become clear that it is needed to restrict the environment to *balanced sutured Heegaard diagrams*. For the reader that is familiar to such approach for the construction of Floer homology, the main argument is that we need $|\alpha| = |\beta|$ and their independence in first homology, because otherwise the theory developed on the tori $\mathbb{T}_\alpha$ and $\mathbb{T}_\beta$ of the symmetric product $\text{Sym}^g(\Sigma)$ is not well defined and it is not possible to proceed with the construction of the Floer homology. Since we are going to use Lipshitz's cylindrical reformulation for the construction of Floer homology and not Whitey discs, we directly define sutured Heegaard diagrams with these requirements without justifying them further. For details about the mentioned construction, one can see Section 4.2 of [Alt13].

Definition 2.30 (Sutured Heegaard diagrams). We define a *sutured Heegaard diagram* (also called *balanced sutured Heegaard diagram*) to be a tuple (Σ, α, β) , where Σ is a compact and oriented 2-manifold with border and $\alpha = \{\alpha_1, \dots, \alpha_n\}, \beta = \{\beta_1, \dots, \beta_n\}$ are two sets of pairwise disjoint simple closed curves in the interior of Σ , such that $\{[\alpha_1], \dots, [\alpha_n]\}$ and $\{[\beta_1], \dots, [\beta_n]\}$ are independent sets in $H_1(\Sigma, \mathbb{Q})$.

Given a sutured Heegaard diagram, we can construct an unique sutured manifold (Y, Γ) in a similar fashion of what we saw for 3-manifolds: thicken up Σ to $\Sigma \times I$ and smoothly attach 3-dimensional 2-handles along the circles $\alpha_i \times \{0\}$ and $\beta_j \times \{1\}$; we then obtain a 3-manifold with boundary and we take $\Gamma = \partial\Sigma \times \{1/2\}$.

The result (Y, Γ) is indeed a sutured manifold, as it holds the following result.

Lemma 2.31 ([Alt13, Lemma 4.16]). *Let (Σ, α, β) be a sutured Heegaard diagram and let (Y, Γ) be the object obtained with the above construction. Since the circles in α and the circles in β are independent in $H_1(\Sigma)$, every connected component A of $R(\Gamma) = \Sigma \setminus \Gamma$ has non-empty boundary. Hence (Y, Γ) is a sutured manifold.*

Moreover, it is possible to prove the following result, from which it follows that we are going to work with balanced sutured manifolds.

Lemma 2.32 ([Alt13, Lemma 4.17 and 4.18]). *Let (Σ, α, β) be a sutured Heegaard diagram and let (Y, Γ) be the sutured manifold represented by it. Then (Y, Γ) is a balanced sutured manifold.*

Moreover, we can see that that (almost) any balanced sutured manifolds admits a sutured Heegaard diagram.

Proposition 2.33 ([Juh06, Proposition 2.14]). *Let (Y, Γ) be a balanced sutured manifold. Then, there exists a sutured Heegaard diagram (Σ, α, β) that represents it.*

As we saw in the case of 3-manifolds, also here all the definition are well posed as the next result explains.

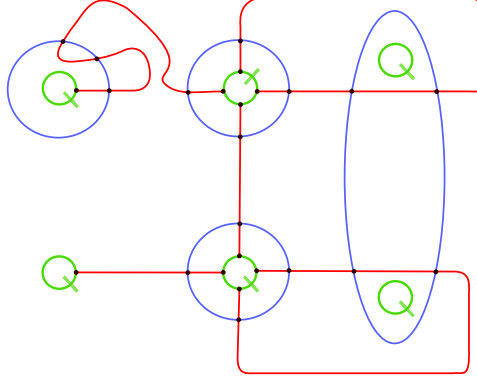


Figure 2.9: An example of a bordered sutured Heegaard diagrams for a bordered sutured manifold with no β -arcs.

Proposition 2.34 ([Alt13, Proposition 4.20]). *Two balanced diagrams represent the same balanced sutured manifold if and only if the diagrams are connected by a sequence of Heegaard moves.*

Remark 2.35 (Re-visitation of Heegaard diagrams for closed manifolds). As anticipated, we can see how we can re-visit Heegaard diagrams for closed 3-manifolds as a special case of sutured Heegaard diagrams.

Let Y be a closed, connected and oriented 3-manifold. We construct a sutured 3-manifold as follows: consider a 3-ball $B_1 \subset Y$ and choose an oriented simple curve $s_1 \subset \partial B_1$; we can then define a sutured manifold $(Y', \{s_1\})$, where $Y' = Y \setminus \text{Int}(B_1)$.

Therefore, enlarging our point of view to the sutured language, we can now think of the basepoint introduced for pointed Heegaard diagrams as the suture s_1 once we shrunk the ball B_1 down to a point.

We will see in Section 3.3 that indeed $\widehat{\text{HF}}$ agrees with SHF in the sense above presented, obtaining

$$\widehat{\text{HF}}(Y) \equiv \text{SFH}(Y', \{s_1\}).$$

2.4.3 Heegaard diagrams for bordered sutured manifolds

We now construct Heegaard diagrams for bordered sutured manifold with α - and β -arcs, following Section 3.1 of [Zib20].

Definition 2.36 (Bordered sutured Heegaard diagram). Let $(Y, \Gamma, \mathcal{Z}_\alpha, \phi_\alpha, \mathcal{Z}_\beta, \phi_\beta)$ be a bordered sutured manifold with α - and β -arcs and considered the sutured Heegaard diagram (Σ, α, β) of the underlying sutured manifold. Then, a *bordered sutured Heegaard diagram* for the bordered sutured manifold is obtained by (Σ, α, β) by adding the graphs of the arc diagrams to it, and we denote it by $(\Sigma, \alpha, \beta, \mathcal{Z}_\alpha, \phi_\alpha, \mathcal{Z}_\beta, \phi_\beta)$.

We can explain in more details what the above definition means exactly. Recall that $\phi_\alpha : G(\mathcal{Z}_\alpha) \hookrightarrow \overline{R_-(\Gamma)}$ is an embedding such that $\phi_\alpha(\mathcal{Z}_\alpha) \subset \Gamma$; we can consider the embedding ϕ_α to be such that $\phi_\alpha(G(\mathcal{Z}_\alpha))$ misses the 3-dimensional 2-handles

corresponding to the α -circles: it suffices to slide them off $\mathcal{S}^0 \times \mathbb{D}^2 \subset R_-(\Gamma)$. We do the same for $G(\mathcal{Z}_\beta)$. Therefore, we have that the line segments are embedded in the border of Y (which corresponds to the sutures), and we have some α -segments (namely $\phi_\alpha(G(\mathcal{Z}_\alpha) \setminus Z)$) and β -segments (namely $\phi_\beta(G(\mathcal{Z}_\beta) \setminus Z)$) between the sutures. We call them respectively α -arcs and β -arcs. Moreover, we put a *basepoint* (i.e. a marked point), in every open component of the boundary minus the image of Z . We usually draw the sutures in green.

As in the previous cases, we have the following result.

Proposition 2.37 ([Zar09, Proposition 4.5]). *Any sutured bordered manifold has a compatible bordered sutured Heegaard diagram.*

Moreover, given any two diagrams which represent the same bordered sutured manifolds, we can connect them with a sequence of

- *Isotopies of α -circles and β -circles and isotopies of α -arcs that do not move the endpoints,*
- *handleslides of a β -circle in over another β -circle,*
- *handleslides of any α -curve over an α -circle*
- *stabilisations in the interior of Σ .*

2.5 Heegaard diagrams for knots

In Section 3.4, we are going to see a version of Heegaard Floer homology developed to study knots embedded in $\mathcal{S}^3$, called *knot Floer homology* and denoted by $\widehat{\text{HFK}}(\mathcal{S}^3, K)$ (where K is the knot). In this section, we present Heegaard diagrams for knots, following Section 1.3 of [Hom19].

For HFK, we need *doubly pointed Heegaard diagrams*, i.e. diagrams with two basepoints that we will denote by z and w . The need for two basepoints can seem a bit random at first glance, but it will be justified with the sutured perspective in Remark 2.42.

Definition 2.38 (Doubly pointed Heegaard diagram). *A doubly pointed Heegaard diagram for a knot $K \subset \mathcal{S}^3$ is a tuple $(\Sigma, \alpha, \beta, w, z)$ where w, z are basepoints in $\Sigma \setminus (\alpha \cup \beta)$, such that*

1. (Σ, α, β) is a Heegaard diagram for $\mathcal{S}^3$;
2. K is the union of arcs a and b , where
 - a is an arc in $\Sigma \setminus \alpha$ connecting w to z , pushed slightly into H_1 ,
 - b is an arc in $\Sigma \setminus \beta$ connecting z to w , pushed slightly into H_2 .

Example 2.39 (Trefoil). The following is an example for the left-handed trefoil.

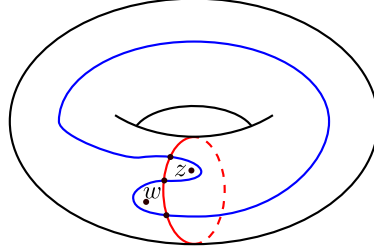


Figure 2.10: Genus 1 doubly pointed Heegaard diagram for the left-handed trefoil

Lemma 2.40. *Every knot $K \subset \mathcal{S}^3$ admits a doubly pointed Heegaard diagram.*

Proof ([Hom19]). There is an explicit way to construct such a diagram for a generic knot K . Let D be a knot diagram for K (i.e., a projection of K on the plane with the information of "over"/"under" on the crossings); let n_c be the number of crossing of D . If we forget about the crossings' "over"/"under" data for a moment, we have a close, non-simple curve that divides the plane in $n_c + 2$ regions, $n_c + 1$ that are bounded and one that is not bounded (the "external region"). Let Σ be the surface of tubular, regular neighbour of such curve in $\mathbb{R}^3$; therefore Σ is a surface of genus $n_c + 1$ which looks like a thickened K . For each of the bounded regions in the complement of D , we put a β -circle on Σ that goes around the respective hole of the surface. See Figure 2.11 for an example of this construction on the left-handed trefoil.

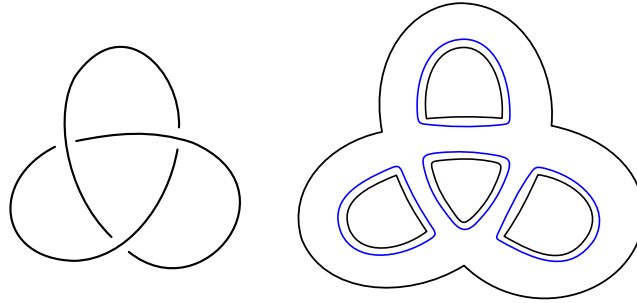


Figure 2.11: Construction of a genus 4 doubly pointed Heegaard diagram for the left-handed trefoil: Σ and β -circles.

We need now a way to encode the crossings' data on Σ . Recalling how α -circles give instruction for the gluing of discs on the Heegaard surface, it is natural to encode such data with the α -circles as indicated in Figure 2.12.

We have now a genus $n_c + 1$ surface with $n_c + 1$ circles around the holes, the β -circles, and n_c circles around the crossings, the α -circles. We then add the last α -circle, corresponding to a meridian of K and we place the basepoint w on one side of it and the basepoint z on the other side. Notice that the choice of the side on which we put w determines the orientation of K .

See Figure 2.13 for the result of this construction for the left-handed trefoil. $\square$

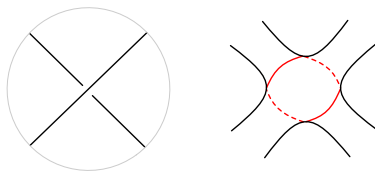
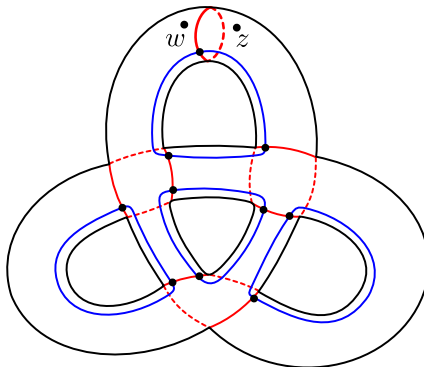
Figure 2.12: How the α -circles hold the crossing data.

Figure 2.13: Genus 4 doubly pointed Heegaard diagram for the left-handed trefoil.

We can now define doubly pointed isotopies and doubly pointed handleslides just as we defined the pointed ones: isotopies are required to miss both w and z and neither w nor z is allowed to be in the pair of pants involved in the handleslide. We then have an analogue of Corollary 2.17, which is proven in [OS04].

Proposition 2.41 ([OS04, Proposition 3.5]). *Let $(\Sigma, \alpha, \beta, w, z)$ and $(\Sigma', \alpha', \beta', w', z')$ be two doubly pointed Heegaard diagrams for a knot $K \subset \mathcal{S}^3$. Then, it exists a finite sequence of doubly pointed isotopies, doubly pointed handleslides and stabilisations that connects the two diagrams.*

Remark 2.42 (Heegaard diagrams for knots as particular case of sutured Heegaard diagrams). We can motivate the construction done above with the theory developed for sutured manifolds. As a reference, one can see Section 9 of [Juh06].

Recall the sutured manifold $(Y_K, \{s_1, s_2\})$ associated to a knot K , defined in Remark 1.65.

Therefore, we can think about the two basepoints placed in the construction of doubly pointed Heegaard diagrams as these two sutures, shrunk down to a point. We will see in Section 3.4 that also the construction for knot Heegaard Floer homology agrees in this way with sutured Floer homology.

2.6 Heegaard diagrams for tangles

We are now going to see how we can construct a Heegaard diagram for tangles, following Chapter II.2 of [Zib17].

We start by defining what a Heegaard diagram is in this case. As in Section 2.5, we can justify the following construction with the bordered sutured point of view: we do so in Subsection 2.6.2

Definition 2.43 (Heegaard diagram for a tangle). Consider a tangle T with n open components and m closed components. A *Heegaard diagram for the tangle T* consists of the tuple $(\Sigma, \alpha = \alpha^c \cup \alpha^a, \beta)$, where:

- an oriented surface Σ_g of genus g with $2(n + m)$ boundary components. We denote the collection of these border components by $\mathcal{Z}$, which are partitioned into $(n + m)$ pairs;
- a set α^c of $(g + m)$ pairwise disjoint circles $\alpha_1, \dots, \alpha_{g+m}$ on Σ_g . These are called α -circles;
- a set α^a of $2n$ pairwise disjoint arcs $\alpha_1^a, \dots, \alpha_{2n}^a$ on Σ_g which are disjoint from α^c and whose endpoints lie on $\mathcal{Z}$. These are called α -arcs;
- a set β of $(g + m + n - 1)$ pairwise disjoint circles $\beta_1, \dots, \beta_{g+m+n-1}$ on Σ_g . These are called β -circles.

We write $\alpha := \alpha^c \cup \alpha^a$ to indicate all the α -curves. Moreover, we impose the following conditions on the data above:

1. contracting all boundary components turns α^a into a single circle. This means that on each component of $\mathcal{Z}$, we could either have no or exactly two endpoints of α^a ;
2. define the surface $S_{\alpha^c}(\Sigma_g)$ as the surface obtained by surgery along the curves in α^c . We want it to be a disjoint union of m anuli, each of whose boundary is a pair in $\mathcal{Z}$, and a 2-sphere with $2n$ boundary components. We denote by $\mathcal{Z}^\alpha$ the set of circles in $\mathcal{Z}$ which meet the α -curves;
3. define the surface $S_\beta(\Sigma_g)$ as the surface obtained by surgery along the curves in β . We want it to be a disjoint union of $(n + m)$ anuli, each of whose boundary is a pair in $\mathcal{Z}$.

Remark 2.44 (Boundary components in a Heegaard diagram for a tangle). Recall the bordered sutured structure given to a tangle in Remark 1.72. We can then interpret the $2(n + m)$ boundary components of a Heegaard diagram for a tangle in the same way, i.e. two oppositely oriented meridional circles around each closed components and a single suture around each tangle end.

In our work, we are going to focus on 4-ended tangles without any closed component. It could be useful to see the definition of a Heegaard diagram in this case.

Definition 2.45 (Heegaard diagram for a 4-ended tangle without closed components). A *Heegaard diagram for a 4-ended tangle without closed components* consists of the tuple $(\Sigma, \alpha = \alpha^c \cup \alpha^a, \beta)$, where:

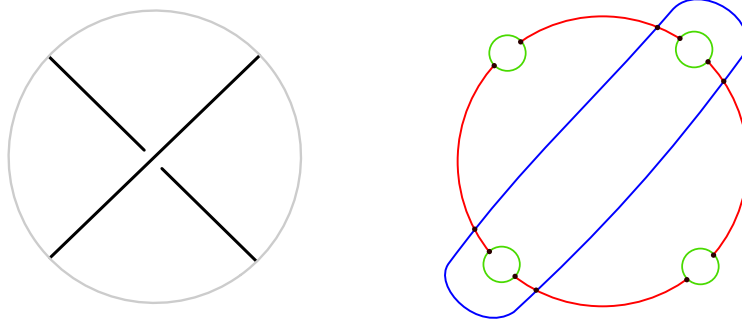


Figure 2.14: Heegaard diagram for the 1-crossing tangle.

- an oriented surface Σ_g of genus g with 4 boundary components. We denote the collection of these border components by $\mathcal{Z}$, which are partitioned into 2 pairs;
- a set α^c of g pairwise disjoint circles $\alpha_1, \dots, \alpha_g$ on Σ_g . These are called α -circles;
- a set α^a of 4 pairwise disjoint arcs on Σ_g , denoted by a, b, c, d , which are disjoint from α^c and whose endpoints lie on $\mathcal{Z}$. These are called α -arcs;
- a set β of $(g + 1)$ pairwise disjoint circles $\beta_1, \dots, \beta_{g+1}$ on Σ_g . These are called β -circles.

We write $\alpha := \alpha^c \cup \alpha^a$ to indicate all the α -curves. Moreover, we impose the following conditions on the data above:

1. contracting all boundary components turns α^a into a single circle. This means that on each component of $\mathcal{Z}$, we could either have no or exactly two endpoints of α^a ;
2. define the surface $S_{\alpha^c}(\Sigma_g)$ as the surface obtained by surgery along the curves in α^c . We want it to be a 2-sphere with 4 boundary components. We denote by $\mathcal{Z}^\alpha$ the set of circles in $\mathcal{Z}$ which meet the α -curves;
3. define the surface $S_\beta(\Sigma_g)$ as the surface obtained by surgery along the curves in β . We want it to be a disjoint union of 2 annuli, each of whose boundary is a pair in $\mathcal{Z}$.

Example 2.46 (Heegaard diagram for the 1-crossing tangle). Let T be the 1-crossing tangle (see Figure 2.14); then we can draw a genus 0 Heegaard diagram as it is shown in Figure 2.14.

There is one way to think about the construction of this genus 0 Heegaard diagram which can be more intuitive. Consider our tangle embedded in B^3 ; we want in some way to encode all the information of our tangle on the surface of this ball. We start by considering a tubular neighbourhood of our tangle, call it $\nu(T)$, and we drill

it out of B^3 . We then obtain $B^3 \setminus \nu(T)$, which we can imagine as a ball with a cave system that clearly encodes the data of the tangle. We now consider the boundary of it, which is a sphere with 4 punctures on it the four-punctured sphere $\mathcal{S}_4^2$. We can think about it as if we have the entrances of our previous cave system, but we do not know which entrance is linked to which; this information is given by the β -circle that we embed on $\mathcal{S}_4^2$, which says that the north-east suture is linked with the south-west sutures in a "straight way": in fact, if we embed the disc which has the β -circle as boundary, then we get a separation of the inside of the sphere which agrees with the initial cave system that we wanted to encode on $\mathcal{S}_4^2$ (we will see better in Subsection 2.6.1 how the construction of the β -circle works for more intricate tangles). On top of this we add 4 α -arcs, which represent the fixed circle S on which the tangle ends sit.

As we had Corollary 2.17 for pointed Heegaard diagrams, we have also the following result in the tangle case.

Proposition 2.47. *Let T be a tangle. Given any two diagrams for the same tangle, one can be obtained from the another by a sequence of the following moves:*

- *isotopies of α - or β -circles and isotopies of α -arcs that leave the endpoints fixed;*
- *handleslides of a β -circle over another β -circle;*
- *handleslides of an α -curve over an α -circle;*
- *stabilisations in the interior of Σ .*

Moreover, using the knowledge of Example 2.46, we obtain the following lemma.

Lemma 2.48. *Let T be a tangle without closed components, then it has a Heegaard diagram.*

There is actually a version of the above lemma for tangles with closed components; in that case we have to use *ladybugs* to get the right final diagram. For the details, one can see Example 2.5 of [Zib17].

Proof ([Zib17]). To prove the statement, we are going to see how it is possible to glue Heegaard diagrams of different tangles together. Once that that is proven, we just need to cut our tangle into 4-ended tangles with a single crossing each; then by gluing together all the diagrams for those using the bordered sutured structures, we obtain the whole diagram.

The way to glue together Heegaard diagrams is quite straightforward: when we want to re-attach two endpoints, we glue together the two corresponding sutures of the diagrams. When we do so, we need to pay attention to the α -arcs that have endpoints on those sutures: we need to glue them, pairing them as it is indicated in Figure 2.15 for the case of a tangle with two crossings.

Further details for the gluing can be found in Section 4.6 of [Zar11]. $\square$

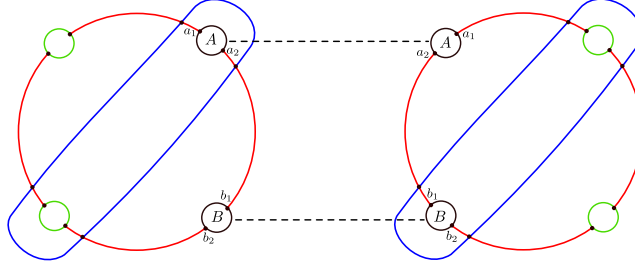


Figure 2.15: How to glue tangles.

Remark 2.49. • In the case of a 4-ended tangle with n crossings, we can notice that from the gluing the α -arcs we are going to obtain $n - 1$ α -circles and 4 α -arcs. In this way, the final diagram is a proper diagram for a 4-ended tangle.

- Therefore, in the case of a 4-ended tangle with n crossings, the final diagram that we obtain is going to have genus $n - 1$. This makes this construction not very useful in practice, as many tangles actually admit some lower genus diagram (for instance, any rational tangle admits a genus 0 diagram) that are more practical in the actual computations of tangle Floer homology.
- It is also possible to cut our tangle in more convenient pieces, for instance we can try to cut it leaving untouched the biggest "rational pieces" that we can find. This is something that can also help with the input of [nicepy], as one can read in Appendix A.

In order to compute combinatorially the invariant HFT, we will need *peculiar Heegaard diagrams*, a slight modification of the diagrams above defined. We define this new notion of diagram for 4-ended tangles with eventually closed components in agreement with Definition 2.4 of [Zib20].

Definition 2.50 (Peculiar Heegaard diagram). Let T be a 4-ended tangle and $(\Sigma, \alpha = \alpha^c \cup \alpha^a, \beta)$ be a Heegaard diagram that represents it. A *peculiar Heegaard diagram* for T , denoted by $(\Sigma', \alpha^c \cup \{\mathcal{S}\}, \beta, \{p_i\}_{i=1}^4, \{q_j\}_{j=1}^4, \{z_k\}_{k=1}^m, \{w_l\}_{l=1}^m)$, is the diagram obtained from $(\Sigma, \alpha = \alpha^c \cup \alpha^a, \beta)$ after the following local modification around the sutures:

- we collapse the four sutures of $\mathcal{S}_4^2$, obtaining a α -circle which is the union of the four α -arcs (which is isotopic to the fixed circle $\mathcal{S}$);
- we add two marked points for every collapsed sutured on the two sides of $\mathcal{S}$, we denote by p_i 's the ones placed on the front of the sphere and by q_j 's the ones placed on the back of the sphere, for $i, j = 1, \dots, 4$;
- for every collapsed suture, we connect the respective p_i and q_j with an arc that intersects only $\mathcal{S}$ and in only one point (see Figure 2.16);



Figure 2.16: Collapsing of a suture and placing of the linked p_i and q_j .

- we contract all the other boundary components (i.e. the couples representing a closed component of T) to points, which we call z_k, w_l for $k, l = 1, \dots, m$ in a way that z_k and w_k correspond to the two sutures of the same closed components.

We then refer to the p_i 's, the q_j 's, the z_k 's and the w_l 's as *basepoints* of the diagram; the motivation for the use of this name will be clear in Subsection 4.4.4.

The main advantage of these diagrams is that the Heegaard surface is a closed surface, allowing us to avoid any bordered Heegaard Floer theory. In particular, we can also notice that in a peculiar Heegaard diagram, the number of α -circles and the number of β -circle coincide.

We need to adapt Heegaard moves for peculiar diagrams.

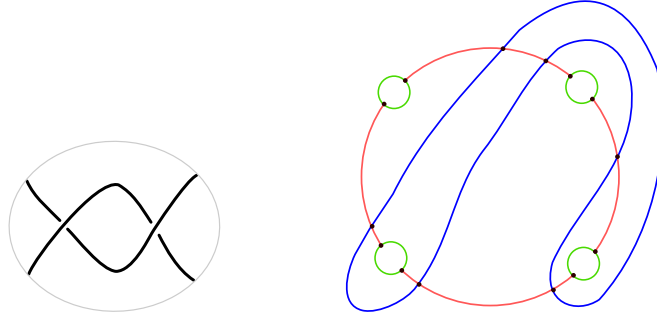
Definition 2.51 (Peculiar Heegaard moves). The allowed Heegaard moves for peculiar diagrams are:

1. isotopies of the α - and β -circles that do not touch the basepoints and the arcs that connect these,
2. handleslides of β -circles over β -circles and handleslides of α -circles over α -circles different than $\mathcal{S}$,
3. stabilisations.

2.6.1 Heegaard diagrams for rational tangles

Starting from the diagram of the 1-crossing, it is possible to construct the diagram for any rational tangle with the use of *Dehn twists*. In this way, we can draw Heegaard diagrams on genus 0 surfaces.

Let us take Q_2 : this is Q_1 (the single crossing) to which we added a twist on the right. Consider the diagram for Q_1 given in Figure 2.14. We want to "add the extra twist" to this diagram. In the tangle, the twist is done between the north-east and the south-east ends, with south-east going over north-east. We are going to do the same on the diagram. As we now have that the south-west suture is linked with the south-west one, the β -circle is going to contain these two. Moreover, as south-east has gone over north-east, we have to let pass the β -circle behind the north-east suture (i.e., on the back of $\mathcal{S}_4^2$), obtaining the diagram in Figure 2.17.

Figure 2.17: Q_2 and a Heegaard diagram for it.

Remark 2.52 (Genus 0 Heegaard diagrams for rational tangles). We can immediately notice that, as a consequence of the above construction, every rational tangle admits a genus 0 Heegaard diagram.

In fact, it actually holds that a 4-ended tangle is rational if and only if it has a genus 0 Heegaard diagram. Indeed, if we consider a genus 0 Heegaard diagram for a 4-ended tangle, it has no α -circles and just a single β -circle. By definition, we know that performing surgery along this β -circle gives us two cylinders, so it separates two punctures from the other two. Therefore the tangle represented is rational. The other direction is proven by the above construction.

The above construction is something that works for any added twist on a tangle, but it can be a bit tricky when the number of twists grows. There is another construction which can be more useful in practice, which was implemented to help to construct inputs for the program.

Suppose that we want to construct a genus 0 diagram for a rational tangle $Q_{p/q}$; we start by laying down 4 sutures and the 4 α -arcs between them on the plane. Recall that we call the sites by $\{a, b, c, d\}$, starting from the site between north-west and south-west and going counter-clock wise. We place then the intersection points on the α -arcs: we put p points on the sites b and d and q points on a and c . It is time to lay down the β -circle; for this we need to take the sign of p/q into account.

- In the case of $p/q > 0$, we start from the north-west sutures and we find the first intersection point on the right of the suture with the first point below the suture; then we link them with a straight blue edge which stays on the front component of $\mathcal{S}_4^2$. We now consider the second intersection point on the right on the suture and the second point below it and we link them in the same way. Once we linked the first p couples we also finished the points on the site d , hence we start to consider the points on the site c ; in the same way, after q couples we will need to move on the site b as the points on site a will be finished. The last couple to link will be the one "around" the south-east suture. Now we need to link the points also on the back of the sphere, i.e. outside of the rectangle; we do it in a similar fashion as before, by starting from the north-east suture: we take the two intersection points that are the closest to the suture and we link them. The last couple will be the one "around" the south-west suture.

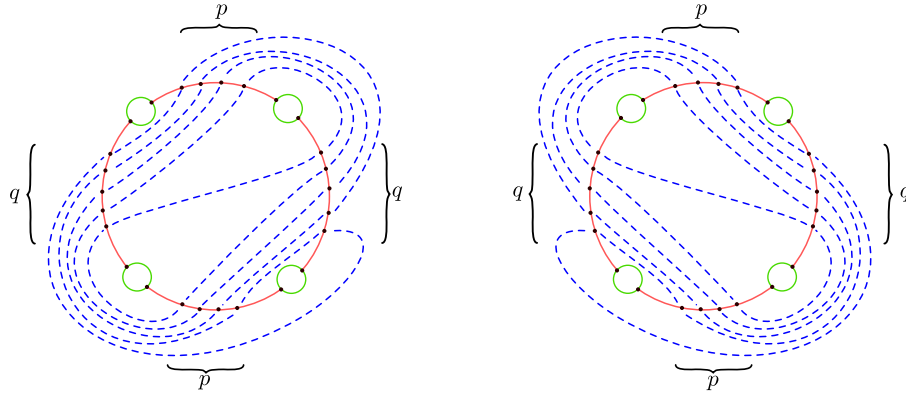


Figure 2.18: The construction for a genus 0 Heegaard diagram for the rational tangle $Q_{p/q}$ with $p/q > 0$ (on the right) and with $p/q < 0$ (on the left).

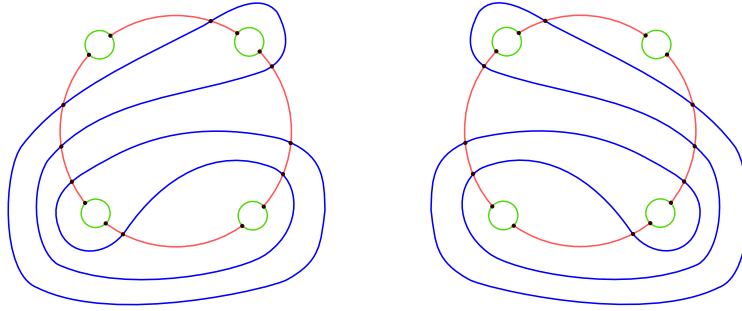


Figure 2.19: Final diagrams for $Q_{1/3}$ (left) and $Q_{-1/3}$ (right).

- In the case of $p/q < 0$, the procedure is almost the same and we just need to start from different sutures. For the internal edges, we will start from the north-east suture; for the external edges we will start from the north-west suture.

It is possible to visualise the general construction in Figure 2.18 and in Figure 2.19 it is possible to see the examples for $Q_{\pm 1/3}$.

Remark 2.53. • In order to remember which one is the right construction, we can notice that in the case $p/q > 0$ the internal edges have positive slopes and that in case $p/q < 0$ they have negative slope.

- The coprimality of p and q guarantees that that we will have an unique β -circle in the final result of the construction.

2.6.2 Heegaard diagrams for tangles as particular case of sutured Heegaard diagrams

As we did in Remark 2.42, we are going to motivate the construction of Heegaard diagrams for tangles with the bordered sutured perspective.

As a reference, one can see Section II.2 of [Zib17].

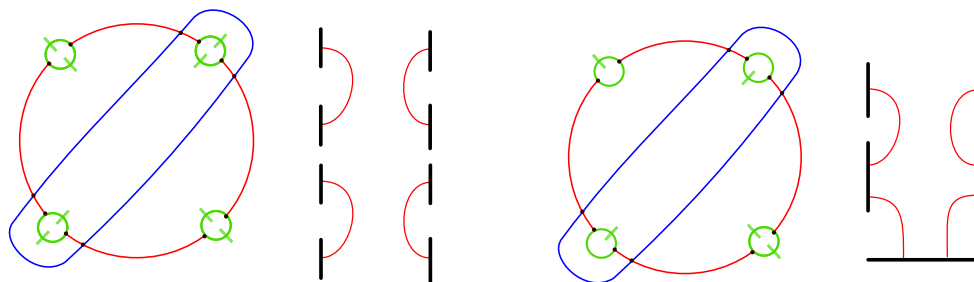


Figure 2.20: Two bordered sutured structures on the tangle Heegaard diagram of Q_1 : a trivial one (on the left) and a more interesting one (on the right).

Starting from a tangle Heegaard diagram, we can construct a bordered sutured Heegaard diagram. In fact, the α -arcs on the boundary define such a structure, we just need to decide where to "open up" the sutures, i.e. which are the segments of the arc diagram $\mathcal{Z}_\alpha$. We indicate with a green tick the position on which we cut a suture, we need to put at least one on each suture in order to have a well-defined bordered sutured diagram. In particular, remark that these ticks are going to have the role of basepoints in bordered sutured Floer homology (more details in 4.4.3).

We can see an explicit example of the construction explained above.

Example 2.54. Consider for instance the tangle Heegaard diagram for the 1-crossing tangle presented in Figure 2.14. One choice to make it a bordered sutured Heegaard diagram could be to put two ticks for suture, one on the back and one on the front; this choice would yield the diagram in the left part of Figure 2.20. By removing some of these ticks, we can get a more interesting bordered sutured structure, as the one showed in the right part of Figure 2.20.

Moreover, we will see in Subsection 3.5.4 how HFT inherits a useful gluing property from bordered sutured Floer homology.

Chapter 3

Heegaard Floer homologies

Now that we have defined Heegaard diagrams for all the topological objects that we are interested in, we can build our way to Heegaard Floer homologies. We see in details the construction of $\widehat{\text{HF}}$, the one for the closed 3-manifold case, which we then adapt to the more general context of sutured 3-manifolds and to the case of 4-ended tangles. After defining the chain complex $\widehat{\text{CF}}$ on which we compute $\widehat{\text{HF}}$, it will be clear that the difficulty of this homology lies in the construction of the boundary map: this is the main motivation for our interest in *nice Heegaard diagrams*, that will be discussed in extension in Chapter 4. In the last section, we construct tangle Heegaard Floer homology, the main focus of this thesis work. The description of the invariant yielded by it is going to be of use in Chapter 5, where we will see the invariants obtained from the computation of HFT for some families of tangles.

3.1 Hat version of Heegaard Floer homology $\widehat{\text{HF}}$

We start the construction of the hat version of Heegaard Floer homology for 3-manifolds, denoted by $\widehat{\text{HF}}$. Traditionally, this was done with the use of *symmetric products*, i.e. products in which the order of the elements in a tuple does not matter. However, following closely what is done in [SW06], we are going to use Robert Lipshitz's cylindrical reformulation of $\widehat{\text{HF}}$, which allows us to completely avoid the discussion on symmetric products. This reformulation is presented in [Lip06], where further details can be found. Details on the construction of $\widehat{\text{HF}}$ via symmetric products can be found in [OS06], Sections from 4 to 8, or in [Hom19], Section 2.2.

For this section, one can use Section 2 of [SW06] as reference.

We start with the construction of the chain complex $\widehat{\text{CF}}(\Sigma, \alpha, \beta, z)$, the complex with coefficients in $\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$ on which $\widehat{\text{HF}}$ is computed. For the entirety of this section, let $(\Sigma, \alpha, \beta, z)$ be a genus g pointed Heegaard diagram for some closed 3-manifold Y .

Definition 3.1 (Generator). A *generator* $\mathbf{x}$ in $(\Sigma, \alpha, \beta, z)$ is a collection of g points of Σ , denoted by $\mathbf{x} = \{x_1, \dots, x_g\}$, such that for any $1 \leq i \leq g$ we have that

- $x_i \in \alpha_i$;
- exists an unique j such that $x_i \in \beta_j$.

We are therefore asking that we have one (and only one) point on each α -curve and on each β -curve.

This name come to the fact that they are going to be the generators of the chain complex $\widehat{\text{CF}}(\Sigma, \alpha, \beta, z)$, as it is explained in the next definition.

Definition 3.2 ($\widehat{\text{CF}}(\Sigma, \alpha, \beta, z)$). Let $(\Sigma, \alpha, \beta, z)$ be a Heegaard diagram. We define $\widehat{\text{CF}}(\Sigma, \alpha, \beta, z)$ to be the vector space over $\mathbb{F}_2$ generated by all the generators $\mathbf{x}$ of $(\Sigma, \alpha, \beta, z)$.

Now that we have defined $\widehat{\text{CF}}$, we are going to focus on the construction of the main ingredient that we are missing: the boundary map

$$\partial : \widehat{\text{CF}} \longrightarrow \widehat{\text{CF}}.$$

This turns out to be also the difficult bit in this construction, as we basically will need to count the number of points in certain moduli spaces of holomorphic embeddings of domains in our diagram. Once we will reduce ourselves to the nice Heegaard diagram case, we will see that this differential can be computed completely in a combinatorial way; however for now we remain in the more general case.

3.1.1 Domains

The first step is to define *domains* in Heegaard diagrams, i.e. linear combinations of regions delimited by the intersections of α -curves and β -curves on Σ .

Definition 3.3 (Region, 2-chain). A *region* is a connected component of $\Sigma \setminus (\alpha \cup \beta)$. A formal sum of region with integer coefficients is called a *2-chain*.

In general, for a 2-chain $\phi = \sum_i a_i \cdot \phi_i$ we will denote by $\partial\phi$ its boundary

$$\partial\phi = \sum_i a_i \cdot \partial\phi_i.$$

Moreover, we denote by $\partial(\phi)|_\alpha = \partial(\phi) \cap \alpha$ the α -edges on the boundary of ϕ and by $\partial(\phi)|_\beta = \partial(\phi) \cap \beta$ the β -edges.

Definition 3.4 (Domains and $\pi_2(\mathbf{x}, \mathbf{y})$). Let now $\mathbf{x}, \mathbf{y}$ be two generators. We are going to denote by $\pi_2(\mathbf{x}, \mathbf{y})$ the collection of all 2-chains ϕ such that $\partial(\partial(\phi)|_\alpha) = \mathbf{y} - \mathbf{x}$, all the elements of this set are called *domains*. We also say that $\partial\phi$ *connects* $\mathbf{x}$ to $\mathbf{y}$ when $\phi \in \pi_2(\mathbf{x}, \mathbf{y})$.

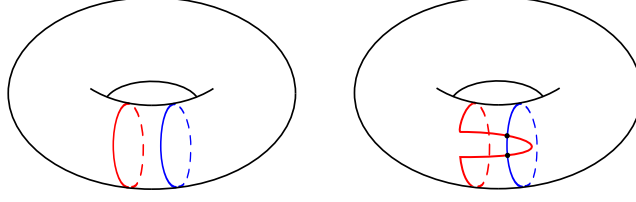


Figure 3.1: An example of a non-admissible diagram and how it is possible to make it admissible.

We can define an additive structure on the set $X = \{\pi_2(\mathbf{x}, \mathbf{y}) \mid \mathbf{x}, \mathbf{y} \text{ generators}\}$ in the following way: let ϕ, ϕ' be two domains such that $\phi = \sum_i a_i \cdot \phi_i$ connects $\mathbf{x}$ to $\mathbf{y}$ and $\phi' = \sum_j a'_j \cdot \phi'_j$ connects $\mathbf{y}$ to $\mathbf{z}$; then we can sum them to get a domain which connects $\mathbf{x}$ to $\mathbf{z}$ as

$$\phi + \phi' := \sum_i a_i \cdot \phi_i + \sum_j a'_j \cdot \phi'_j \in \pi_2(\mathbf{x}, \mathbf{z}).$$

This operation gives rise to an addition

$$\pi_2(\mathbf{x}, \mathbf{y}) \times \pi_2(\mathbf{y}, \mathbf{z}) \xrightarrow{+} \pi_2(\mathbf{x}, \mathbf{z}).$$

Given a point $p \in \Sigma \setminus (\alpha \cup \beta)$, we define the *algebraic intersection number* $n_p(\phi)$ to be the coefficient of the region containing p in ϕ ; we say that a domain ϕ is *positive* if $n_p(\phi) \geq 0$ for all points $p \in \Sigma \setminus (\alpha \cup \beta)$.

Moreover, we define $\pi_2^0(\mathbf{x}, \mathbf{y})$ to be the collection of domains such that have algebraic intersection number 0 on the basepoint z (i.e. the collection of domains that do not contain the basepoint). In symbols,

$$\pi_2^0(\mathbf{x}, \mathbf{y}) = \{\phi \in \pi_2(\mathbf{x}, \mathbf{y}) \mid n_z(\phi) = 0\}.$$

Remark 3.5. The addition is also well defined for

$$\pi_2^0(\mathbf{x}, \mathbf{y}) \times \pi_2^0(\mathbf{y}, \mathbf{z}) \xrightarrow{+} \pi_2^0(\mathbf{x}, \mathbf{z}).$$

Definition 3.6 (Periodic domains). Let $\mathbf{x}$ be a generator. We call $\phi \in \pi_2^0(\mathbf{x}, \mathbf{x})$ a periodic domain if for each i the segments of α_i appear with the same multiplicity on $\partial\phi$ and the same happens for any β_i .

More informally, we are asking that $\partial\phi$ can be expressed as a sum of the α -circles and β -circles.

Definition 3.7 (Admissible Heegaard diagram). We say that a Heegaard diagram is *weakly admissible* if for any generator $\mathbf{x}$ and any non-zero periodic domain $\phi \in \pi_2^0(\mathbf{x}, \mathbf{x})$, we have that ϕ has positive and negative coefficients. This property is sometimes also called *weak admissibility*.

It is possible to see an example of a non-admissible diagram in Figure 3.1.

The property to be admissible for Heegaard diagrams is going to be required when the 3-manifold Y is such that $b_1(Y) > 0$.

Not always it exists a domain that connects two given generators $\mathbf{x}$ and $\mathbf{y}$. It is not easy to graphically understand whether or not $\pi_2(\mathbf{x}, \mathbf{y})$ is empty, but we can develop a method to check this a bit more easily.

Definition 3.8 ($\varepsilon(\mathbf{x}, \mathbf{y})$). Let $\mathbf{x}, \mathbf{y}$ be two generators, explicitly written in coordinates as $\mathbf{x} = \{x_1, \dots, x_g\}$ and $\mathbf{y} = \{y_1, \dots, y_g\}$. We are now going to construct two paths from $\mathbf{x}$ to $\mathbf{y}$, such that one is composed only by α -arcs and one only by β -arcs. We define the path a as a union of g arcs in $\alpha_1 \cup \dots \cup \alpha_g$, say $a = \gamma_1 \cup \dots \cup \gamma_g$, such that $\partial a = y_1 + \dots + y_g - x_1 - \dots - x_g$. In the same way, let $b = \delta_1 \cup \dots \cup \delta_g \subset \beta_1 \cup \dots \cup \beta_g$ be a path with $\partial b = y_1 + \dots + y_g - x_1 - \dots - x_g$. Therefore the difference $a - b$ is a closed 1-cycle in Σ and we define

$$\varepsilon(\mathbf{x}, \mathbf{y}) := [a - b] \in H_1(Y, \mathbb{Z}).$$

We can notice that $\varepsilon(\mathbf{x}, \mathbf{y})$ is independent on the choice of the paths a and b .

As anticipated before, this following criterion explains how we can use $\varepsilon(\mathbf{x}, \mathbf{y})$ to determine whether we have or not a domain connecting $\mathbf{x}$ and $\mathbf{y}$; this is going to be quite useful for the study of Heegaard diagrams.

Proposition 3.9 (Criterion for existence of connecting domain ([Lip06, Lemma 2.2])). *Let $\mathbf{x}, \mathbf{y}$ be two generators of a genus g Heegaard diagram (Σ, α, β) of a 3-manifold Y . Consider two paths a, b as described above; then*

$$\varepsilon(\mathbf{x}, \mathbf{y}) = 0 \iff \pi_2(\mathbf{x}, \mathbf{y}) \neq \emptyset.$$

This shows is that the set of generators of a Heegaard diagram naturally fall into equivalence classes labelled by elements in an affine space over $H_1(Y; \mathbb{Z})$.

Sketch of proof ([Lip06]). ($\Rightarrow$) The main idea to prove this direction is to exploit the fact that a 2-chain $\phi \in H_2(Y)$ can be thought as a domain in $\pi_2(\mathbf{x}, \mathbf{y})$ as well. When $g \geq 2$, we have to exploit the following isomorphism

$$H_2(Y) \cong \frac{H_2((\Sigma \times I) \cup \sqcup_{\alpha \times I} (\mathbb{D}^2 \times I) \sqcup_{\beta \times I} (\mathbb{D}^2 \times I))}{\langle e_1 + \dots + e_n \rangle},$$

where $e_1, \dots, e_n$ are the regions of the diagram. To see this, one can use Mayer-Vietoris on the decomposition

$$Y = \left((\Sigma \times I) \cup \sqcup_{\alpha \times I} (\mathbb{D}^2 \times I) \sqcup_{\beta \times I} (\mathbb{D}^2 \times I) \right) \sqcup \left(\sqcup_0 B^3 \sqcup_1 B^3 \right)$$

exploiting similar arguments to the ones used in the proof of Lemma 2.6.

Start with $\varepsilon(\mathbf{x}, \mathbf{y}) = 0 = [a - b] \in H_1(Y)$. This means that $a - b = \partial \phi$ for some ϕ . Such ϕ can be pushed on Σ (via the isomorphism above indicated), obtaining a domain in $\pi_2(\mathbf{x}, \mathbf{y})$ as we wanted.

($\Leftarrow$) For the converse, assume that $\pi_2(\mathbf{x}, \mathbf{y}) \neq 0$. This means that we have a domain ϕ such that $\partial(\partial(\phi)|_{\alpha}) = \mathbf{y} - \mathbf{x}$ and $\partial(\partial(\phi)|_{\beta}) = \mathbf{y} - \mathbf{x}$; we define

$$\varepsilon(\mathbf{x}, \mathbf{y}) = [\partial(\partial(\phi)|_{\alpha}) - \partial(\partial(\phi)|_{\beta})] \in H_1(Y).$$

Hence, we clearly have $\varepsilon(\mathbf{x}, \mathbf{y}) = [\partial\phi] = 0$. $\square$

Remark 3.10. There are some additional details connected to the notion of $\varepsilon(\mathbf{x}, \mathbf{y})$ that we are omitting for the sake of the exposition. The main notion that we are missing is the one of $\text{Spin}^{\mathbb{C}}$ -structure, but as we are not going to need it, we gloss over it. Further details on this can be found in Section 6 of [OS06].

3.1.2 The Maslov index

We now define the *Maslov index*, a central notion in the construction of the boundary map of $\widehat{\text{CF}}$.

For this, we follow Lipshitz cylindrical reformulation presented in [Lip06], as it is done in [SW06].

Fix a domain ϕ and two generators $\mathbf{x}, \mathbf{y}$ such that $\phi \in \pi_2^0(\mathbf{x}, \mathbf{y})$. We want to define what a *holomorphic embedding* of ϕ is. Let S be a surface with boundary, with $2g$ marked points $(X_1, \dots, X_g, Y_1, \dots, Y_g)$ on ∂S , such that the X -points and the Y -points alternate. The $2g$ arcs on ∂S in the complement of the marked points are divided into two groups, call them A and B , each containing g arcs, such that the A arcs and the B arcs alternate.

Let p_1 and p_2 be the projection maps from $\Sigma \times \mathbb{D}^2$ onto the two factors of the product; we consider $\mathbb{D} \subset \mathbb{C}^2$ and we call $e_1 \subset \partial\mathbb{D}$ the arc that connects $-i$ to i and $e_2 \subset \partial\mathbb{D}$ the one from i to $-i$.

Let $u : S \rightarrow \Sigma \times \mathbb{D}^2$ be the map in the following diagram

$$\begin{array}{ccc} S & \xrightarrow{\quad u \quad} & \Sigma \times \mathbb{D}^2 \\ & & \begin{array}{l} \nearrow p_1 \rightarrow \Sigma \\ \searrow p_2 \rightarrow \mathbb{D}^2 \end{array} \end{array}$$

such that the following properties are satisfied:

- $p_1 \circ u$ is equal to ϕ in $H_2(\Sigma)$;
- in the second homology group $H_2(\mathbb{D}^2, \partial\mathbb{D}^2)$, $[p_2 \circ u] \equiv g\mathbb{D}^2$;
- $p_1 \circ u(\{X_1, \dots, X_g\}) = \{x_1, \dots, x_g\}$ and $p_2 \circ u(\{X_1, \dots, X_g\}) = \{-i\}$;
- $p_1 \circ u(\{Y_1, \dots, Y_g\}) = \{y_1, \dots, y_g\}$ and $p_2 \circ u(\{Y_1, \dots, Y_g\}) = \{i\}$;
- $p_1 \circ u(A) \subset \alpha$ (more precisely, every A -arc is sent to some α -arc) and $p_2 \circ u(A_i) \subset e_1$ for any A -arc A_i ;

- $p_1 \circ u(B) \subset \beta$ (more precisely, every B -arc is sent to some β -arc) and $p_2 \circ u(B_i) \subset e_2$ for any B -arc B_i .

Definition 3.11 (Holomorphic embedding for a domain). For a domain ϕ and two generators $\mathbf{x}, \mathbf{y}$ such that $\phi \in \pi_2^0(\mathbf{x}, \mathbf{y})$, we call (if it exists) a map with the properties of the map u above described a *holomorphic embedding of ϕ* (sometimes also called *holomorphic representative*).

Remark 3.12 (On the definition of S). In the construction of $\widehat{\text{HF}}$ done via Whitney discs, the surface with boundary S is generated starting from a representation f of some domain ϕ , where $f : \mathbb{D}^2 \rightarrow \text{Sym}^g(\Sigma)$, as

$$S = \{(c, d) \in \mathbb{D}^2 \times \Sigma \mid d \in f(c)\}.$$

Therefore, the projection $p_2 : S \rightarrow \mathbb{D}^2$ is a g -fold branched covering map and the projection $p_1 : S \rightarrow \Sigma$ defines a 2-chain in Σ . Then, we proceed by taking two points a, b on $\partial\mathbb{D}^2$ which map to $\mathbf{x}$ and $\mathbf{y}$ and the $2g$ marked points $(X_1, \dots, X_g, Y_1, \dots, Y_g)$ on ∂S are obtained as the inverse images of a, b under p_2 . The construction then continues as we presented above.

Definition 3.13 (Trivial disc). We call a *trivial disc* a component of S that is mapped to a point by $p_1 \circ u$.

Now fix complex structures on Σ and $\mathbb{D}^2$ and take the product complex structure on $\Sigma \times \mathbb{D}^2$.

Remark 3.14. To do things properly, now we would have to talk about *almost-complex structures* and *perturbations*. However, for the main topic of this thesis, these are not needed and we will gloss over these details. However, as one can see in [Lip06] or in Theorem 3.4 of [SW06], the hypothesis needed for our work are satisfied once a choice of a generic perturbation on the α -circles and on the β -circles is made.

Consider now all the holomorphic embeddings u described above for a fixed domain ϕ . They form a *moduli space*, denoted by $\mathcal{M}(\phi)$, that we claim to be a finitely dimensional, smooth manifold. We will not prove such properties of $\mathcal{M}(\phi)$, however one can find in [OS01a], Proposition 3.9, a proof of the smoothness.

Definition 3.15 (Maslov index). We can now define the *Maslov index* for a domain ϕ . It is denoted by $\mu(\phi)$, and it is defined to be the expected dimension of $\mathcal{M}(\phi)$.

At first glance, this index seems to be quite difficult to compute. However, it turns out that it can be computed combinatorially in terms of the *Euler measure* and the *point measures*, which are defined as follows.

Definition 3.16 (Point measure). For a generator $\mathbf{x} = \sum_i x_i$ and a domain ϕ , the point measure on a single x_i , denoted $\mu_{x_i}(\phi)$, is defined to be the average of the coefficients of the four regions around x_i in ϕ (see Figure 3.2).

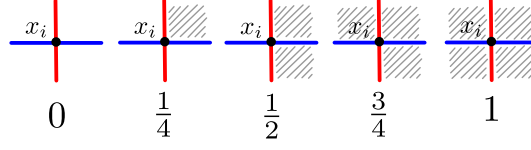


Figure 3.2: Point measure $\mu_{x_i}(\phi)$ (the shadowed part of the drawing represents ϕ).

The *point measure* $\mu_{\mathbf{x}}(\phi)$ is defined as

$$\mu_{\mathbf{x}}(\phi) = \sum_{i=1}^g \mu_{x_i}(\phi).$$

Definition 3.17 (Euler measure). If we fix a metric on Σ which makes all the α and β circles geodesic, intersecting each other with right angles, then the *Euler measure* $e(\phi)$ is defined to be $1/2\pi$ of the integral of the curvature on ϕ .

Remark 3.18 (Combinatorial formula for Euler measure). As shown by Lipshitz in [Lip06] (Corollary 4.10), the Euler measure can be computed combinatorially in the following way. Let us consider a region D ; then we have that

$$e(D) = \chi(D) - \frac{1}{4} \# \{\text{acute corners}\} + \frac{1}{4} \# \{\text{obtuse corners}\},$$

where $\chi(D)$ is the Euler characteristic of D (this number can be computed as $\chi(D) = k_0 - k_1 + k_2 - k_3 + \dots$, where k_i is the number of i -cells in D as a CW-complex). To better explicit what we mean by "acute corners" and "obtuse corners", look at Figure 3.3.

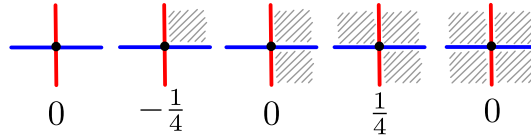


Figure 3.3: Acute and obtuse corners and their contribute for Euler measure, where the shadowed part of the drawing represents ϕ .

Remark now that the Euler measure is additive. Therefore, if we consider a domain $\phi = \sum_i r_i D_i$, we can compute its Euler measure as

$$e(\phi) = \sum_i r_i \left(\chi(D) - \frac{1}{4} \# \{\text{acute corners}\} + \frac{1}{4} \# \{\text{obtuse corners}\} \right).$$

Remark 3.19. If D is a $2n$ -gon region, then it is topologically a disk and hence $\chi(D) = 1$. Therefore, $e(D) = 1 - (2n)/4 = 1 - n/2$.

We have then the following formula, proven by Lipshitz in [Lip06], that explain exactly how to compute the Maslov index in terms of the point measure and the Euler measure.

Proposition 3.20 (Lipshitz Formula ([Lip06])). *For a domain $\phi \in \pi_2(\mathbf{x}, \mathbf{y})$, the Maslov index is given by*

$$\mu(\phi) = e(\phi) + \mu_{\mathbf{x}}(\phi) + \mu_{\mathbf{y}}(\phi).$$

Remark 3.21 (Additivity of Maslov index ([Lip06])). It is possible to prove that the Maslov index is additive, i.e.

$$\mu(\phi_1 + \phi_2) = \mu(\phi_1) + \mu(\phi_2).$$

This can be seen with some analytical arguments, see for instance [Lip06], or with some combinatorial arguments closer to the setting presented here, see Theorem 3.3 of [Sar06].

3.1.3 Hat version of Heegaard Floer homology

We can almost define the boundary map in the chain complex $\widehat{\text{CF}}$. The last piece that is missing, which is also the only non-combinatorial part of $\widehat{\text{HF}}$, is a *count function* that counts the points of a manifold constructed starting from $\mathcal{M}(\phi)$.

Definition 3.22 ($\mathbb{R}$ -action on $\mathcal{M}(\phi)$). If ϕ is non-trivial, then $\mathcal{M}(\phi)$ admits a free $\mathbb{R}$ -action coming from the one-parameter family of holomorphic automorphisms of $\mathbb{D}^2$ which preserves $\pm i$ and the boundary arcs e_1, e_2 . One way to see this $\mathbb{R}$ -action is to use the Riemann mapping theorem to change the unit disc to the infinite strip $[0, 1] \times i\mathbb{R} \subset \mathbb{C}$, where e_1 corresponds to $\{1\} \times i\mathbb{R}$ and e_2 to $\{0\} \times i\mathbb{R}$. Then the $\mathbb{R}$ -action corresponds to vertical translations.

We let $\widetilde{\mathcal{M}}(\phi) = \mathcal{M}(\phi)/\mathbb{R}$. If $\mu(\phi) = 1$, then $\widetilde{\mathcal{M}}(\phi)$ is a compact zero dimensional manifold (see [OS01a], Theorem 3.18).

Therefore, when $\mu(\phi) = 1$, we can then define a *count function* that counts the points of $\widetilde{\mathcal{M}}(\phi)$.

Definition 3.23 (Count function $c(\phi)$). Let $\phi \in \pi_2^0(\mathbf{x}, \mathbf{y})$ be a domain such that $\mu(\phi) = 1$. We define the *count function* $c(\phi)$ to be the number of points in $\widetilde{\mathcal{M}}(\phi)$ counted modulo 2.

Remark 3.24 (On the relation of $\mu(\phi)$ and $c(\phi)$). As we stated above, $\mu(\phi)$ is the expected dimension of $\mathcal{M}(\phi)$. In particular, even if $\mu(\phi) = 1$, we cannot say for sure that ϕ has some holomorphic embeddings and that $\widetilde{\mathcal{M}}(\phi)$ has at least one point; $\mathcal{M}(\phi)$ can be empty even if $\mu(\phi) = 1$. On the other hand, if ϕ does not have any holomorphic embedding and $\mu(\phi) = 1$, then $c(\phi) = 0$.

Thanks to this count function, we can define the boundary map of $\widehat{\text{CF}}$.

Definition 3.25 (Boundary map of $\widehat{\text{CF}}$). The *boundary map* in the chain complex $\widehat{\text{CF}}$ is

$$\partial \mathbf{x} = \sum_{\mathbf{y}} \left(\sum_{\substack{\phi \in \pi_2^0(\mathbf{x}, \mathbf{y}) \\ \text{s.t. } \mu(\phi)=1}} c(\phi) \cdot \mathbf{y} \right).$$

It is shown in [Lip06] (Lemma 8.1) that indeed this is a chain complex. Therefore, we can then define hat Heegaard Floer homology for a Heegaard diagram.

Definition 3.26 ($\widehat{\text{HF}}(\Sigma, \alpha, \beta, z)$). Given a Heegaard diagram $(\Sigma, \alpha, \beta, z)$, we define the *Floer homology group* of the diagram to be the homology group of the chain complex $(\widehat{\text{CF}}(\Sigma, \alpha, \beta, z), \partial)$, which is denoted by $\widehat{\text{HF}}(\Sigma, \alpha, \beta, z)$.

Remark 3.27. All the theory developed so far is quite *combinatorial* (in the sense that it could be easily handled by a calculator) except for this count function above defined.

We will solve this problem using *nice Heegaard diagrams*: in fact, in the case of a diagram with the property of being *nice* (notion that we will define in Chapter 4), we will see that also this last piece of the theory becomes combinatorial and the invariant can be computed by a calculator.

As one can read in Section 8 of [OS06], the following result holds.

Theorem 3.28 ([OS06, Theorem 1]). *Let $(\Sigma, \alpha, \beta, z)$ and $(\Sigma', \alpha', \beta', z')$ be two Heegaard diagrams representing the same closed 3-manifold Y . Then $\widehat{\text{HF}}(\Sigma, \alpha, \beta, z)$ and $\widehat{\text{HF}}(\Sigma', \alpha', \beta', z')$ are isomorphic.*

Therefore, we can define Heegaard Floer homology for 3-manifolds.

Definition 3.29 ($\widehat{\text{HF}}(Y)$). We define $\widehat{\text{HF}}(Y) = \widehat{\text{HF}}(\Sigma, \alpha, \beta, z)$, for some choice of Heegaard diagram of Y .

In fact, in [Lip06], is proved that the chain complex $\widehat{\text{CF}}(\Sigma, \alpha, \beta, z)$ that we constructed is indeed an invariant for the closed 3-manifold Y .

3.2 A first example: $\widehat{\text{HF}}(L(p, q))$

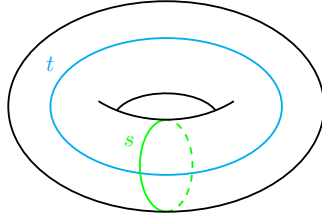
We present here a first example of computation of Heegaard Floer homology. This is a quite simple example: in fact, we see that it is not necessary to compute this complex boundary map of the chain complex. However, it could help to understand how it is possible to work with these objects.

As real example, we compute the Heegaard Floer homology of the general Lens space $L(p, q)$. Recall the definition of $L(p, q)$, given already in Example 1.17 as the quotient of S^3 by the action of $\mathbb{Z}/p\mathbb{Z}$ previously defined.

Our goal is to prove that

$$\widehat{\text{HF}}(L(p, q)) = (\mathbb{F}_2)^p.$$

We start by finding a Heegaard diagram for such space. As already stated, it is possible to see that $L(p, q)$ has a genus 1 Heegaard diagram (Σ, α, β) . We denote s and t the generators of $H_1(\Sigma) = \mathbb{Z}^2$, according to the following drawing.

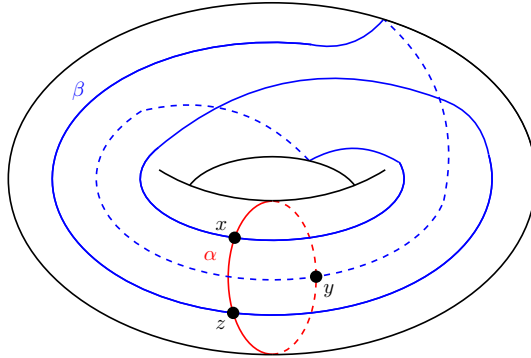


With this notation, we have that the curves of the diagram are described as $[\alpha] = s$ and $[\beta] = qs + pt$ and they are such that intersect at p points (hence we have p generators of $\widehat{\text{CF}}$).

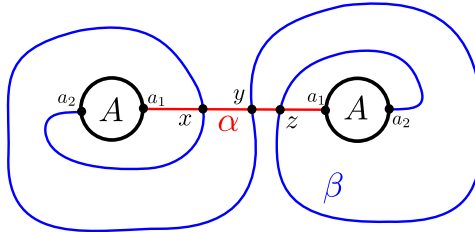
3.2.1 Particular case: $\widehat{\text{HF}}(L(3, 1))$

We consider first a more explicit example, the one with $p = 3$ and $q = 1$. Therefore here we have that $[\alpha] = s$ and $[\beta] = s + 3t$ and we have 3 generators of $\widehat{\text{CF}}$, which we call x, y and z . We have different ways to imagine such diagram; not all of them are actually useful in order to do our computations of $\widehat{\text{HF}}$.

For example, the representation of the embedded torus in $\mathbb{R}^3$ that we already saw in Example 2.4 is not particularly helpful.

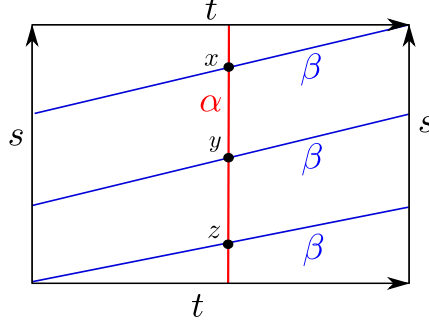


We could more easily use the following representation, where we represented the torus as the plane in which we cut out two discs and we connect the two borders with an handle in the third dimension; but when we work with the general case $L(p, q)$, also this one becomes not very helpful.



The most convenient representation turns out to be the 2-dimensional representation of the torus as a rectangle with identified borders. Here we can draw the Heegaard

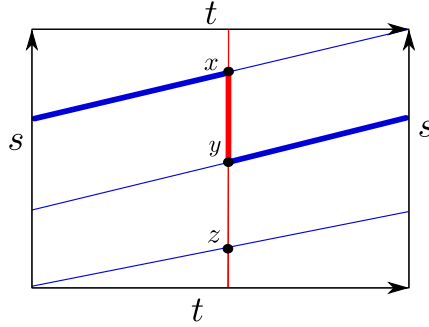
diagram and it is easy to understand what are the regions obtained, also for the general case $L(p, q)$ as we will see in the next Subsection.



We have then three regions. If we draw them individually, we see that each of these regions has all three generators as vertices, with one that is repeated two times; therefore they do not connect two generators. We want now understand the differentials of $\widehat{\text{CF}}$ for this diagram. As first question, we try to understand if exists regions that connect two generators. To do so, we exploit Proposition 3.9 to show that π_2^0 is empty for all the couples of generators. For instance, we compute $\varepsilon(x, y)$. First of all, we need to compute $H_1(L(3, 1))$; for $(\star)$ we have that

$$H_1(L(3, 1)) \cong \frac{H_1(\Sigma)}{\langle [\alpha], [\beta] \rangle} \cong \frac{\langle s, t \rangle}{\langle s, s + 3t \rangle} \cong \frac{\mathbb{Z}}{3\mathbb{Z}}.$$

We need now two paths, $a \subset \mathbb{T}_\alpha$ and $b \subset \mathbb{T}_\beta$ that link x to y ; we choose them as in the following drawing.



It is clear that $[a - b] = [t] = 1 \in H_1(L(3, 1))$, therefore $\varepsilon(x, y) \neq 0$. We can conclude that there are not domains that connect x and y , i.e. $\pi_2(x, y) = \emptyset$. The same reasoning works for all the combinations of generators. It follows that there are not differentials in $\widehat{\text{CF}}$ for this diagram. Since we have three generators for our chain complex, we conclude that $\widehat{\text{HF}}(L(3, 1))$ is 3-dimensional, hence

$$\widehat{\text{HF}}(L(3, 1)) = (\mathbb{F}_2)^3.$$

3.2.2 General case

We now deal with the general case. We begin by drawing the Heegaard diagram in the following way: consider the 2-dimensional torus representation and subdivide it in p horizontal stripes and q vertical stripes; then draw α on the vertical edges of the rectangle and β along the diagonals of the rectangles given by the stripes-subdivision. Notice that we have exactly p regions: in fact, starting from the left, they can only go "strictly" to the right and they are path-connected since they are connected by hypothesis and the torus is path-connected; therefore to count them is sufficient to count how many starting points of paths there are immediately on the right of the curve α drawn on the left, which are exactly p . We want now to understand how are the differentials here, as above we want to use Proposition 3.9. We start then by computing $H_1(L(p, q))$:

$$H_1(L(p, q)) \cong \frac{H_1(\Sigma)}{\langle [\alpha], [\beta] \rangle} \cong \frac{\langle s, t \rangle}{\langle s, qs + pt \rangle} \cong \frac{\langle s, t \rangle}{\langle s, pt \rangle} \cong \frac{\mathbb{Z}}{p\mathbb{Z}}.$$

We take now two distinct generators x_i, x_j . Clearly any path $b \subset \beta$ goes from left to right completely at least one time and strictly less than p times (as $x_i \neq x_j$); therefore $0 \neq \varepsilon(x_i, x_j) = k \in \mathbb{Z}/p\mathbb{Z}$ and $\pi_2(x_i, x_j) = \emptyset$ for any pair of generators. As in the particular case, we have found that there cannot be differentials in $\widehat{\text{CF}}$ and since we have p generators the Heegaard Floer homology is going to be p -dimensional. We can then conclude that

$$\widehat{\text{HF}}(L(p, q)) \cong (\mathbb{F}_2)^p.$$

3.3 Sutured Floer homology SHF

We introduce sutured Floer homology, a generalisation of $\widehat{\text{HF}}$ constructed for sutured manifolds.

As reference, one can look at Section 9 of [Juh06] and Section 6 of [Juh13].

The construction for the chain complex is the same as the one for the closed case, defining in the same way generators, domains (as well as $\pi_2(\mathbf{x}, \mathbf{y})$), admissible diagrams and the Maslov index. In particular, we require sutured diagrams to be admissible. This is not a restriction, as it holds the following lemma.

Lemma 3.30 ([Juh06, Proposition 3.15]). *Every sutured Heegaard diagram is isotopic to an admissible one.*

We then proceed defining $\text{CF}(\Sigma, \alpha, \beta)$ to be the vector space over $\mathbb{F}_2$ generated by all the generators $\mathbf{x}$ of (Σ, α, β) , exactly as in the closed case.

We need to construct the boundary map. We start by noticing that for any border component of Σ , we have only one region of $\Sigma \setminus (\alpha \cup \beta)$ that contains it (recall that we do not allow α -arcs in the sutured context).

We can then regard this diagram as a multi-pointed Heegaard diagram (in the sense of $\widehat{\text{HF}}$), i.e. we place a basepoint z_i in any region that contains a border component. We define the boundary in the same way of the closed case, posing

$$\partial \mathbf{x} = \sum_{\mathbf{y}} \left(\sum_{\substack{\phi \in \pi_2(\mathbf{x}, \mathbf{y}) \\ \text{s.t. } n_{z_i}(\phi) = 0 \forall i \\ \text{and } \mu(\phi) = 1}} c(\phi) \cdot \mathbf{y} \right);$$

and, thanks to Theorem 7.2 of [Juh06], we can see that the one defined is indeed a chain complex.

Proposition 3.31 ([Juh06, Theorem 7.2]). *$(\text{CF}(\Sigma, \boldsymbol{\alpha}, \boldsymbol{\beta}), \partial)$ is a chain complex.*

Definition 3.32 ($\text{SHF}(\Sigma, \boldsymbol{\alpha}, \boldsymbol{\beta})$). Let $(\Sigma, \boldsymbol{\alpha}, \boldsymbol{\beta})$ be an admissible sutured Heegaard diagram. We define the *sutured Floer homology group* for the diagram to be the homology group of the chain complex $(\text{CF}(\Sigma, \boldsymbol{\alpha}, \boldsymbol{\beta}), \partial)$ and we denote it by $\text{SHF}(\Sigma, \boldsymbol{\alpha}, \boldsymbol{\beta})$.

Moreover, we can see that this definition leads us to an invariant of the sutured manifold Y .

Proposition 3.33 ([Juh06, Theorem 7.5]). *Let $(\Sigma, \boldsymbol{\alpha}, \boldsymbol{\beta})$ and $(\Sigma', \boldsymbol{\alpha}', \boldsymbol{\beta}')$ be two admissible sutured Heegaard diagrams for the same balanced sutured manifold. The holds*

$$\text{SHF}(\Sigma, \boldsymbol{\alpha}, \boldsymbol{\beta}) \cong \text{SHF}(\Sigma', \boldsymbol{\alpha}', \boldsymbol{\beta}').$$

Therefore, we can define sutured Floer homology for a generic balanced sutured manifold.

Definition 3.34 ($\text{SHF}(Y, \Gamma)$). Let (Y, Γ) be a balanced sutured manifold. We define $\text{SHF}(Y, \Gamma) = \text{SHF}(\Sigma, \boldsymbol{\alpha}, \boldsymbol{\beta})$, where $(\Sigma, \boldsymbol{\alpha}, \boldsymbol{\beta})$ is an admissible sutured Heegaard diagram for (Y, Γ) .

Remark 3.35 ($\widehat{\text{HF}}$ in terms of SHF). As anticipated, we can re-visit $\widehat{\text{HF}}$ in terms of SHF. Let Y be a closed 3-manifold. Recall the construction for $(Y', \{s_1\})$ done in Remark 2.35. Then, as it is proven in Proposition 9.1 of [Juh06], it holds

$$\widehat{\text{HF}}(Y) \cong \text{SHF}(Y', \{s_1\}).$$

Therefore, we can see $\widehat{\text{HF}}$ as a special case of SHF.

3.4 Knot Floer homology HFK

We can now generalise the construction that we did for $\widehat{\text{CF}}$ to the case of knots in $\mathcal{S}^3$ in a natural way. This theory can actually be generalised to knots in any 3-manifold, as a reference one can look at [OS04] or [Ras03].

As reference for this section, one can look at Section 10 of [OS06].

Definition 3.36 (Knot chain complex $(C(K), \partial_K)$). Let $K \subset \mathcal{S}^3$ be a knot and $(\Sigma, \alpha, \beta, w, z)$ be a compatible doubly pointed Heegaard diagram.

We define $C(K)$ to be the vector space over $\mathbb{F}_2$ generated by the intersection points $\mathbf{x} \in \mathbb{T}_\alpha \cap \mathbb{T}_\beta$. For a generic choice of an (almost) complex structure, we let

$$\partial_K : C(K) \longrightarrow C(K)$$

to be given by

$$\partial_K(\mathbf{x}) = \sum_{\mathbf{y}} \left(\sum_{\substack{\phi \in \pi_2(\mathbf{x}, \mathbf{y}) \text{ s.t. } \mu(\phi)=1 \\ \text{and } n_z(\phi)=n_w(\phi)=0}} c(\phi) \cdot \mathbf{y} \right).$$

We then have the following result, whose proof can be found in [OS04].

Theorem 3.37. $(C(K), \partial_K)$ is a chain complex.

Moreover, its homology $\widehat{\text{HFK}}(C(K), \partial_K)$ is the same given any choice of doubly pointed Heegaard diagrams representing some knot K .

Therefore, we can define the knot Floer homology of a knot in $\mathcal{S}^3$.

Definition 3.38 ($\widehat{\text{HFK}}(\mathcal{S}^3, K)$). Let $K \subset \mathcal{S}^3$ be a knot. We define its knot Floer homology as

$$\widehat{\text{HFK}}(\mathcal{S}^3, K) := \widehat{\text{HFK}}(C(K), \partial_K)$$

for some choice of Heegaard diagrams representing K .

Remark 3.39. With the above description, it is clear how $\widehat{\text{HFK}}$ is a special case of sutured Floer homology, in the sense of what said in Remark 2.42.

3.5 Tangle Floer homology HFT

In this section, we build the invariant HFT for 4-ended tangles.

To fully understand its construction, presented by Claudius Zibrowius in [Zib17] and [Zib20], one would have to have a discuss a particular type of $\mathcal{A}_\infty$ -category: namely, the wrapped Fukaya category for the 4-punctured sphere, denoted by $\text{Fuk}(\mathcal{S}_4^2)$. We can give a rough idea of what this object is for the reader that knows already the algebraic fundamentals: $\text{Fuk}(\mathcal{S}_4^2)$ is defined to be the $\mathcal{A}_\infty$ -category in which the objects are the Lagrangians on a 4-punctured sphere (equipped with a certain 1-form) and the morphisms corresponds to intersection points of these; then the higher composition maps are defined by "counting some polygons" that these Lagrangians draw on $\mathcal{S}_4^2$. More details about this can be found, besides than in Zibrowius' articles, in [Abo06], [Abo+11] and [Fuk93].

In this thesis, the choice to omit this part of the construction of HFT is made: for the scope of the work, it is sufficient to talk about curves on $\mathcal{S}_4^2$ without having to deal with the above presented theory. However, we point out when the results presented are proven with arguments around $\text{Fuk}(\mathcal{S}_4^2)$.

As general reference, one can use Chapter III of [Zib17].

As first step, we adapt the notion of *generator* to the case of tangle Heegaard diagrams (non peculiar).

Definition 3.40 (Generators tangle Heegaard diagram). Let T be a tangle with n open components and m closed components and let $(\Sigma, \alpha = \alpha^c \cup \alpha^a, \beta)$ be a genus g Heegaard diagram representing it.

A *generator* in $(\Sigma, \alpha = \alpha^c \cup \alpha^a, \beta)$ is a tuple $\mathbf{x} = (x_1, \dots, x_{g+m+n-1})$, of points $x_1, \dots, x_{g+m+n-1} \in \alpha \cap \beta$ with the properties that:

- there is exactly one point on each α - and β -circle,
- there is at most one point on each α -arc.

We denote by $\mathbb{T}$ the set of all the generators of the Heegaard diagram.

We restrict now to the case of 4-ended tangles without closed components. Here, generators have exactly one point on a α -arc. Therefore, given $\mathbf{x}$ a generator, we can denote by $s(\mathbf{x})$ the only site of the tangle Heegaard diagram that it stays on. We denote by $\mathbb{T}^s$ the set of all generators which have a point on the site s . Therefore, we have a partition

$$\mathbb{T} = \bigsqcup_{s \text{ site of } T} \mathbb{T}^s.$$

We then define domains, the sets $\pi_2(\mathbf{x}, \mathbf{y})$ and periodic domains as we did for closed 3-manifolds. Consequentially, we can define admissible tangle Heegaard diagrams. From here on, we restrict ourselves to admissible tangle Heegaard diagrams; as showed in Lemma II.2.8 of [Zib17], this can be done without loss of generality, as it is always possible to obtain an admissible diagram starting from a non admissible one. Moreover, we define the Maslov index $\mu(\phi)$ for a domain and the count function $c(\cdot)$ exactly in the same way.

We move now to the setting of peculiar Heegaard diagrams. Here, given a domain ϕ , we define $n_x(\phi)$ to be the multiplicity of it at a basepoint $x = p_i, q_j, z_k, w_l$. We set moreover

$$n_p(\phi) = \sum_{i=1}^4 n_{p_i}(\phi),$$

$$n_q(\phi) = \sum_{j=1}^4 n_{q_j}(\phi).$$

We have now all the objects needed to define a grading on the domains of a peculiar Heegaard diagram, called the *Alexander grading*. This definition is placed here a bit out of context: this grading, among others, is needed for the definition of CFT ^{∂} . Even though we are not going to talk about these gradings, this one is required as input in [nicepy] (when working with tangles) and hence we present it briefly.

Definition 3.41 (Alexander grading). Let T be a 4-ended tangle with n closed components and consider an admissible peculiar Heegaard diagram for T ; call t_1, t_2 the open components and $t_3, \dots, t_{n+2}$ the closed components. We give an orientation to the open components.

Moreover, let $\{(i_1, o_1), (i_2, o_2)\}$ be an ordered partition of $\{1, 2, 3, 4\}$ into pairs, constructed as follows. We associate numbers from 1 to 4 to the tangle ends (starting from north-west and going counter-clock wise); then i_1 is the number corresponding to the inward pointing end of the string t_1 and o_1 the number of the outward pointed end. The same for the second couple and the open component t_2 .

Let ϕ be a domain, $\phi \in \pi_2(\mathbf{x}, \mathbf{y})$ for two generators; We define the *Alexander grading relative to the open components of ϕ* , denoted by $A_{t_k}(\phi)$ for $k = 1, 2$ as

$$A_{t_k}(\phi) := n_{p_{o_k}}(\phi) + n_{q_{o_k}}(\phi) - n_{p_{i_k}}(\phi) - n_{q_{i_k}}(\phi).$$

We define the *Alexander grading relative to the closed components of ϕ* , denoted by $A_{t_h}(\phi)$ for $h = 3, \dots, n+2$ as

$$A_{t_h}(\phi) := n_{w_h}(\phi) - n_{z_k}(\phi).$$

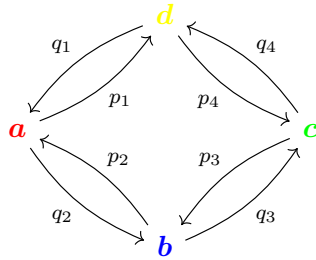
3.5.1 Peculiar modules for 4-ended tangles

In this subsection, we make an intermediate step in the construction of HFT: we talk about the *peculiar module* for a 4-ended tangle T , denoted by $\text{CFT}^\partial(T)$.

As a reference, see Section III.1 of [Zib17].

Definition 3.42 (Quiver). We call a *quiver* an oriented graph that allows loops and multiple arrows between two vertices.

We are going to work with the following quiver, which we denote by $\mathcal{Q}$.



Definition 3.43 (Peculiar algebra $\mathcal{A}^\partial$). We define the *peculiar algebra $\mathcal{A}^\partial$* to be $\mathbb{F}_2[\mathcal{Q}]$, the *path algebra* (also called *quiver algebra*) over $\mathcal{Q}$, with relations $p_i q_i = 0 = q_i p_i$

$$\mathcal{A}^\partial = \frac{\mathbb{F}_2[\mathcal{Q}]}{p_i q_i = 0 = q_i p_i}.$$

This is the $\mathbb{F}_2$ -vector space generated by all the paths in $\mathcal{Q}$, where the multiplication $\mu_{\mathcal{A}^\partial}$ is the concatenation of paths extended bilinearly to all the vector space. By

"concatenating", we mean that given two paths v_1, v_2 , then $v_2 \cdot v_1$ is the concatenation of the two paths if the head of v_1 coincides with the tail of v_2 and zero otherwise. Remark in particular that we have the constant path on any of the vertices, which we indicate with the letter ι ; for instance

$$\iota_a \curvearrowright \mathbf{a}.$$

We call $\mathcal{I}^\partial$ the ring of idempotents of $\mathcal{A}^\partial$ generated by the impotent elements of $\mathcal{A}^\partial$, i.e.

$$\mathcal{I}^\partial = \mathbb{F}_2 \langle \iota_a, \iota_b, \iota_c, \iota_d \rangle$$

To simplify the notation a bit, we can denote the product as

$$\begin{aligned} p_i \cdot p_{i+1} \cdots p_{i+n} &= p_{i(i+1)\dots(i+n)} \\ q_j \cdot q_{j-1} \cdots q_{j-m} &= q_{j(j-1)\dots(j-m)} \end{aligned}$$

where we take the indices modulo 4 and with an offset of 1.

Moreover, we set

$$\begin{aligned} p &= p_1 + p_2 + p_3 + p_4 \\ q &= q_1 + q_2 + q_3 + q_4. \end{aligned}$$

With this notation, we get that $pq = 0 = qp$ and that, for example, $p^2 = p_{12} + p_{23} + p_{34} + p_{41}$. With this new notation, we can forget about the indices 1, 2, 3, 4: in fact, we can write for instance $a \xrightarrow{p^3} b$, which means "the only composition of 3 p -maps that links a and b , i.e. p_{341} ."

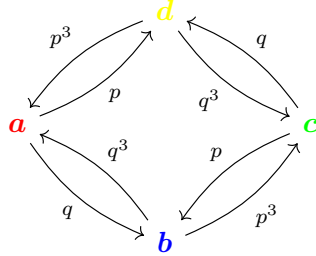
Definition 3.44 (Curved chain complex over $\mathcal{A}^\partial$). A *curved chain complex over $\mathcal{A}^\partial$* is defined by a pair (X, ∂) , where

- X is a left $\mathcal{I}^\partial$ -module, and
- ∂ is a morphism $X \xrightarrow{\partial} \mathcal{A}^\partial \otimes_{\mathcal{I}^\partial} X$ such that $\partial^2 = (p^4 + q^4) \otimes \text{id}_X$, where we define ∂^2 as the following composition of maps

$$\begin{array}{c} X \xrightarrow{\partial} \mathcal{A}^\partial \otimes_{\mathcal{I}^\partial} X \xrightarrow{\text{id}_{\mathcal{A}^\partial} \otimes \partial} \mathcal{A}^\partial \otimes_{\mathcal{I}^\partial} \mathcal{A}^\partial \otimes_{\mathcal{I}^\partial} X \xrightarrow{\mu_{\mathcal{A}^\partial} \otimes \text{id}_X} \mathcal{A}^\partial \otimes_{\mathcal{I}^\partial} X \\ \searrow \hspace{10em} \nearrow \\ \partial^2 \end{array}$$

We denote by $\text{Mod}_{p^4+q^4}^{\mathcal{A}^\partial}$ the category of all the curved chain complexes over $\mathcal{A}^\partial$: the objects are curved chain complexes and, given $(X_1, \partial_1), (X_2, \partial_2)$ two curved chain complexes, the morphisms between them are chain maps, i.e. maps $f : X_1 \rightarrow X_2$ that commute with the differentials in the following sense $\partial_2 \circ f = f \circ \partial_1$.

Example 3.45 (A first curved chain complex). We can see a first example of a curved chain complex (X, ∂) , where X is



We can see that it is a curved chain complex. Firstly, we have that for all vertices $x \neq y$ holds

$$\sum_z (x \rightarrow z \rightarrow y) = 0 :$$

in fact, either we have $x \xrightarrow{p^m} z \xrightarrow{q^n} y$ or $x \xrightarrow{q^n} z \xrightarrow{p^m} y$; hence the composition is zero. Secondly, we have that for any vertex x holds

$$\sum_z (x \rightarrow z \rightarrow x) = p^4 + q^4 :$$

in fact, for any vertex we have one arrow pair whose composition is p^4 and one whose composition is q^4 .

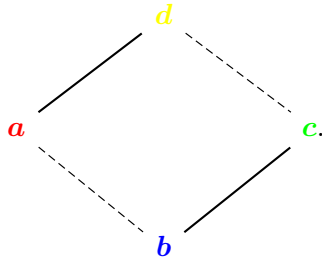
We can still slim the notation a bit without adding confusion. Let n, m be positive integers such that $n + m = 4$; then for any two generators x, y we denote

$$\begin{array}{c} x \xrightarrow{p^n} y \\ x \xleftarrow{p^m} y \end{array} \quad \equiv \quad x \text{ ————— } y$$

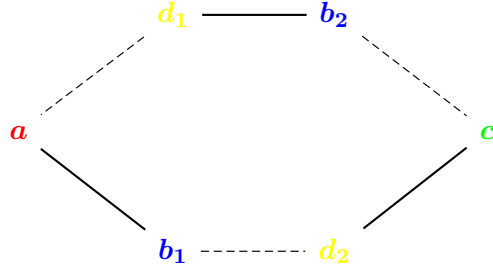
and

$$\begin{array}{c} x \xrightarrow{q^n} y \\ x \xleftarrow{q^m} y \end{array} \quad \equiv \quad x \text{ - - - - - } y.$$

With this notation, the last example is written as

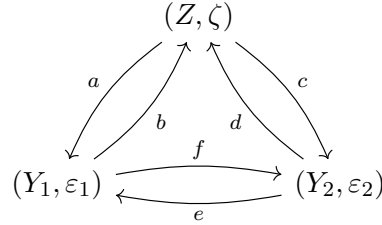


Example 3.46. Another example of curved chain complex is the following



We have two main results that help us to work with curved chain complex: the Cancellation Lemma and the Clean-Up Lemma. These two lemmas allow us to "simplify" curved chain complexes. The proofs of both can be found in [Zib20], Lemma 1.22 and Lemma 1.24.

Lemma 3.47 (Cancellation Lemma ([Zib20, Lemma 1.22])). *Let (X, ∂) be a curved chain complex of the following form*



where (Z, ζ) , (Y_1, ε_1) and (Y_2, ε_2) are pre-curved chain complexes, i.e. a curved chain complex for which we do not require that $\partial^2 = (p^4 + q^4) \cdot \text{id}$, and f is an isomorphism with inverse g .

Then (X, ∂) is chain homotopic to the curved chain complex $(Z, \zeta + bgc)$.

Lemma 3.48 (Clean-Up Lemma ([Zib20, Lemma 1.24])). *Let (X, ∂) be a curved chain complex. Then, for any morphism $h \in \text{Mor}((X, \partial), (X, \partial))$ with the properties that*

- $h^2 = 0$,
- $h(h\partial + \partial h) = 0$,
- $(h\partial + \partial h)h = 0$,

it holds that (X, ∂) is chain isomorphic to $(X, \partial + (h\partial + \partial h))$.

In Definition III.1.11 of [Zib17], the author introduced the *the peculiar module of a tangle T* . We do not explain how the construction works in details: the main thing that we are missing is a discussion on three different gradings defined on these algebraic objects; for instance, the Alexander grading presented in Definition 3.41 is one of them. With these gradings, one can define *peculiar modules*, a particular class of curved chain complexes endowed with these additional structures. However, as we present this topic only in broad terms in order to highlight its Floer homology structure, we talk about peculiar modules and curved chain complexes as if they were the same thing.

Definition 3.49 ($((\text{CFT}^\partial(T), \partial))$). Let T be a 4-ended tangle without closed components and consider an admissible peculiar Heegaard diagram for T ; call $\mathbb{T}$ the set of generators of the diagram.

We define $\text{CFT}^\partial(T)$ to be $\mathbb{F}_2$ -module freely generated by elements in $\mathbb{T}$.

We can turn $\text{CFT}^\partial(T)$ into a left $\mathcal{I}^\partial$ -module by defining for any $\mathbf{x} \in \mathbb{T}$ the following multiplication

$$\iota_i \cdot \mathbf{x} = \begin{cases} \mathbf{x} & \text{if } s(\mathbf{x}) = i, \\ 0 & \text{otherwise.} \end{cases}$$

Moreover, we need to define a boundary map on $\text{CFT}^\partial(T)$; this is done in similar fashion to the closed case.

Definition 3.50 (Boundary map for $\text{CFT}^\partial(T)$). Given a generator $\mathbf{x}$, we define

$$\partial \mathbf{x} = \sum_{\mathbf{y}} \left(\sum_{\substack{\phi \in \pi_2(\mathbf{x}, \mathbf{y}) \\ \text{s.t. } n_p(\phi), n_q(\phi) \geq 0 \\ \text{and } \mu(\phi) = 1}} c(\phi) \cdot \iota_{s(\mathbf{x})} \cdot p^{n_p(\phi)} q^{n_q(\phi)} \cdot \iota_{s(\mathbf{y})} \otimes_{\mathcal{I}^\partial} \mathbf{y} \right).$$

The following result says that we have indeed define a peculiar module in the sense above presented and that it is an invariant for the tangle T .

Theorem 3.51 ([Zib17, Theorem III.1.12]). $\text{CFT}^\partial(T)$ is indeed a well-defined peculiar module. Furthermore, its chain homotopy type is an invariant of the tangle T .

3.5.2 The multicurve invariant HFT

We now link the algebraic theory developed around $\text{Mod}_{p^4+q^4}^{\mathcal{A}^\partial}$ and CFT^∂ and the more geometric perspective of curves in $\mathcal{S}_4^2$. To do so, we need to talk about *multicurves* on $\mathcal{S}_4^2$.

We discuss this according to what is done in Section 1.2 of [Zib19] and in Section 4.2 of [Aub22].

Definition 3.52 (Loop or immersed curved with local system). A *loop* or *immersed curve with local system* on the 4-punctured sphere $\mathcal{S}_4^2$ is defined as a pair (γ, X) , where

- γ is an immersion of an oriented circle into $\mathcal{S}_4^2$, which represents a non-trivial primitive element¹ of $\pi_1(\mathcal{S}_4^2)$;
- $X \in \text{GL}_n(\mathbb{F}_2)$ for some integer n .

¹Recall that an element h of a group G is said to be primitive if it does not exists $g \in G$ such that for some $n \in \mathbb{N}_{\geq 2}$ holds $h = g^n$.

We consider γ up to isotopy, X up to matrix similarity and (γ, X) up to simultaneous orientation reversal of γ and inversion of the matrix X . Given a loop (γ, X) , we say that γ is the *underlying curve* of the loop and that X is its *local system*.

Definition 3.53 (Multicurve). We define a *multicurve* on $\mathcal{S}_4^2$ to be an unordered set of loops with local systems $L = \{(\gamma_i, A_i)\}_{i=1}^k$. Given a multicurve L and a loop γ , we define $A_L(\gamma)$ as the direct sum² of the following matrices

$$A_L(\gamma) = \bigoplus_{\gamma=\gamma_i} A_i \oplus \bigoplus_{-\gamma=\gamma_i} A_i^{-1},$$

where with $\gamma = \gamma_i$ we mean that they are isotopic and $-\gamma$ is γ with the orientation reversed. We then say that two multicurves $L = \{(\gamma_i, A_i)\}_{i=1}^k$ and $L' = \{(\gamma'_j, A'_j)\}_{j=1}^l$ are *isotopic* if for any loop γ the matrices $A_L(\gamma)$ and $A_{L'}(\gamma)$ are similar.

Remark 3.54. It is clear from the definition how it is possible to go from a multicurve to a collection of immersed curves and vice versa. Hence, we can freely talk about collection of immersed curves instead of multicurve if it is advantageous.

Theorem 3.55 ([Zib19, Theorem 1.11]). *We can associate to any peculiar module (X, ∂) a multicurve $L(X, \partial) = \{(\gamma_i, A_i)\}_{i \in I}$ in a way such that if another peculiar module (X', ∂') is given with $L(X', \partial') = \{(\gamma'_j, A'_j)\}_{j \in J}$, then we have that (X, ∂) and (X', ∂') are homotopic if and only if the multicurves are the same.*

In other words, we have a 1-to-1 correspondence

$$\frac{\left\{ \text{Multicurves on } \mathcal{S}_4^2 \right\}}{\text{homotopy}} \xleftrightarrow{1:1} \frac{\left\{ \text{Objects in } \text{Mod}_{p^4+q^4}^{\mathcal{A}^\partial} \right\}}{\text{chain homotopy}}.$$

A more detailed version of Theorem 3.55 is one of the main result present in [Zib20], as one can read in its Section 0.2. It comes from an equivalence between the category of peculiar modules and the compact part of the wrapped Fukaya category of $\mathcal{S}_4^2$. We are going to see that this is a central result for HFT since, by virtue of it, it is possible to classify peculiar modules in terms of multicurves.

We can see explicitly how this correspondence works with some illustrative examples; a more detailed explanation can be found in Definition 1.10 of [Zib19].

Let $(X, \partial) \in \text{Mod}_{p^4+q^4}^{\mathcal{A}^\partial}$ be a curved chain complex and consider the 4-punctured sphere $\mathcal{S}_4^2$ with the sites a, b, c, d as previously defined. We have that every generator in X corresponds to an intersection point on some site: a red generator (labelled "a") is a point on the site a , a blue one is on the site b , and so on. Then we have

²Recall that the direct sum of matrices is defined as

$$\oplus_{i=1}^n X_i = \begin{pmatrix} X_1 & 0 & \dots & 0 & 0 \\ 0 & X_2 & \dots & 0 & 0 \\ 0 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 0 & X_n \end{pmatrix}.$$

that a solid line that connects two generators is an segment on the front of $\mathcal{S}_4^2$ and a dashed line is a segment on the back of it. See Figure 3.4 for two examples example.

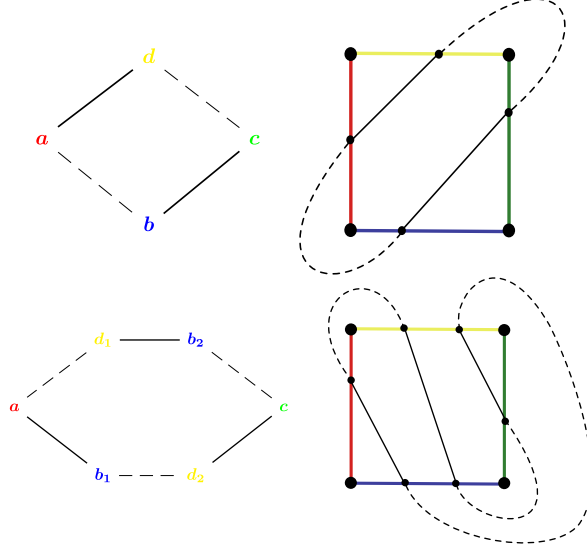


Figure 3.4: Examples of the correspondence between multicurves and curved chain complexes.

We can finally define the invariant HFT in terms of $\text{CFT}^\partial(T)$.

Definition 3.56 (HFT). Given an oriented 4-ended tangle T , we define $\text{HFT}(T)$ as the multicurve associated to $\text{CFT}^\partial(T)$.

Surprisingly, we can classify the multicurves yielded by HFT with only two families: the *rational curves* and the *special curves*, also called *irrational curves*. For this classification, we use the covering space of $\mathcal{S}_4^2$ presented in Section 1.4: given a closed curve $\gamma : \mathcal{S}^1 \rightarrow \mathcal{S}_4^2$, we can lift it to a curve $\tilde{\gamma}$ in $\mathbb{R}^2/\mathbb{Z}^2$ as

$$\begin{array}{ccc} I & \xrightarrow{\tilde{\gamma}} & \mathbb{R}^2/\mathbb{Z}^2 \\ \downarrow \scriptstyle 0 \equiv 1 & & \downarrow \scriptstyle \eta \\ \mathcal{S}^1 & \xrightarrow{\gamma} & \mathcal{S}_4^2. \end{array}$$

Definition 3.57 (Rational and special curves). Let p/q be an element of $\mathbb{QP}^1$. We define the *rational curve of slope p/q* to be the primitive loop $\mathfrak{r}(p/q) \subset \mathcal{S}_4^2$ that under the covering map η lifts (up to homotopy) to a straight line of slope p/q . We denote it by $\mathfrak{r}_A(p/q)$ if we equip it with the local system A .

Let now i_1, i_2 be two distinct tangle ends that lie on a line of slope $p/q \in \mathbb{QP}^1$ in $\mathbb{R}^2$ with the lattice given by $\mathbb{Z}^2$. We can divide this line into intervals of equal length with the points of the lattice; we mark every $(2n)$ -th interval of the line for a fixed integer $n > 0$. We consider a small push-off of this line, done in a way that the result intersects only the marked intervals and each of them exactly once. We

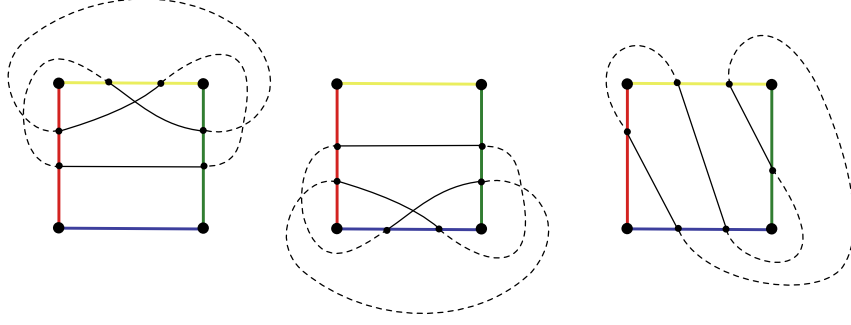


Figure 3.5: Three linear loops. Starting from the left: $\mathfrak{s}_2(0; 1, 4)$, $\mathfrak{s}_2(0; 2, 3)$ and $\mathfrak{r}(-2)$.

can define the *special curve of slope p/q of length n through the punctures i_1, i_2* to be the primitive loop $\mathfrak{s}_n(p/q; i_1, i_2) \subset \mathcal{S}_4^2$ that under the covering map η lifts (up to homotopy) to the curve described above.

If a primitive loop in $\mathcal{S}_4^2$ is either rational or special we call it *linear with slope p/q* .

The following theorem explains in which sense the multicurves obtained via HFT can be classified by these two families.

Theorem 3.58 ([Zib19, Theorem 0.5]). *For a 4-ended tangle T , the underlying curve of each component of $\text{HFT}(T)$ is either rational or special. Moreover, if it is special, its local system is equal to an identity matrix.*

We can slim a bit more the notation of the rational and special curves. First, we lose the data of the local system. In fact, by the above theorem, for the special curves it is always going to be trivial. For rational curves it could in principle be non trivial; however, to today's knowledge of the author, none computation of HFT yielded a rational curve with non trivial local system. We can still push the notation a bit further with the help of next result. Notice that, for each slope p/q and a fixed $n > 0$, there are exactly two choices for the pair of punctures (i_1, i_2) . The next theorem tells us that given any $\text{HFT}(T)$, the number of such components agrees.

Theorem 3.59 ([LMZ20, Theorem 3.9]). *Let $i, j, k, l = 1, 2, 3, 4$, $p/q \in \mathbb{Q}P^1$ and $n \in \mathbb{N}$. Then, for any 4-ended tangle T , the number of components of the form $\mathfrak{s}_n(p/q; i, j)$ and $\mathfrak{s}_n(p/q; k, l)$ in $\text{HFT}(T)$ agree.*

We can therefore introduce the following notations for special curves: we denote by $\mathfrak{s}_{2n}(p/q)$ the data of the presence of $\mathfrak{s}_n(p/q; i, j)$ and $\mathfrak{s}_n(p/q; k, l)$.

Moreover, recalling the definition of mirror image $m(\cdot)$ of a tangle, the following result explains how it affects the tangle Floer homology of a given tangle.

Proposition 3.60 ([Zib20, Proposition 5.4]). *Let T be a tangle with tangle Floer homology*

$$\text{HFT}(T) = \sum_{i=1}^n \mathfrak{r}(p_i/q_i) + \sum_{j=1}^m \mathfrak{s}_{2n}(p_j/q_j).$$

Then, it holds

$$\text{HFT}(m(T)) = m(\text{HFT}(T)) = \sum_{i=1}^n \mathfrak{r}(-p_i/q_i) + \sum_{j=1}^m \mathfrak{s}_{2n}(-p_j/q_j).$$

3.5.3 Naturality on rational tangles

We can explore in particular the case of rational tangle: in fact, the invariant HFT behaves particularly well on these, managing even to detect them.

In [Zib20], Zibrowius proved the following result.

Theorem 3.61 ([Zib20, Theorem 6.2]). *A 4-ended tangle T is rational if and only if $\text{HFT}(T)$ consists of single rational component carrying the unique one-dimensional local system. Moreover, $\text{HFT}(Q_{p/q}) = \mathfrak{r}(p/q)$.*

Remark 3.62. This result, justify once more our choice to lose the data of the local system for rational curves.

Moreover, in [Zib19], the author was able to prove a stronger version of the above result, which allows us to know how adding vertical or horizontal (half-) twists to any 4-ended tangle affects HFT. Recall the definition of the two operators τ_1 and τ_2 given in Section 1.4.

Theorem 3.63 ([Zib19, Theorem 2.1]). *Let T be a 4-ended tangle and $\tau \in \langle \tau_1, \tau_2 \rangle$. Then*

$$\text{HFT}(\tau T) = \tau \text{HFT}(T).$$

Recalling Remark 1.38, we can see an application of this last result, as it is going to be a useful argument in Chapter 5. Let T be a tangle and denote its tangle Floer homology as

$$\text{HFT}(T) = \sum_{i=1}^n \mathfrak{r}(p_i/q_i) + \sum_{j=1}^m \mathfrak{s}_{2n}(p_j/q_j).$$

Assume that we want to add a horizontal or vertical half-twists to it. The final tangle T' can be described as

$$T' = T + Q_{\pm 1}$$

if the half twist is horizontal and

$$T' = T * Q_{\pm 1}$$

if it is vertical. Let us say that it is horizontal; then, we obtain

$$\text{HFT}(T') = \text{HFT}(T + Q_{\pm 1}) = \sum_{i=1}^n \mathfrak{r}(p_i/q_i \pm 1) + \sum_{j=1}^m \mathfrak{s}_{2n}(p_j/q_j \pm 1).$$

In the vertical case, we obtain

$$\text{HFT}(T') = \text{HFT}(T * Q_{\pm 1}) = \sum_{i=1}^n \mathfrak{r}(p_i/(q_i \pm 1)) + \sum_{j=1}^m \mathfrak{s}_{2n}(p_j/(q_j \pm 1)).$$

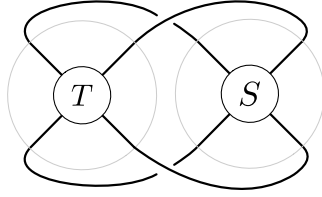


Figure 3.6: Knot obtained as the gluing of two 4-ended tangles.

3.5.4 Motivations from bordered sutured point of view

The construction of this version of the Heegaard Floer homology is based on *bordered sutured Floer homology*. Even though we do not explore such theory, we can still use it to motivate what we presented for tangles. As anticipated in Section 1.7, the main interesting property of bordered sutured Floer homology that is inherited by HFT is this gluing property. We can state a particular case of Theorem 0.2 of [Zib20] to understand better what this means.

Theorem 3.64 ([Zib20, Theorem 0.2]). *Let T_1 and T_2 be two 4-ended tangles without closed components and such that their gluing as showed in Figure 3.6.*

Then the knot Floer homology $\widehat{\text{HFK}}(\mathcal{S}^3, K)$ can be computed from $\text{CFT}^\partial(T_1)$ and $\text{CFT}^\partial(T_2)$.

Further details can be found in the Introduction of [Zib20].

Chapter 4

The Nicefication algorithm

In this chapter, we introduce *nice Heegaard diagrams*, understanding how we can compute Heegaard Floer homology in this case. The notion of nice Heegaard diagram was presented in [SW06] as follow.

Definition 4.1 (Nice Heegaard diagram). We say that a pointed Heegaard diagram $\mathcal{H} = (\Sigma, \alpha, \beta, z)$ for a 3-manifold Y is *nice* if any region that does not contain z is either a bigon or a rectangle.

The main motivation of our interest in nice Heegaard diagram is summed up in the next two theorems, proved by Sarkar and Wang in [SW06]. This chapter is devoted to prove them and to generalise them in the cases of our interest.

Theorem 4.2 (Sarkar-Wang, 2006 ([SW06, Theorem 1.1])). *Given a nice Heegaard diagram of a closed oriented three-manifold Y , we have that its Heegaard Floer homology $\widehat{\mathrm{HF}}(Y)$ can be computed combinatorially.*

Similarly, for a knot $K \subset \mathcal{S}^3$, we have that its Heegaard Floer knot homology $\widehat{\mathrm{HFK}}(\mathcal{S}^3, K)$ can be computed combinatorially in a nice Heegaard diagram.

Theorem 4.3 (Sarkar-Wang, 2006 ([SW06, Theorem 1.2])). *Every closed oriented 3-manifold Y admits a nice Heegaard diagram.*

Every knot K in $\mathcal{S}^3$ admits a nice Heegaard diagram.

In particular, Theorem 4.3 is proven by the existence of the *Nicefication Algorithm*, an algorithm able to convert any pointed Heegaard diagram to a nice Heegaard diagram via isotopies and handleslides. This was developed by Sarkar and Wang in [SW06] and implemented in the course of this thesis work by the author.

4.1 Holomorphic discs in nice Heegaard diagrams

In this section, we prove Theorem 4.2. This is done by looking at the count function $c(\cdot)$ in the nice case, proving that it is not trivial only when ϕ is a special type of domain, called *empty embedded n -gon*.

As a reference, one can look at Section 3 of [SW06].

Let $\mathcal{H} = (\Sigma, \alpha, \beta, z)$ be a nice admissible Heegaard diagram for a 3-manifold Y . Choose a product complex structure on $\Sigma \times \mathbb{D}^2$.

Definition 4.4 (Empty embedded $2n$ -gon). Let $\mathbf{x} = \{x_1, \dots, x_g\}$ and $\mathbf{y} = \{y_1, \dots, y_g\}$ be generators in $\mathcal{H}$. A domain $\phi \in \pi_2^0(\mathbf{x}, \mathbf{y})$ is called an *empty embedded $2n$ -gon* if it has the following properties:

1. it only has 0 and 1 as coefficients;
2. it is topologically an embedded disc with $2n$ vertices on its boundary;
3. for each vertex v holds $\mu_v(\phi) = 1/4$ (meaning that, if $D_1, \dots, D_4$ are the region around v , only one of them has 1 as coefficient in ϕ);
4. it does not contain x_i or y_i for any i in its interior.

Remark 4.5 (For any empty embedded $2n$ -gon ϕ , $\mu(\phi) = 1$). Any empty embedded $2n$ -gon $\phi \in \pi_2^0(\mathbf{x}, \mathbf{y})$ is detected by the boundary map of $\widehat{\text{CF}}$. In fact, by Remark 3.19 we have that $e(\phi) = 1 - n/2$. Moreover, each of the corners has to be an x_i or an y_i and at every other x_j and y_j the point measure is zero, i.e. $\mu_{x_j}(\phi) = 0 = \mu_{y_j}(\phi)$. Therefore $\mu_{\mathbf{x}}(\phi) + \mu_{\mathbf{y}}(\phi) = (2n)/4 = n/2$ and $\mu(\phi) = e(\phi) + \mu_{\mathbf{x}}(\phi) + \mu_{\mathbf{y}}(\phi) = 1$.

To prove Theorem 4.2, we exploit two other results proven in [SW06]. The first one asserts that in a nice Heegaard diagram the only domains that contribute in the boundary map of $\widehat{\text{CF}}$ are empty embedded bigons and empty embedded squares.

Theorem 4.6 ([SW06, Theorem 3.3]). *Let $\phi \in \pi_2^0(\mathbf{x}, \mathbf{y})$ be a domain in a nice Heegaard diagram such that $\mu(\phi) = 1$. If ϕ has a holomorphic embedding, then ϕ is an empty embedded bigon or an empty embedded square.*

The second one, which we present slightly differently from what is done in [SW06], states that every empty embedded bigons and every empty embedded square have non trivial count function.

Theorem 4.7 ([SW06, Theorem 3.4]). *If $\phi \in \pi_2^0(\mathbf{x}, \mathbf{y})$ is an empty embedded bigon or an empty embedded square, then $\mu(\phi) = c(\phi) = 1$.*

Therefore, we can already prove Theorem 4.2 as a consequence of Theorems 4.6 and 4.7.

Proof of Theorem 4.2 ([SW06]). Let ϕ be a domain in a nice Heegaard diagram with $\mu(\phi) = 1$. If ϕ does not have a holomorphic embedding, then $c(\phi) = 0$ and ϕ is not detected by the boundary map of $\widehat{\text{CF}}$. If ϕ has some holomorphic embedding, then by Theorem 4.6 we obtain that ϕ is either an empty embedded bigon or an empty embedded square and by Theorem 4.7 we get $c(\phi) = 1$.

Moreover, again by Theorem 4.7, every empty embedded bigon and every empty embedded square has count function equal to 1.

Therefore, the count function is combinatorial in a nice Heegaard diagram and consequentially also the computation of $\widehat{\text{HF}}$ can be done combinatorially: it sufficient to count the empty embedded bigons and the empty embedded squares of the nice Heegaard diagram. $\square$

We devote the rest of this section to the proofs of Theorems 4.6 and 4.7.

In order to prove Theorem 4.6, we need to discuss the consequences for a domain ϕ to have at least one holomorphic embedding u . We have the following proposition, whose proof can be found in [SW06].

Proposition 4.8 ([SW06, Proposition 2.4]). *Let ϕ be a domain in $\pi_2^0(\mathbf{x}, \mathbf{y})$ for some generators $\mathbf{x}$ and $\mathbf{y}$. If ϕ has a holomorphic embedding, then ϕ is a positive domain. In particular, if $c(\phi) \neq 0$, then ϕ is a positive domain.*

Moreover, as one can read in Section 2.3 of [SW06], also the following lemma holds.

Lemma 4.9 ([SW06]). *If ϕ has a holomorphic embedding u , then the number of branch points of $p_2 \circ u$ is given by $\mu_{\mathbf{x}}(\phi) + \mu_{\mathbf{y}}(\phi) - e(\phi)$.*

In such situation, the Maslov index can be computed in a more practical way. In fact, as one can deduce by manipulating the formula given in the statement of Proposition 4.2 of [Lip06] and the formulas given in its proof, we have that

$$\begin{aligned} \mu(\phi) &= 2e(\phi) + g - \chi(S) \\ &= e(\phi) + b + \frac{1}{2}(g - t), \end{aligned} \tag{4.1}$$

where

- b denotes the number of branch points of $p_1 \circ u$;
- t denotes the number of trivial discs (Definition 3.13); notice that they correspond to the coordinates x_i of $\mathbf{x}$ with $\mu_{x_i}(\phi) = 0$.

With this acquired knowledge, we can prove Theorem 4.6.

Proof of Theorem 4.6 ([SW06]). Let $\phi = \sum_i a_i D_i$ be a domain, where the D_i 's are regions that do not contain basepoints. Let u be one holomorphic embedding for ϕ ; then for Proposition 4.8, we have that $a_i \geq 0$, for any i . Moreover, by the niceness of the diagram we have that each D_i is a bigon or a square; therefore for Remark 3.19 we have $e(D_i) \geq 0$ and hence $e(\phi) = \sum_i a_i \cdot e(D_i) \geq 0$. So, by Lipshitz's formula

$$1 = \mu(\phi) = e(\phi) + \mu_{\mathbf{x}}(\phi) + \mu_{\mathbf{y}}(\phi),$$

we obtain that $0 \leq \mu_{\mathbf{x}}(\phi) + \mu_{\mathbf{y}}(\phi) \leq 1$. We are now going to see that there are only two possibilities for the values of $\mu_{\mathbf{x}}(\phi)$ and $\mu_{\mathbf{y}}(\phi)$: either $\mu_{\mathbf{x}}(\phi) = \mu_{\mathbf{y}}(\phi) =$

$1/2$ or $\mu_{\mathbf{x}}(\phi) = \mu_{\mathbf{y}}(\phi) = 1/4$. After proving this, we will use Lipshitz's cylindrical reformulation to understand what bordered surface maps into $\Sigma \times \mathbb{D}^2$ and represents ϕ , proving that it is either a "bigon" or a "square".

We need the following definition in order to continue the proof.

Definition 4.10. We say that ϕ *hits the α -circle α_i* if $\partial\phi$ is non-zero on some part of α_i .

We have the following observations.

Observation 4.11. 1. For any $x_i \in \mathbf{x}$ and $y_i \in \mathbf{y}$, we can assume that $x_i, y_i \in \alpha_i$.

2. Since $\phi \neq \Sigma$ (because ϕ does not contain the basepoint z), it has to hit at least one α -circle; without loss of generality, assume that ϕ hits α_1 . Therefore, we can deduce that $\mu_{x_1}(\phi), \mu_{y_1}(\phi) \geq 1/4$: in fact, by definition, if $\phi \in \pi_2^0(\mathbf{x}, \mathbf{y})$, then $\partial(\partial\phi|_{\alpha}) = \mathbf{y} - \mathbf{x}$; since $\partial\phi|_{\alpha_1} \neq 0$, we have that x_1 and y_1 are vertices of ϕ and that $\mu_{x_1}(\phi), \mu_{y_1}(\phi) \geq 1/4$ (they are vertices at least one time for ϕ).
3. Assume that ϕ does not hit α_i , for some $i \geq 2$. Then $x_i = y_i$, otherwise it would not be true that $\partial(\partial\phi|_{\alpha}) = \mathbf{y} - \mathbf{x}$.

Moreover, they must lie outside the domain ϕ . In fact, if they were to be on the boundary, we would have that $\mu_{x_i}(\phi) = \mu_{y_i}(\phi) \geq 1/2$, if they were to be on the interior, we would have that $\mu_{x_i}(\phi) = \mu_{y_i}(\phi) = 1$. In both cases, we would have $\mu_{x_i}(\phi) = \mu_{y_i}(\phi) \geq 1/2$ and hence we would have

$$1 < \mu_{\mathbf{x}}(\phi) + \mu_{\mathbf{y}}(\phi) = \mu_{x_1}(\phi) + \mu_{y_1}(\phi) + \mu_{x_i}(\phi) + \mu_{y_i}(\phi),$$

which is an absurd.

4. ϕ can hit at most two α -circles: if it was to hit more than 2, we would have that $\mu_{\mathbf{x}}(\phi) + \mu_{\mathbf{y}}(\phi) \geq 3/4$.
5. Remark also that $e(\phi) \in 1/2\mathbb{N}$ (since Euler measure is additive and $e(D) = 1/2$ for D a bigon and $e(D) = 0$ for D a square, see Remark 3.19).

Therefore, we can only have the following cases regarding ϕ .

- **Case 1:** ϕ hits two α -circles. In such case, we have that the following properties hold:
 - without loss of generality, we assume ϕ to hit α_1 and α_2 ;
 - $\mu_{x_1}(\phi) = \mu_{x_2}(\phi) = \mu_{y_1}(\phi) = \mu_{y_2}(\phi) = 1/4$;
 - ϕ consists of squares, since for Lipshitz's formula $e(\phi) = 0$;
 - there are $(g - 2)$ trivial discs (corresponding to $x_3 = y_3, \dots, x_g = y_g$).
- **Case 2:** ϕ hits only one α -circle. In such case, we have two possible sub-cases, distinguishing on the sum $\mu_{\mathbf{x}}(\phi) + \mu_{\mathbf{y}}(\phi)$
 - * **Sub-case 2a:** $\mu_{\mathbf{x}}(\phi) + \mu_{\mathbf{y}}(\phi) = 1/2$. In this case, we have that:

- ϕ hits only α_1 ;
 - $\mu_{x_1}(\phi) = \mu_{y_1}(\phi) = 1/4$;
 - ϕ consists of squares and exactly one bigon (since for Lipshitz's formula $e(\phi) = 1/2$);
 - there are $(g-1)$ trivial discs (corresponding to $x_2 = y_2, \dots, x_g = y_g$).
- * **Sub-case 2b:** $\mu_{\mathbf{x}}(\phi) + \mu_{\mathbf{y}}(\phi) = 1$. In this case, we have that:
- ϕ hits only α_1 ;
 - $\mu_{x_1}(\phi) = \mu_{y_1}(\phi) = 1$;
 - ϕ consists of squares (since for Lipshitz's formula $e(\phi) = 0$);
 - there are $(g-1)$ trivial discs (corresponding to $x_2 = y_2, \dots, x_g = y_g$).

Our aim will now be to understand the surface S which maps to $\Sigma \times \mathbb{D}^2$ (using the cylindrical reformulation by Lipshitz) in these three cases.

Observation 4.12. We can make some observations on the three cases found, reducing ourselves to look at only the first two of them.

- Let us consider the first case. This corresponds to a map from S to Σ , where S has $g-2$ trivial disc components and, using (ϖ) , $\chi(S) = g-1$ (recall that $e(\phi) = 0$).

Let F be $F = S \setminus \{\text{trivial discs}\}$; then F is a double branched cover over $\mathbb{D}^2$ and we claim that $\chi(F) = 1$ and that it has 1 branch point. To see that $\chi(F) = 1$, one can prove that "to remove a disc" from some surface A makes the Euler characteristic of the surface to go down by one (using the Inclusion-Exclusion principle for $\chi(\cdot)$); since we are removing one disc for every trivial disc component, we get that $\chi(F) = (g-1) - (g-2) = 1$. To see that F has 1 branch point, recall that for Lemma 4.9 we have that the number of branch points is given by $\mu_{\mathbf{x}}(\phi) + \mu_{\mathbf{y}}(\phi) - e(\phi) = 1$.

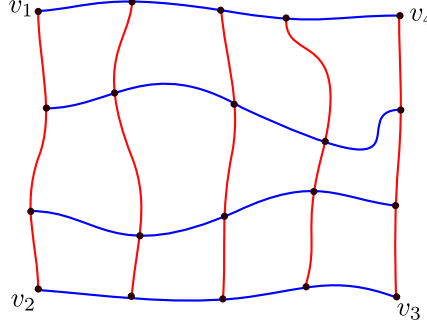
Therefore F is a disc with 4 marked points on its boundary (which correspond to x_1, y_1, x_2, y_2). We then say that F is a *square* and we call the marked points *corners of F*.

- Consider now the other two cases. In both, S has $(g-1)$ trivial disc components and we call again $F = S \setminus \{\text{trivial discs}\}$. In this case, F is just a single cover over $\mathbb{D}^2$ and the number of branch points has to be 0.

Consider now Sub-case 2b. If this case were to be possible, we would have thanks to Lemma 4.9 that the number of branch points would be 1: absurd. Therefore, the third case is not possible.

Consider then Sub-case 2a. Here, F is a disc with 2 marked points on its boundary (corresponding to x_1 and y_1). We then say that F is a *bigon* and we call the marked points *corners of F*.

We have then proven that only Case 1 and 2a can occur, we will refer to them as Case 1 and Case 2.

Figure 4.1: Pre-image of α -edges and β -edges for a square.

We want now to study the map $f = p_1 \circ u|_F : F \rightarrow \Sigma$. In both the cases that we have, ϕ is the image of F via f and the trivial discs are mapped to $x_3 = y_3, \dots, x_g = y_g$ (and also $x_2 = y_2$ for the second case), that are outside of ϕ . Notice that for both cases, f has no branch point. Therefore, it is a local diffeomorphism even at the boundary of F . We can also prove that there is for every corner v of F , we have that $f(v)$ has exactly one preimage: it is sufficient to notice that $\mu_{x_i}(\phi) = \mu_{y_i}(\phi) = 1/4$ in the non-trivial x_i 's.

All is left to prove is that f is an embedding, i.e. that is not only a local diffeomorphism but an actual diffeomorphism: once that is proven, we will also obtain that ϕ is an embedded square (Case 1) or an embedded bigon (Case 2). We prove this fact case by case.

Case 1. Here we have a map $f : F \rightarrow \Sigma$, where F is a square (i.e. a disc with 4 marked points on the boundary), we denote the corners by v_1, v_2, v_3, v_4 .

In order to prove that f is an embedding, we first study the preimage of the α - and β -edges of the boundary and the interior of ϕ via f . Exploiting the fact that f is a local diffeomorphism, we get that these pre-images are 1-manifolds with boundary; we can therefore still call them α - and β -edges. For the same argument, the α -edges are only allowed to intersect β -edges and the other way around; moreover, in any corner v_i we have an intersection between a α -edge and a β -edge. Remark that any preimage of a square region is still a square region, therefore the α - and β -edges cut F into squares; this means in particular that we cannot have an edge to form a closed loop γ : if that was the case, we would have that the Euler measure of $A = F \setminus \{\text{inside region of } \gamma\}$ is negative and hence A couldn't be tiled by squares. With the same reasoning we obtain that an edge cannot intersect two times the same edge of the other colour; this holds in particular on the border of F , meaning that an edge cannot start and finish on the same edge. Therefore, we have that the edges are straight lines inside F : we can take the α -edges to be vertical edges that cut F into vertical rectangles and the β -edges to be horizontal edges that cut F into horizontal rectangles, as it is showed in Figure 4.1.

Let us call *vertices* the intersection points between α - and β -edges, including as well the corners. Thanks to the observations done above, we can reduce ourselves to prove that two different vertices are mapped to two different points by f : this will

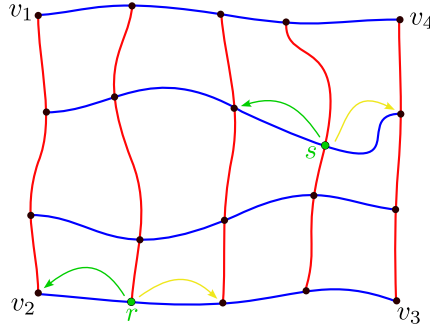


Figure 4.2: Points found with the "walk" on β_i , finishing in the wrong case (we reach the border first starting from s) in yellow and in the right case (we reach the border first starting from r) in green.

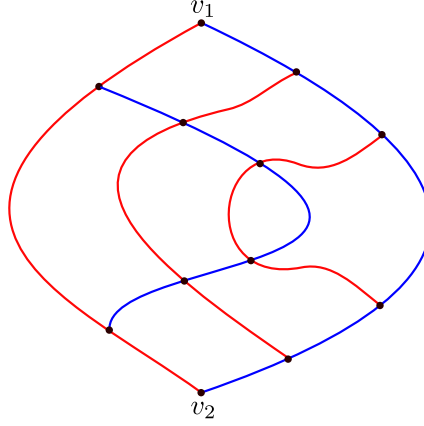
then imply that f is an embedding.

Let $r, s \in F$ be two distinct vertices and assume by absurd that $f(r) = f(s)$. We consider the following two cases:

- (a) at least one between r and s is in ∂F ,
- (b) both r and s are in the interior of F .

We prove the contradiction only in (a), since the second can be reduced to the first. Assume for instance that $r \in b_r \subset \partial F$, where b_r is a β -edge and call b_s the β -edge on which stays s ; the case $r \in a_r \subset \partial F$ for a_r an α -arc can be proven analogously. Let β_i be the β -circle in Σ on which stay $f(r) = f(s)$; we choose an orientation for β_i and we explore it until we reach another intersection point with some α -circle. We look at the pre-images of such point via f , which are at least 2 since we are virtually moving from both r and s staying on the β -edges b_r, b_s in F ; if at least one of them is contained in an α -edge $a \subset \partial F$ we stop, otherwise we continue to explore the intersection points on β_i following the orientation. At one point we will reach a point of this type; let t, u be two pre-images of such intersection point on β_i such that $t \in a \subset \partial F$. We have two possible cases: $t \in b_r$ or $t \in b_s$; if the second holds, we can just repeat the walk on β_i reversing the chosen orientation. In particular, remark that t is a corner; therefore we have found $f(t) = f(u)$ for two distinct vertices, one of which is a corner: this is a contradiction to the embedding assumption near the corners. $\nexists$ See Figure 4.2 for a visual idea.

We still need to show how to reduce case (b) to case (a). Assume both r and s are in the interior of F , call a_r, a_s the α -edges on which they stay and let α_i be the α -circle in Σ that contains $f(r) = f(s)$; as before, we choose an orientation and we look at successive points of intersection with β -circles. For the same argument of above, we have at least two pre-images in F for each of these intersection points: one on the a_r and one on the a_s . We proceed to explore α_i until one of the pre-images of some intersection point falls on ∂F ; then we conclude with the proof of case (a).

Figure 4.3: Pre-image of α -edges and β -edges for a bigon

Case 2. In this case we have a map $f : F \rightarrow \Sigma$, where F is a square (i.e. a disc with 2 marked points on the boundary), we denote the corners by v_1, v_2 .

We first set up the environment as before: the pre-images of α and β -edges are 1-manifold with border, one edge can intersect only another edge of another colour and we call vertices the intersection points; in any corner v_i we have an intersection between a α -edge and a β -edge. Once more, we cannot have any closed loops and the tiling on $\phi \subset \Sigma$ induces a tiling with squares and one bigon on F .

Differently from before, the border of F is the union of one α -edge $\bar{a}$ and one β -edge $\bar{b}$ and the other edges have to enter and leave F through the same border edge of the other colour. However, the way that they do so is quite particular: they are "nested". Consider for instance the α -edges of F different than $\bar{a}$: if we look at the connected regions of $F \setminus \{a \mid a \text{ is a } \alpha\text{-edge different than } \bar{a}\}$, then only one region can be a bigon, the closest to $\bar{b}$. The same holds for the β -edges. Therefore F is tiled as showed in Figure 4.3, with one bigon in the middle and squares around.

As in the previous case, to show that f is an embedding it is enough to show that it is an embedding restricted to vertices; this is done in the same fashion of Case 1.

Hence, both in Case 1 and Case 2, f is an embedding and the theorem is proven. $\square$

Proof of Theorem 4.7 ([SW06]). Let $\phi \in \pi_2^0(\mathbf{x}, \mathbf{y})$ be an empty embedded square; then by Remark 4.5 it holds $\mu(\phi) = 1$. We want to prove that ϕ has an unique holomorphic embedding u , this would imply that $c(\phi) = 1$. We first construct the branched cover F of $\mathbb{D}^2$ that is sent diffeomorphically to ϕ . We take F to be a disc with four marked points on its boundary; then, given a complex structure on Σ , we can give a holomorphic structure to F : the cross-ratio of the four points on its boundary determines an one-parameter family of positions of the branch point in $\mathbb{D}^2$ which gives that cross-ratio. Hence, there is a holomorphic branched cover $g : F \rightarrow \mathbb{D}^2$ satisfying the boundary conditions, which is unique up to re-parametrization: therefore, we proved that ϕ has a holomorphic representative u . Moreover, for what

we have done in the proof of Theorem 4.6, we know that u determines uniquely the topological type of F ; thus u is the unique holomorphic embedding of ϕ .

The case for $\phi \in \pi_2^0(\mathbf{x}, \mathbf{y})$ an empty embedded bigon is done analogously with F a disc with two marked points on its boundary.

□

4.2 The nicefication algorithm

We are now going to see how we can construct a nice Heegaard diagram starting from a generic admissible one; for this purpose, only isotopies and handle slides on the β -curves are going to be needed. The algorithm that it is presented was developed by Sarkar and Wang in Section 4 of [SW06], and we are going to follow their exposition quite closely. For the implementation, some changes were made to the algorithm in order to make it more effective; these are going to be discussed in Section 4.5.

From here on, we are going to call *good regions* bigons and squares and *bad regions* all the others. Schematically, the algorithm is the following:

Step 1: killing all non-disc regions;

Step 2: making all but one region to be bigons and squares.

Let $\mathcal{H} = (\Sigma, \alpha, \beta, w)$ be a pointed Heegaard diagram.

4.2.1 Step 1: killing all non-disc regions

We want to only have disc regions in our diagram. We divide this first step in two sub-steps.

Sub-step 1.1: making any circle intersects some other circle

First of all, we want to ensure that any α -circle intersects at least one β -circle and that any β -circle intersects at least one α -circle.

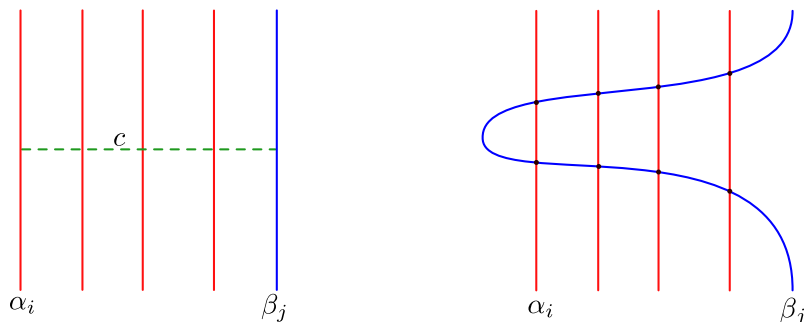
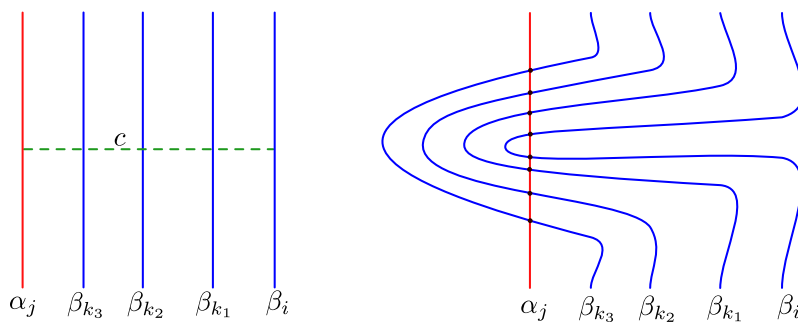
Let α_i be an α -circle that does not intersect any β -circle. We can then find an arc c that connects α_i to some β_j , satisfying the following requests:

1. it intersects β_j only at its end;
2. it does not intersect any other β -circle.

Then, we operate a finger move on β_j along c , to make α_i intersect it. See Figure 4.4 for a visual idea.

Let now β_i be a β -circle that does not intersect any α -circle. we can then find an arc c that connects β_i to some α_j , satisfying the following requests:

1. it intersects α_j only at its end;
2. it does not intersect any other α -circle.

Figure 4.4: Making all the α -circles intersect some β -circle.Figure 4.5: Making all the β -circles intersect some α -circle.

Let $\beta_{k_1}, \dots, \beta_{k_n}$ be the others β -circles intersected by c , in addition to β_i . Then, we operate $n + 1$ "nested" finger moves along c , to make β_i intersect α_j : one on β_i and the others on the β_k 's. See Figure 4.5 for a visual idea.

Operating these two moves for all the circles that do not intersect any other circle, we make sure that now any α -circle intersects at least one β -circle and that any β -circle intersects at least one α -circle.

Sub-step 1.2: making all the regions discs

We can notice that, since $\Sigma \setminus \alpha$ is a punctured sphere, all the regions are planar surfaces. Let D be a non-disc region, i.e. a region that has more than one boundary component. For what we did in the previous sub-step, we have that every boundary component has to contain both α and β -edges (if not, it would mean that there is a circle that does not intersect any other circle). To eliminate the boundary components in excess, we operate some finger moves on β -circles, as it is showed in Figure 4.6.

After sufficiently many iterations, we remain with only disc regions.

4.2.2 Step 2: making all but one region bigons or squares

Let now $\mathcal{H} = (\Sigma, \alpha, \beta, w)$ be a pointed Heegaard diagram with only disc regions; we are now going to make all the regions but one bigons or squares, using finger moves

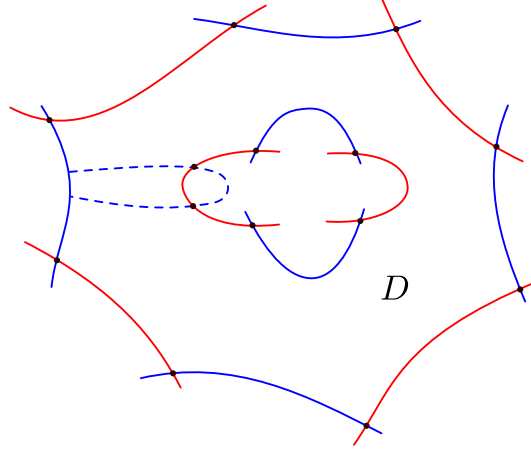


Figure 4.6: Killing the non-disc regions. The dotted arcs indicate the finger moves that we need to do on the β -circles.

and handleslides. It is important to remark that with the following algorithm we generate only disc regions in our diagram.

We are going to need some definitions for this step of the algorithm.

Definition 4.13 (Distance of a region). Given a region D of $\mathcal{H}$, let $w' \in D$ be a point in its interior. We define the *distance of D* , denoted by $d(D)$, to be the smallest number of intersection points between the β -circles and an arc connecting w and w' in the complement of the α -circles.

In particular, we call D_0 the region that contains the basepoint w ; this is the only region of distance 0.

Remark that the definition of distance is well posed, as by the definition of Heegaard diagram we have that $\Sigma \setminus \alpha$ is connected.

Definition 4.14 (Badness of a region). For a $2n$ -gon disc region D , we define the *badness of D* , denoted by $b(D)$, to be $b(D) = \max\{n - 2, 0\}$.

Definition 4.15 (Distance of $\mathcal{H}$). We define the *distance of the diagram* to be the maximum distance of a bad region in the diagram. We denote it by $d(\mathcal{H})$.

Definition 4.16 (Distance d_* complexity). Consider a distance d_* , $0 < d_* \leq d(\mathcal{H})$. We define the *distance d_* complexity of $\mathcal{H}$* , denoted by $c_{d_*}(\mathcal{H})$ to be the tuple

$$c_{d_*}(\mathcal{H}) = \left(\sum_{i=1}^m b(D_i), -b(D_1), -b(D_2), \dots, -b(D_m) \right),$$

where $D_1, \dots, D_m$ are all the bad regions of distance d_* , ordered such that $b(D_1) \geq b(D_2) \geq \dots \geq b(D_m)$.

We call the first term the *total badness of distance d_* of $\mathcal{H}$* and we denote it by $b_{d_*}(\mathcal{H})$. If there are no distance d_* bad regions, then $c_{d_*}(\mathcal{H}) = (0)$. We order the set of distance d_* complexities lexicographically.

The algorithm is based on the following result, which states in details how we can simplify our diagram to a nicefied version of it only by finger moves and handleslides.

Theorem 4.17. *For a distance d pointed Heegaard diagram $\mathcal{H}$ with only disc regions, if $c_d(\mathcal{H}) \neq (0)$, we can modify $\mathcal{H}$ by isotopies and handleslides to get a new Heegaard diagram $\mathcal{H}'$ with only disc regions, satisfying $d(\mathcal{H}') \leq d(\mathcal{H})$ and $c_d(\mathcal{H}') < c_d(\mathcal{H})$.*

To see the proof of the above theorem, we explain how a full cycle of the algorithm works, showing that for any possible case that we find we have indeed that $c_d(\mathcal{H}') < c_d(\mathcal{H})$.

We order the bad regions of distance d as in the definition of the distance d complexity. Now we look at the *worst region* of the diagram, i.e. the region of distance d with the smallest badness, which we denote by D_m . Since it is a bad region, it is a $(2n)$ -gon with $n \geq 3$.

Pick an adjacent region D_* with distance $d - 1$ having a common β -edge with D_m (remark that by how the assignment of the distance is done, we can only have that regions that are neighbour via a β -edge can only have distance $d - 1, d$ or $d + 1$). Let b_* be (one of) their common β -edge(s). We order the α -edges of D_m counterclockwise starting at b_* and we denote them by $a_1, a_2, \dots, a_n$. We try now to make a finger move on b_* into the D_m and out of D_m through a_2 , as indicated in Figure 4.7 when D_m is an octagon.

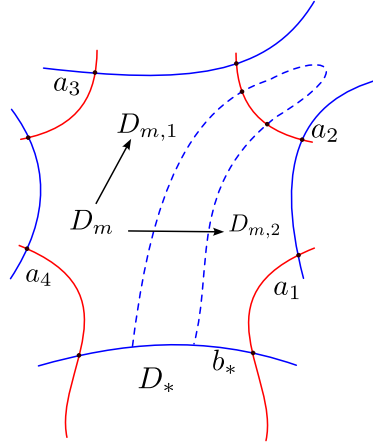


Figure 4.7: Initial finger move done on b_* , exiting from D_m through a_2 .

If we reach a square region of distance $\geq d$, we push through with our finger, going outside of this region via the opposite α -edge, as in Figure 4.8. Note that doing a finger move through regions of distance $\geq d$ does not change the distance of any of the bad regions, since they all have distance $\leq d$: therefore the minimal path connecting the basepoint of the diagram to the bad regions $\overline{D}$ cannot pass through this square region, otherwise we would have that the distance of $\overline{D}$ is $> d$.

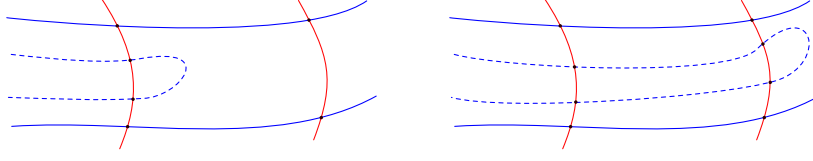


Figure 4.8: When a square region of distance $\geq d$ is reached (figure on the left), we push through it (figure on the right).

We continue to push up our finger move through all the squares of distance $\geq d$, until we reach one of the following four cases:

1. a bigon region is reached;
2. a region with distance $\leq d - 1$ is reached;
3. a bad region with distance d other than D_m , i.e., D_i with $i < m$, is reached;
4. we come back to D_m .

We will now study each case, showing how for every one of them we have that the statement of Theorem 4.17 holds. Remark that the above cases are all the possible cases that we can encounter.

Case 1: a bigon region is reached

This is the case in which we reached a bigon that can have any distance. Remark that all regions in between D_m and this bigon are square regions with distance $\geq d$. We then stop the finger move inside the bigon region, separating the bigon into a square and a new bigon, as in Figure 4.9.

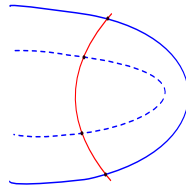


Figure 4.9: A bigon is reached by the finger move.

Denote the new Heegaard diagram obtained by $\mathcal{H}'$. We have $b(D_{m,1}) = b(D_m) - 1$ and $D_{m,1}$ is still a region of distance d . Since $D_{m,2}$ is a square and is good, we get $b_d(\mathcal{H}') = b_d(\mathcal{H}) - 1$. We have also increased the badness of D_* , but it is a distance $d - 1$ region and hence is not a problem. Notice that we did not increase the distance of any other bad region since we did not pass through any region of distance $\leq d - 1$ and all bad regions has distance $\leq d$. Therefore $d(\mathcal{H}') \leq d(\mathcal{H})$ and $c_d(\mathcal{H}') < c_d(\mathcal{H})$; hence the theorem is proved in this case.

Case 2: a region with distance $\leq d - 1$ is reached

Let D' be the region with distance $< d$ we reached with our finger move; denote $d(D') = d'$. Let $\mathcal{H}'$ be the new Heegaard diagram. Notice that D' might be a bigon or also a square region. We call the new generated regions as in Figure 4.10.

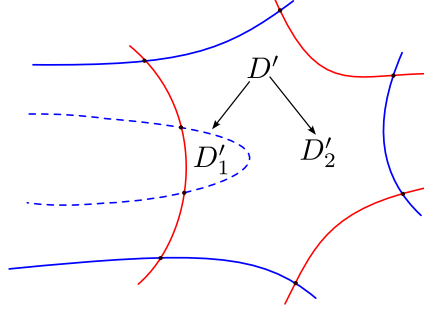


Figure 4.10: A smaller distance region is reached.

As in the previous case, we have that $b(D_{m,1}) = b(D_m) - 1$ and that $D_{m,2}$ is good. Our finger move splits D' into two regions: the bigon region D'_1 and the other region D'_2 . The latter is going to be a bad region if D' was not a bigon; hence, when D' is a square or a bad region, D'_2 will be a bad region of distance $d' < d$. We have increased the badness of D_* (which is a distance $d - 1$ region) and we might have increased the distance d' complexity (as we replaced D' with a badder region, and hence the total badness of distance d' may be increased), but we have $d(\mathcal{H}') \leq d(\mathcal{H})$ and $c_d(\mathcal{H}') < c_d(\mathcal{H})$. Therefore Theorem 4.17 is proved also in this second case.

Case 3: a bad region with distance d other than D_m is reached

This third case is the worst of the four, as we do not decrease the total badness of distance d but we only "shift" the badness to some other bad region of distance d that has a larger badness of D_m and hence it come first in $c_d(\mathcal{H})$. It is very similar to the second case, but we need to be a bit more careful in proving the theorem.

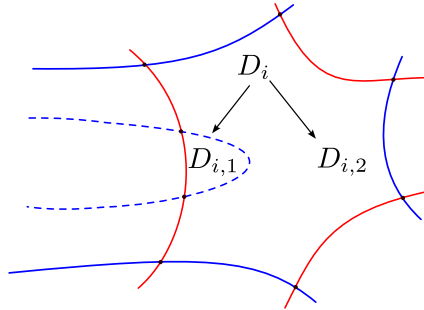


Figure 4.11: Another distance d bad region is reached.

In this case, we reach some distance d bad region D_i with $i < m$. Denote by $D_{i,1}$

and $D_{i,2}$ the two parts of D_i separated by our finger as in Figure 4.11. $D_{i,1}$ is a bigon and hence good, while $D_{i,2}$ is a bad region of distance d . We have $b(D_{i,2}) = b(D_i) + 1$ and $b(D_{m,1}) = b(D_m) - 1$, thus the total badness of distance d remains the same. But we are decreasing the distance d complexity since we are moving the badness from a later bad region to an earlier bad region:

$$\begin{aligned} c_d(\mathcal{H}') &= (b_d(\mathcal{H}), -b(D_1), \dots, -b(D_i) - 1, \dots, -b(D_m) + 1) \\ &< (b_d(\mathcal{H}), -b(D_1), \dots, -b(D_i), \dots, -b(D_m)) = c_d(\mathcal{H}). \end{aligned}$$

Remark that the position of D_i inside $c_d(\mathcal{H})$ may have changed since the badness increased, but it could only have moved more "to the left" of the tuple. We have also increased the badness of D_* , but it is a distance $d - 1$ region and hence is not a problem. Hence for the new Heegaard diagram $\mathcal{H}'$, we have $d(\mathcal{H}') = d(\mathcal{H})$ and $c_d(\mathcal{H}') < c_d(\mathcal{H})$ and the statement is proved.

Case 4: we come back to D_m

This case is very different from the previous three. We divide it into two sub-cases, according to which edge the finger move is coming back through.

Sub-case 4.1: we come back through an adjacent edge

This sub-case is drawn in Figure 4.12. Without loss of generality, we assume the finger comes back via a_1 .

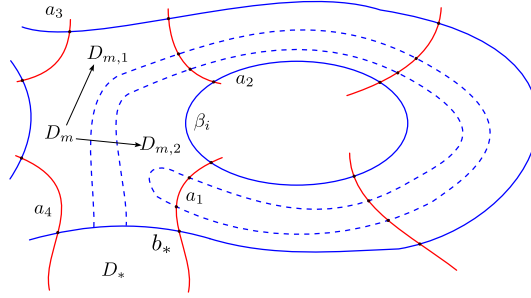


Figure 4.12: Coming back via an adjacent edge with the finger move. The finger is denoted by the dotted arc.

In this case, we see the full copy of some β -circle, say β_i , on the right side along our long finger: in fact, we have pushed through linked squares and doing this it is not possible to "change" the β -circle that we have on the right, as it would mean that we passed through an hexagon (or something bigger). Suppose $b_* \subset \beta_j$. Note that $i \neq j$: otherwise we would have had $b_* \subset \beta_i$ and we would have reached either D_m or D_* at an earlier time as we would have passed by the entirety of β_i . Now, instead of doing the finger move, we perform a handleslide of β_j over β_i , as we can see in Figure 4.13.

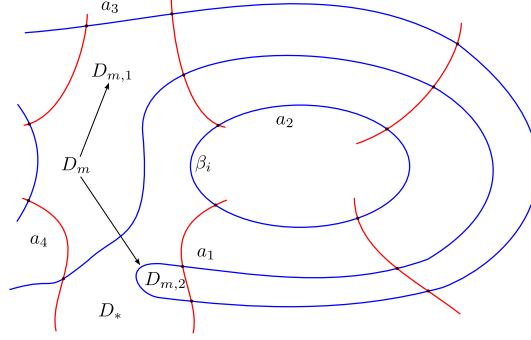


Figure 4.13: Coming back via an adjacent edge, performing the handleslide. The dotted arc denotes β_i after the handleslide.

Notice that after the handle slide, we are not increasing the distance of any bad region. In fact, that would happen only if the basepoint was "inside" of β_i , but this is not possible: if this was to be true, then we would find that D_* was inside of β_i , but this is a contradiction.

As in the previous cases, we have increased the badness of D_* ; this is not a problem since it is a distance $d - 1$ region. Moreover, $D_{m,2}$ is a bigon region and $b(D_{m,1}) = b(D_m) - 1$. Thus for the new Heegaard diagram $\mathcal{H}'$ after the handleslide, the total badness of distance d is decreased by 1; we have $d(\mathcal{H}') \leq d(\mathcal{H})$ and $c_d(\mathcal{H}') < c_d(\mathcal{H})$. Hence the theorem is proved also in this case.

Sub-case 4.2: we come back through a non-adjacent edge

Suppose that we return through a_k with $3 < k \leq n$ (recall that D_m is a $(2n)$ -gon). Then, instead of the finger move through a_2 , we do a finger move on b_* through a_3 as it is drawn in Figure 4.14.

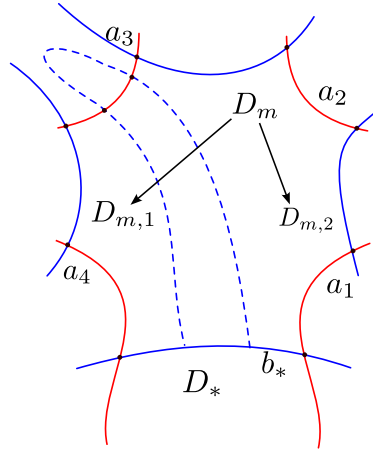


Figure 4.14: Finger move done on b_* , exiting from D_m through a_3 .

If we reach one of the first three cases, we can do the finger move and we are still decreasing the distance d complexity. In fact, we have that $b(D_{m,1}) = b(D_m) - 2$ and

$b(D_{m,2}) = 1$, therefore the total complexity of distance d is going down by a factor of 1; we are increasing the badness of D_* , but it is a distance $d - 1$ region.

Suppose instead that we come back to D_m via some α -edge a_i . We claim that $3 < i < k$. Certainly we can not come back via a_3 , as we only push through squares and we go out from them via the only other α -edge that they have. The finger can not come back via a_k since the chain of squares from a_k is connected to a_2 . The last possibility is that either $i > k$ or $i < 3$. If that was the case, we could get two simple closed curves c_1 and c_2 , as shown in Figure 4.15.

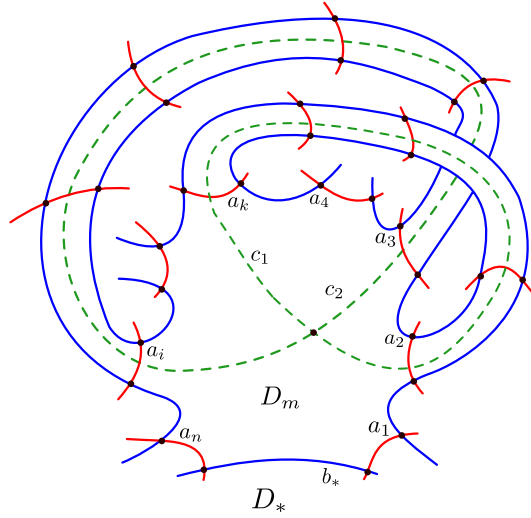


Figure 4.15: There are no crossing fingers. The fingers are not showed here. Instead, the two dotted green circles denote the cores of the two fingers.

Then c_1 and c_2 intersect transversely at exactly one point and they are in the complement of the β curves. The complement of the β curves is a punctured sphere, attach discs to the holes to get a sphere: then, as homology classes, we get $[c_1] \cdot [c_2] = 1$. But $H_1(S^2) \cong 0$: this is a contradiction $\nexists$. Thus we must have $3 < i < k$.

Now, instead of the finger move through a_3 , we do another finger move through a_4 . Iterating the same arguments as above, we see that we either end up with a finger which does not come back, or we get some finger that starts at a_j and comes back via a_{j+1} . Then:

- if the finger does not come back, we reduce it to the previous cases and the theorem follows;
- if there is a finger which starts at a_j and comes back at a_{j+1} , we went around a full β -circle and we can do a handleslide similar to the one in Sub-case 4.1. In this case, we have that $b(D_{m,1}) = \max\{n - j - 1, 0\}$ and $b(D_{m,2}) = \max\{j - 2, 0\}$; hence $b(D_{m,1}) + b(D_{m,2}) \leq (n - j - 1) + (j - 2) = n - 3$ and we decreased the total badness of distance d . We increased the badness of D_* , but it is a distance $d - 1$ region. Thus, for the new Heegaard diagram $\mathcal{H}'$ after the handleslide, we have $d(\mathcal{H}') \leq d(\mathcal{H})$ and $c_d(\mathcal{H}') < c_d(\mathcal{H})$.

Thus we end the proof of Theorem 4.17.

4.3 Admissibility of the nicefied diagram

We now show that if we start with an admissible Heegaard diagram, then the algorithm yields an admissible nice Heegaard diagram. The algorithm uses isotopies and handleslides, therefore it is sufficient to show that they leave the admissible property unaltered.

As a reference, one can look at section 4.2 of [SW06].

We start by proving that isotopies do not affect the admissibility. Let $\mathcal{H}$ and $\mathcal{H}'$ be the Heegaard diagrams before and after the isotopy; we denote the regions modified by the isotopy as in Figure 4.16.

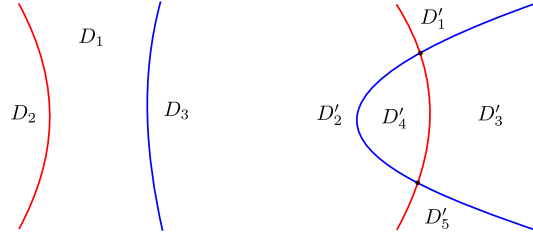


Figure 4.16: Isotopy of the β curve.

Suppose $\mathcal{H}$ is admissible. Consider a periodic domain ϕ' in $\mathcal{H}'$, described as

$$\phi' = c_1 D'_1 + c_2 D'_2 + c_3 D'_3 + c_4 D'_4 + c_5 D'_5 + \cdots.$$

We have the following properties.

- $c_2 - c_1 = c_4 - c_3 = c_2 - c_5$. To see this, we think of the α -curve as if it goes from top to down; we can divide it in three pieces, the first between D'_1 and D'_2 , the second between D'_3 and D'_4 and the third between D'_5 and D'_2 . Since ϕ is a periodic domain, we have that the coefficients of this particular α -circle has to be the same on the border of ϕ , hence we obtain exactly the relations stated above;
- repeating the same argument on the β -curve, we obtain $c_1 - c_3 = c_2 - c_4 = c_5 - c_3$.

Hence $c_1 = c_5$ and $c_4 = c_2 + c_3 - c_1$. Notice that the regions are all the same before and after the isotopy except for those in Figure 4.16; therefore,

$$\phi = c_1 D_1 + c_2 D_2 + c_3 D_3 + \cdots$$

is a periodic domain for $\mathcal{H}$. In fact, the if a is the multiplicity of the α -curve in all ϕ , then we need that $a = c_2 - c_1$, and this is true since we have that $a = c_2 - c_1 = c_4 - c_3 = c_2 - c_5$ in ϕ' . Same goes for the β -curve.

Since $\mathcal{H}$ is admissible, ϕ has both positive and negative coefficients, and so does ϕ' as we only added coefficient with respect to ϕ . Hence $\mathcal{H}'$ is admissible.

We now want to prove the same thing for a handleslide; let us consider the one indicated in Figure 4.17.

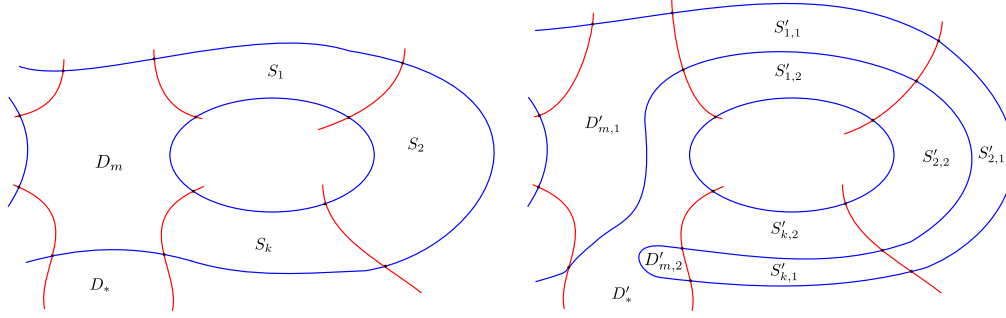


Figure 4.17: Handleslide of the β curve: before (on the left) and after (right).

Suppose that $\mathcal{H}$ is admissible. Let the following be a periodic domain in $\mathcal{H}'$

$$\phi' = c_* D'_* + c_1 D'_{m,1} + c_2 D'_{m,2} + c_{1,1} S'_{1,1} + c_{1,2} S'_{1,2} + \dots + c_{k,1} S'_{k,1} + c_{k,2} S'_{k,2} + \dots$$

By looking at the coefficients for the β -curve that we modified, we get

$$c_1 - c_* = c_{1,1} - c_{1,2} = \dots = c_{k,1} - c_{k,2} = c_2 - c_*.$$

Denote $c_0 = c_1 - c_*$, then we have the relations $c_{i,1} = c_{i,2} + c_0$ and $c_1 = c_2$. We can then claim that

$$\phi = c_* D_* + c_1 D_m + c_{1,1} S_1 + \dots + c_{k,1} S_k + \dots$$

is a periodic domain for $\mathcal{H}$. To see this, we just need to check the relations on the α -edges that we modify with the handleslide. Indeed we have that the differences $c_1 - c_{1,1}$ and $c_1 - c_{k,1}$ are correct because we have them also in $\mathcal{H}'$ (since $c_1 = c_2$). The checks for $c_{i,1} - c_{j,1}$ and $c_1 - c_*$ are immediate.

Since $\mathcal{H}$ is admissible, ϕ has both positive and negative coefficients. Hence ϕ' has both positive and negative coefficients, so $\mathcal{H}'$ is admissible.

Remark 4.18. A stronger statement actually holds: in fact, it can be shown that nice Heegaard diagrams are always admissible (as reference, one can see [LMW08], Corollary 3.2).

4.4 Generalisations of the algorithm

The algorithm as it was presented in Section 4.2 works only for closed Heegaard diagram. However, we can easily generalise it for the different settings that we are interested into, obtaining some generalised versions of the Nicefication algorithm. We explore such generalisations one by one in the next subsections, reaching a version for peculiar Heegaard diagrams that we are particularly interested into.

4.4.1 Algorithm for bordered Heegaard diagrams

In this subsection, we see how the algorithm can be adapted to the case of bordered Heegaard diagrams. As reference, one can look at Chapter 8 of [LOT18].

Recall that in this the bordered setting, we allow Heegaard diagrams to have one border component on which we put a basepoint z : let D_z be the region with the basepoint on its border.

We need to generalise the notion of nice Heegaard diagram for this setting.

Definition 4.19 (Nice bordered Heegaard diagram). We call a bordered Heegaard diagram $(\Sigma, \alpha = \alpha^c \cup \alpha^a, \beta, z)$ *nice* if every region in Σ except for D_z is a bigon or a square. For a boundary region A , this means that it has two α -arcs, one β -arc and one arc of $\partial\Sigma$ as boundary.

Remark 4.20 (No bigons on the boundary). We do not allow boundary regions to have one α -arc and one arc of $\partial\Sigma$ as boundary (i.e. to be "bigons"), as it would mean that there is an α -arc whose does not intersect any β -circle.

Then we have a similar result to Theorem 4.3.

Proposition 4.21 ([LOT18, Proposition 8.2]). *Any bordered Heegaard diagram $(\Sigma, \alpha, \beta, z)$ can be turned into a nice bordered Heegaard diagram via*

- *isotopies of the β -circles not crossing $\partial\Sigma$,*
- *handleslides of β -circles over other β -circles.*

Proof. It is clear that, if we are able to fix the bordered diagram around the boundary, then we can simply apply the nicefication algorithm as it is done in the closed case in order to obtain a nice bordered Heegaard diagram.

We then introduce what we could call a "initial finger move around the border", done as follows. Denote the endpoints of the α -arcs as in Figure 4.18, with the indices increasing if you go around the border counter-clock wise and with the basepoint z between a_1 and a_{4k} . Observe now the boundary of D_z ; we start from the endpoint a_1 and move in the boundary, going away from $\partial\Sigma$, until we reach the first β -arc and we call it b . We now do the following finger move on b : we push it staying close to the border through the α -arc containing a_1 , then through the one containing a_2 and so on, up to intersecting the one containing a_{4k} . Then, in the resulting diagram, the border regions other than D_z are all rectangles.

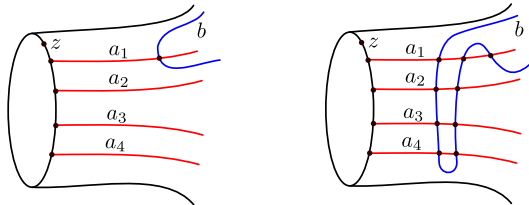


Figure 4.18: Initial finger move around the border to make all the border regions (except the one with the basepoint) squares.

We can now apply the nicefication algorithm, obtaining a nice bordered Heegaard diagram. $\square$

Remark moreover that, always for Corollary 3.2 of [LMW08], the resulting diagram is admissible.

4.4.2 Algorithm for sutured Heegaard diagrams

To generalise the algorithm to the sutured case, we just need to adapt it to a case in which multiple basepoints are allowed, which can be done quite easily. In fact, in such case, we still define the distance of a region A to be the smallest number of intersection points between the β -circles and an arc $c \subset \Sigma \setminus \alpha$ connecting a point in the interior of A and one region containing some basepoint w ; therefore we can define the distance complexities in the same way, and they will still be well-posed definitions. We then apply the algorithm as it was presented in Section 4.2; in particular, we can also prove that the distance complexity is strictly decreasing in the same way as we did for the closed case. Thus, the algorithm works as before.

4.4.3 Algorithm for bordered sutured Heegaard diagrams

We can see how we can put together the generalisations saw for the bordered and the sutured case, obtaining a generalisation of the algorithm for bordered sutured Heegaard diagrams.

As reference, one can look at Section 4.7 of [Zar11].

We first need to define nice Heegaard diagrams in this setting.

Definition 4.22 (Nice ordered sutured Heegaard diagrams). A bordered sutured Heegaard diagram $(\Sigma, \alpha, \beta, \mathcal{Z}_\alpha, \phi_\alpha, \mathcal{Z}_\beta, \phi_\beta)$ is called *nice* if every region of $\Sigma \setminus (\alpha \cup \beta)$ is either:

- a *multiplicity zero region*, which is defined as a region adjacent to $\partial\Sigma \setminus (\phi_\alpha(Z_\alpha) \cup \phi_\beta(Z_\beta))$ (i.e. it has inside one "tick" as we defined them in Subsection 2.6.2),
- a bigon, whose no sides are in $\phi_\alpha(Z_\alpha) \cup \phi_\beta(Z_\beta)$,
- a square, whose at most one of the sides is in $\phi_\alpha(Z_\alpha) \cup \phi_\beta(Z_\beta)$.

Moreover, we request that every connected component of $\Sigma \setminus \alpha$ contains at least one multiplicity zero region.

Then, we can generalised the nicefication algorithm as we did for the last two cases.

Proposition 4.23 ([Zar11, Proposition 4.7.2]). *Any bordered sutured diagram can be made nice via*

- *isotopies of the β -circles not crossing $\partial\Sigma$,*

- isotopies of β -arcs that do not move the endpoints not crossing $\partial\Sigma$,
- handleslides of any β -curve over a β -circle,
- stabilisations.

Proof. We start by applying stabilisations, in order to ensure that any connected component of Σ has both α - and β -curves.

Then, for any border component, we apply to some β -curve the "initial finger move around the border" that we described in Subsection 4.4.1.

Now we can apply the nicefication algorithm as first presented, obtaining a nice bordered sutured Heegaard diagram. In particular, the distance of a region A is defined as the smallest number of intersection points between the β -circles and an arc $c \subset \Sigma \setminus \alpha$ connecting a point in the interior of A and one multiplicity zero region; since we required that every connected component of $\Sigma \setminus \alpha$ contains at least one multiplicity zero region, this notion is well defined also in this case. $\square$

4.4.4 Algorithm for peculiar Heegaard diagrams

We conclude this section with the generalisation for the case of tangles, i.e. the case of peculiar Heegaard diagrams.

As a reference, one can see Section 5.1 of [Zib20].

We first need to define nice Heegaard diagrams in this setting. Recall that we denote by $(\Sigma, \alpha = \alpha^c \cup \{\mathcal{S}\}, \beta, \{p_i\}_{i=1}^4, \{q_j\}_{j=1}^4, \{z_k\}_{k=1}^m, \{w_l\}_{l=1}^m)$ a peculiar Heegaard diagram for a 4-ended tangle with m closed components.

Definition 4.24 (Nice peculiar Heegaard diagram). Consider a peculiar Heegaard diagram $(\Sigma, \alpha, \beta, \{p_i\}_{i=1}^4, \{q_j\}_{j=1}^4, \{z_k\}_{k=1}^m, \{w_l\}_{l=1}^m)$ for some 4-ended tangle T . We need to pick some *special basepoints*: one p_i and one q_j for some $i, j \in \{1, 2, 3, 4\}$ and one between z_k and w_k for each closed component of T . We call a region that does contain a special basepoint a *multiplicity zero region*.

We say that a peculiar Heegaard diagram is *nice* with respect to the choice of the special basepoints above if any region that is not a multiplicity zero region is either a square or a bigon.

Remark 4.25. Notice that, thanks to the definition given above, every connected component of $\Sigma \setminus \alpha$ contains at least one multiplicity zero region.

Therefore, this is a particular case of a multi-pointed closed Heegaard diagram. As already discussed, the following results holds as a consequence of the Nicefication algorithm adapted to this case.

Proposition 4.26. *Every tangle T admits a nice peculiar Heegaard diagram for any choice of special basepoints.*

In particular, any peculiar Heegaard diagram with a choice of special basepoints can be made nice with a series of

- isotopies of the β -circles,
- handleslides of β -circles over other β -circles.

In particular, as we proved for closed Heegaard diagrams with Theorem 4.2, also in this case peculiar module of a tangle can be compute combinatorially.

Theorem 4.27 ([Zib20, Corollary 5.7]). *Let T be a 4-ended tangle and let $\mathcal{H}$ be a nice peculiar Heegaard diagram with some choice of special basepoints for T .*

Then $\text{CFT}^\partial(T)$ can be computed combinatorially via $\mathcal{H}$.

The proof of this result can be found in [Zib20]. Moreover, the Mathematica Software [PQM.m] is exactly the implementation of the algorithm which proves the above theorem.

4.5 Modifications done to the algorithm

When we have a nice Heegaard diagram, we can compute the needed flavor of Heegaard Floer homology combinatorially. In the case of 4-ended tangles, that we studied in Section 3.5, it exists the [PQM.m] Mathematica Software, developed by Claudius Zibrowius in 2017, of which the up-to-date version can be found on his personal [website](#). This software takes as input a nice Heegaard diagram for a 4-ended tangle with some other data related to the tangle (the α -arcs sites, the Alexander grading of the tangle and the multiplicity zero regions of the diagram) and computes the peculiar module CFT^∂ of the tangle.

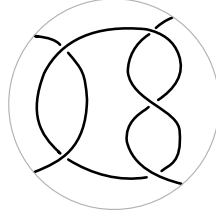
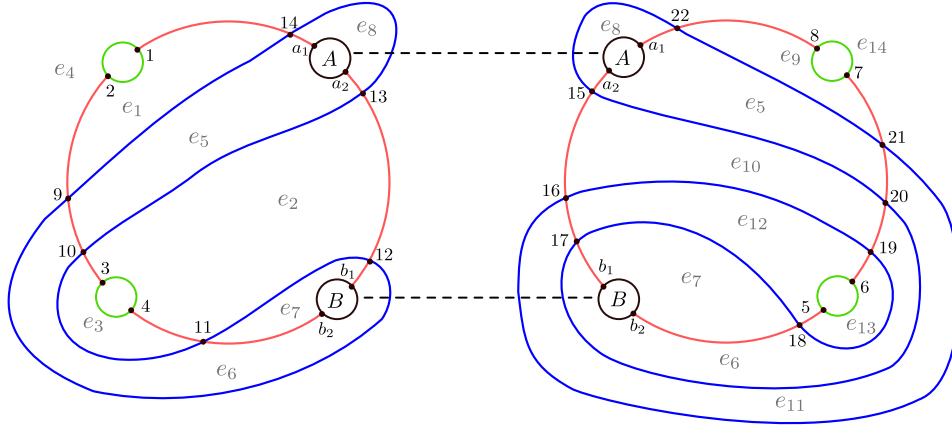
During the implementation of the Nicefication algorithm, some changes were done with the goal to optimise the final result in the sense of the [PQM.m] Software, i.e. aiming to minimise the number of generators of the final result and therefore the running time required to compute the CFT^∂ . The main changes done are the following two:

1. introduction of general handleslides as a move done by the algorithm;
2. exploration of all the possible moves that we can do on the worst region D_m , comparison of them and choice of the best one to do.

We now discuss these changes in details.

4.5.1 General handleslides

With "general handleslide", we mean handleslides as defined in Section 2.2; in fact, these are not considered by the algorithm as it was presented in Section 4.2. This move was introduced in the software during its validation via computation of nice diagrams for Pretzel tangles (whose peculiar modules are known) for the reasons that are now explained. We define the $(2n, -2m - 1)$ -Pretzel tangle to be the sum of two rational tangles: $Q_{1/(2n)} + Q_{-1/(2m+1)}$ for $m, n \in \mathbb{Z}_{>0}$. We consider for instance the $(2, -3)$ Pretzel tangle showed in Figure 4.20. Constructing the Heegaard diagram

Figure 4.19: $(2, -3)$ Pretzel tangle.Figure 4.20: Tangle Heegaard diagram for the $(2, -3)$ Pretzel tangle.

as it is explained in Section 2.6 (i.e. splitting it in its two rational components), we obtain the result showed in Figure 4.20.

Then, we transform the diagram into a peculiar one, taking off the sutures and obtaining what we can see in Figure 4.21.

We then place the two special basepoints, making e_1 and e_3 the two multiplicity zero regions. We can see that we have two bad regions, e_5 and e_6 , which are two octagons of distance 1. Applying the algorithm as it is presented in [SW06], we nicefy the diagram in 4 cycles: for each bad region are required a finger move and a handleslide. The final result has a total of 126 generators. If we allow general handleslide, we can nicefy the diagram with only two cycles with two general handleslides and get a total of 82 generators. The use of general handleslide is an advantage that was observed also in more complex cases, where the difference of number of generators of the final result was more impacting for the running time required by the [PQM.m] Package.

We can further appreciate the efficiency of the general handleslides by observing how they affect the complexities of a diagram. Notice that for the diagram $\mathcal{H}$ in Figure 4.21, the distance 1 complexity (which is the only non-trivial complexity) is

$$c_1(\mathcal{H}) = (4, (-2, \text{Region } 5), (-2, \text{Region } 6)).$$

After the first general handleslide, we obtain the diagram $\mathcal{H}'$, with complexity

$$c_1(\mathcal{H}') = (2, (-2, \text{Region } 5));$$

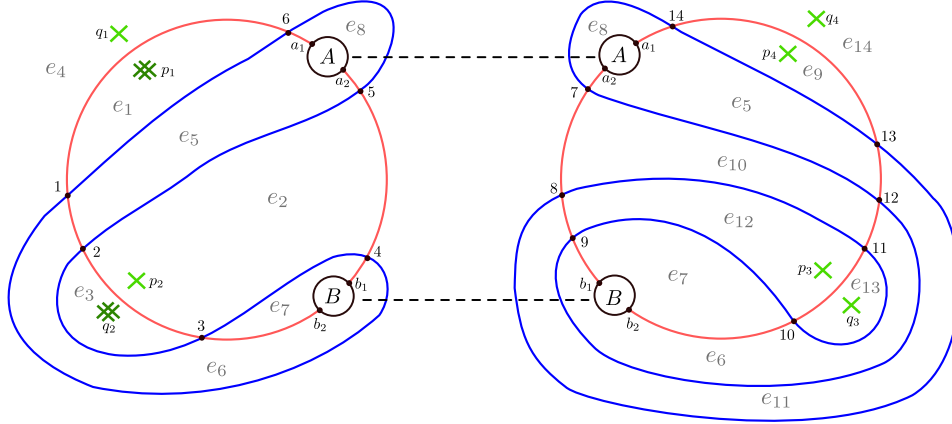


Figure 4.21: Peculiar Heegaard diagram for the $(2, -3)$ Pretzel tangle (the double crosses coloured with a darker green denote the special basepoints).

and after the second handleslide we obtain the nice diagram $\mathcal{H}''$, whose has only trivial complexities. Hence, with a general handleslide we have the possibility to reduce the total badness of the maximum distance by a factor of 2, something that it is not possible with any of the moves in the original algorithm.

Remark 4.28 (Effect of a general handleslide on the distance d complexity). Let $\mathcal{H}$ be a Heegaard diagram of distance d_* . We want to understand how a general handleslide modifies in general $\mathcal{H}$ from the point of view of its niceness, i.e. how it modifies $c_{d_*}(\mathcal{H})$.

The first thing to notice, is that it does not generate any bad region: in fact, a handleslide only generates squares near the β -circle that we are sliding on and it leaves the badness of the regions that it cuts through unchanged.

The critical point is the distance of these regions that it cuts through: if their distance is determined by arcs that are cut by the handleslide, then they may increase by 1 (and potentially also the distance of other regions of the diagram may increase with analogous argument).

However, we do not need to worry about this case, as in the next Subsection we will introduce a new algorithm for comparing different Heegaard diagrams which will avoid this scenario to realise.

Therefore, the general handleslide was included in the pool of possible moves of the algorithm.

4.5.2 Comparison of possible moves

We now have on the table all the possible moves that we can do: finger moves, handleslides and general handleslides. We want now understand how to get the best possible nice Heegaard diagram (i.e. the one with less generators) using these moves: as above mentioned, at every iteration of the algorithm we try to operate all the moves that it is possible to do on the worst region of the diagram and we

generate all the corresponding Heegaard diagrams. Clearly, the best choice would be to try all the possible treads generated by the different moves at every cycle and then compare all the final nice diagrams obtained; however this would not be feasible computationally and we need to compromise. In this first version of the `[nicepy]` Software, the choice is to try to get the best possible diagram after every iteration of the algorithm. For this, we could adopt two different point of views: on one side, we could decide to try to minimise the number of generators, hence choosing to continue to execute the algorithm on the diagram that has less generators; on the other side, we could choose to minimise the number of cycles needed by the algorithm to get a nice Heegaard diagram, hence reducing the number of moves that are done on the initial diagram. It is unclear at the moment which of the two perspectives is the better one or what could be a good balance between them; thus, for now, the choice made is to minimise the number of cycles of the nicefication algorithm. However, this is still an open question that will be analysed in the future.

Therefore, before running the algorithm, we need something to help us compare all these moves and understand which one is better in the sense above described.

Definition 4.29 (Total complexity). Consider a Heegaard diagram $\mathcal{H}$ of distance d_* . We define its *total complexity* as

$$tc(\mathcal{H}) = \sum_{d=1}^{d_*} 3^d \cdot b_d(\mathcal{H}),$$

where $b_d(\mathcal{H})$ is the total badness of distance d defined in Section 4.2.

The total complexity is convenient when we want to compare multiple Heegaard diagrams obtained by applying some move to a common initial diagram and it is a more direct way to compare the effectiveness of these different moves in the attempt to nicefy the diagram.

We quickly describe how the total complexity is affected in the various scenarios that occur in the algorithm. We use the same notation used when talking of the algorithm, i.e. $\mathcal{H}$ is the diagram before the move, $\mathcal{H}'$ is the diagram obtained after the move, D_m is the worst region of distance d_* that we want to modify, $D_{m,1}$ and $D_{m,2}$ are the two regions in which D_m is split by the move, D_* is the neighbourhood region of distance $d_* - 1$ and D'_* is this region after the move.

- In Case 1 (finger move ending in a bigon), we have $b(D_{m,1}) + b(D_{m,2}) \leq b(D_m) - 1$ and $b(D'_*) = b(D_*) + 1$; hence

$$tc(\mathcal{H}') \leq tc(\mathcal{H}) - 3^{d_*} + 3^{d_*-1}.$$

Therefore, in this case holds $tc(\mathcal{H}') < tc(\mathcal{H})$.

- In Case 2 (finger move ending in a region D' of smaller distance $d' < d_*$), we have $b(D_{m,1}) + b(D_{m,2}) \leq b(D_m) - 1$, $b(D') = b(D') + 1$ and $b(D'_*) = b(D_*) + 1$; hence

$$tc(\mathcal{H}') = tc(\mathcal{H}) - 3^{d_*} + 3^{d_*-1} + 3^{d'} \leq tc(\mathcal{H}) - 3^{d_*} + 2 \cdot 3^{d_*-1}.$$

Therefore, in this case holds $tc(\mathcal{H}') < tc(\mathcal{H})$.

- In Case 3 (finger move ending in another bad region D' of distance d_*), we have $b(D_{m,1}) + b(D_{m,2}) \leq b(D_m) - 1$, $b(D') = b(D') + 1$ and $b(D'_*) = b(D_*) + 1$; hence

$$tc(\mathcal{H}') \leq tc(\mathcal{H}) - 3^{d_*} + 3^{d_*} + 3^{d_*-1} = tc(\mathcal{H}) + 3^{d_*-1}.$$

This is the worst case in terms of the total complexity, as it is the only case among the ones of the algorithm in which it could increase.

- In Case 4.1 (handleslide), we have $b(D_{m,1}) + b(D_{m,2}) \leq b(D_m) - 1$ and $b(D') = b(D') + 2$; hence

$$tc(\mathcal{H}') \leq tc(\mathcal{H}) - 3^{d_*} + 2 \cdot 3^{d_*-1}.$$

Therefore, in this case holds $tc(\mathcal{H}') < tc(\mathcal{H})$.

Remark 4.30. We do not discuss how tc changes after a general handleslide, as we will only need the knowledge of the above cases to guarantee that the implementation of the algorithm is going to terminate after a finite number of cycles.

We are ready to run the algorithm on $\mathcal{H}$, a Heegaard diagram of distance d_* . The algorithm picks out the worst region D_m , which we assume to be a $2n$ -gon. The program chooses a blue edge b_* with which D_m is neighbour to a $d_* - 1$ region; let β_* be the β -circle on which b_* stays. Then, it tries to do a (general) handleslide on every β -edge of D_m that stays on a different β -circles, generating k new diagrams $\mathcal{H}_1, \dots, \mathcal{H}_k$ for some $0 \leq k \leq n - 1$. Then we try the finger moves: we numerate the α -edges of D_m by $a_1, \dots, a_n$ as described in the algorithm and we try to go through $a_2, \dots, a_{n-1}$, generating l new diagrams $\mathcal{H}_{k+1}, \dots, \mathcal{H}_{k+s}$ for some $0 \leq s \leq n - 2$. If we find a handleslide, we do not consider it as a move as we already took it in consideration within the general handleslide; moreover, we do not go through a_1 and a_n as the only way to reduce the badness of the region going through those is to do an handleslide (a finger move would left D_m to be a $2n$ -gon).

Remark 4.31. Among all the diagrams $\mathcal{H}_1, \dots, \mathcal{H}_{k+s}$ there is also the diagram that we would obtain by applying the algorithm as it is presented originally; we denote it by $\mathcal{H}^*$.

We now have $k + s$ possible Heegaard diagrams that we need to compare in order to understand which one is the better one to have for the next step. For the comparison, we consider three factors:

- the total complexity of the diagrams $tc(\mathcal{H}_i)$,
- the number of generators of the diagrams,
- the distance d_* complexity c_{d_*} .

The algorithm to compare the different Heegaard diagrams is the following.

- **Step 1.** As first step of the comparison, we exploit the total complexity. We call by $tc_{\min}$ the minimum of all the total complexities, i.e.

$$tc_{\min} = \min \{tc(\mathcal{H}_1), \dots, tc(\mathcal{H}_{k+s})\}.$$

We call $\mathcal{H}_{i_1}, \dots, \mathcal{H}_{i_t}$ the diagrams with minimal total complexity, i.e.

$$tc_{\min} = tc(\mathcal{H}_{i_1}) = \dots = tc(\mathcal{H}_{i_t}).$$

- **Step 2.** We then extract from these diagrams the ones with the minimal number of generators. Let $\#gen_{\min}$ be this number, we call the diagrams which minimise the number of generators $\mathcal{H}_{i_{j_1}}, \dots, \mathcal{H}_{i_{j_s}}$.
- **Step 3.** Among these diagrams, we take the ones that have the smaller distance d_* complexities, we call

$$cd_{*\min} = \min \{cd_*(\mathcal{H}_{i_{j_1}}), \dots, cd_*(\mathcal{H}_{i_{j_s}})\}$$

and let $\mathcal{H}_{i_{j_{k_1}}}, \dots, \mathcal{H}_{i_{j_{k_r}}}$ be these diagrams.

- **Step 4.** Two different cases can occur:
 - (a) $cd_{*\min} < cd_*(\mathcal{H})$ and there are not bad regions of distance greater than d_* in at least one among $\mathcal{H}_{i_{j_{k_1}}}, \dots, \mathcal{H}_{i_{j_{k_r}}}$.
If this is the case, any diagram among $\mathcal{H}_{i_{j_{k_1}}}, \dots, \mathcal{H}_{i_{j_{k_r}}}$ which has not any bad region of distance greater than d_* is nearer to be a nice Heegaard diagram in the sense of the nicefication algorithm as presented in Section 4.2. Thus, we can choose one among them to continue the nicefication.
 - (b) $cd_{*\min} \geq cd_*(\mathcal{H})$ or there is at least one bad region of distance greater than d_* in every diagram $\mathcal{H}_{i_{j_{k_1}}}, \dots, \mathcal{H}_{i_{j_{k_r}}}$. In this case, we cannot conclude as above but we need to take some extra precautions.

We first go back to Step 3: we call $cd_{*\min,2}$ the second lowest distance d_* complexity and we consider the diagrams among $\mathcal{H}_{i_{j_1}}, \dots, \mathcal{H}_{i_{j_s}}$ which have distance d_* complexity $cd_{*\min,2}$. We move to Step 4. If we finish in case (a), we obtain a Heegaard diagram with which we can pass to the next cycle of the nicefication algorithm. If we still finish in case (b), we consider $cd_{*\min,3}$, and so on.

If we finish in case (b) for all $\mathcal{H}_{i_{j_1}}, \dots, \mathcal{H}_{i_{j_s}}$, we go back to Step 2. We call $\#gen_{\min,2}$ the second lowest number of generators and we consider the diagrams among $\mathcal{H}_{i_1}, \dots, \mathcal{H}_{i_t}$ which have such number of generators. Then we move to Step 3 and Step 4. If we finish in case (a), we obtain a Heegaard diagram with which we can pass to the next cycle of the nicefication algorithm. If also in this case we finish always in case (b), then we consider $\#gen_{\min,3}$, and so on.

If we finish in case (b) for all $\mathcal{H}_{i_1}, \dots, \mathcal{H}_{i_t}$, we go back to step 1. We call $tc_{\min,2}$ the second lowest total complexity and we consider the diagrams among $\mathcal{H}_1, \dots, \mathcal{H}_{k+s}$ which have such total complexity. Then we move to Step 2, Step 3 and Step 4. If we finish in case (a), we obtain a Heegaard diagram with which we can pass to the next cycle of the nicefication algorithm. If also in this case we finish always in case (b), then we consider $tc_{\min,3}$, and so on.

Notice that we are going to finish in case (a) at some point, as among the $\mathcal{H}_1, \dots, \mathcal{H}_{k+s}$ there is also $\mathcal{H}^*$ for which the hypothesis of case (a) are satisfied.

We call $\mathcal{H}'$ the diagram obtained by the above algorithm and we continue to run the nicefication on it.

Remark 4.32. Going back to what we said in Remark 4.28, it is now clear that if the chosen diagram $\mathcal{H}'$ is one obtained via a general handleslide, then the above discussion guarantees that we are indeed proceeding towards a nice Heegaard diagram.

Therefore, we have proven the following proposition.

Proposition 4.33. *The [nicepy] Software, implemented starting from the Nicefication algorithm and modified as described above, terminates in a finite number of steps.*

Remark 4.34 (On the utility of the total complexity). The main advantage of the total complexity lies in the ability to better compare the different Heegaard diagrams generated in one cycle of the algorithm: it allows us to understand to some extent where and how the badness is "moved" by the modification done.

We can make an example to better visualise this: consider the diagram showed locally in Figure 4.22; we have that the worst region is an octagon D_m of distance d_* and we let D_* to be the neighbour region of distance $d_* - 1$.

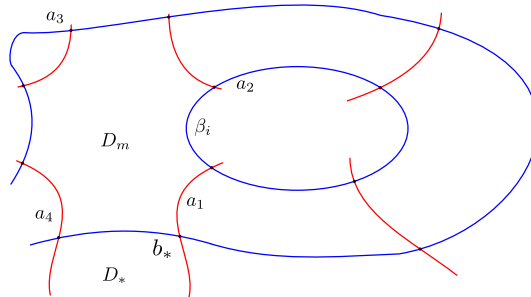


Figure 4.22: Initial situation to study the effectiveness of the total complexity.

Then we compare two possible moves: one finger move ending in a bigon adjacent to D_m (on the left in Figure 4.23) and one (general) handleslide (on the right in Figure 4.23).

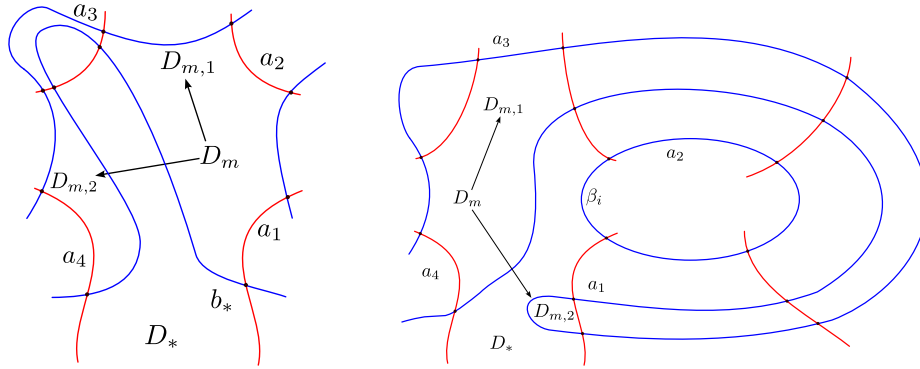


Figure 4.23: Two moves that modify c_{d_*} in the same way but are not equally effective.

Then, the difference between these two moves is not detected by c_{d_*} : in fact, in both cases we have that the contribute of the badness of D_m is reduced by one (in one case $D_{m,2}$ is a bigon, in the other it is a square and $b(D_{m,1}) = b(D_m) - 1$). But the difference is clear if we observe c_{d_*-1} : in one case, the badness of D_* increases by one and in the other it increases by 2. Hence, in this particular case, we prefer the finger move to the handleslide since it allows us to skip an additional cycle later in the algorithm when we will have to make D_* a good region. The total complexity clearly chooses the right move in this sense, allowing us to reduce the number of cycles that the algorithm needs to end.

Chapter 5

Computations

In the course of this master project, some families of tangles were studied in order to try to find some patterns in the invariants yielded. This investigation is meant to be a step further in the research for a more general formula to understand how HFT behaves under the sum of tangles.

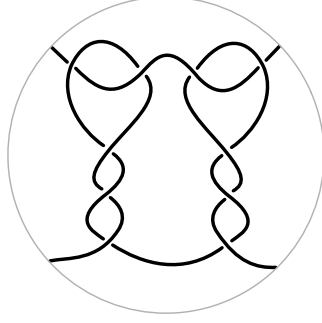
The chosen families from this research were the following two: the family of the sum of a rational tangle with its mirror image, i.e. $Q_{p/q} + Q_{-p/q}$ and the family of the sum of a rational tangle with a number of full vertical twists, i.e. $Q_{p/q} + Q_{1/(2n)}$. In particular, this second family yielded some promising result in terms of a possible pattern that HFT seems to follow, which was presented in the Introduction as Conjecture 0.1. It would be nice to prove this regularity also at a theoretical level, which could lead to an explanation on the behaviour of the rational part of HFT under the sum of tangles.

We see the results obtained in details.

5.1 HFT of $Q_{p/q} + Q_{-p/q}$

We consider a rational tangle $Q_{p/q}$ and its mirror image $Q_{-p/q}$, obtained by swapping all the crossings.

We define the tangle $T_{p/q}$ to be the one obtained by the sum $Q_{p/q} + Q_{-p/q}$ for $p/q \in \mathbb{QP}^1$ and q odd; see Figure 5.1 for the example $T_{3/7}$. The condition on q ensures that we are not going to have closed components in $T_{p/q}$.

Figure 5.1: $T_{3/7} = Q_{3/7} + Q_{-3/7}$.

We now claim that we can reduce ourselves to the computation of HFT for the set $\mathcal{F} = \{T_{p/q} \mid 0 < 2p < q\}$, in the sense that $\text{HFT}(T_{p/q})$ for a generic $T_{p/q}$ can be deduced from $\text{HFT}(F)$ for some $F \in \mathcal{F}$.

To prove this, consider the following intermediate remarks.

Remark 5.1 (We can assume $p > 0$). We can assume $p > 0$ without loss of generality. To prove this, we just have to notice that $m(T_{p/q}) \simeq T_{-p/q}$. Then, for Proposition 3.60, we have that we can obtain $\text{HFT}(T_{-p/q})$ from $\text{HFT}(T_{p/q})$.

Remark 5.2 (We can assume $|p| < q$). We can assume $|p| < q$ without loss of generality. To prove this, we show that $T_{p/q} \simeq T_{(p/q)+n}$ for any $n \in \mathbb{Z}$.

In fact, we can easily see this in terms of continued fractions. Let $[a_1, \dots, a_m]$ be a continued fractions corresponding to p/q . Then, adding $n \in \mathbb{Z}$ means to consider the continued fraction $[a_1 + n, a_2, \dots, a_m]$, i.e. we are adding half-twists on the right of the tangle. Hence we found that

$$T_{(p/q)+n} = Q_{p/q} + Q_n + m(Q_{p/q} + Q_n) = Q_{p/q} + Q_n + Q_{-p/q} + Q_{-n}.$$

Using Corollary 1.52 and the second Reidemeister move, we then obtain that

$$\begin{aligned} T_{(p/q)+n} &= Q_{p/q} + Q_n + m(Q_{p/q} + Q_n) \\ &= Q_{p/q} + Q_n + Q_{-p/q} + Q_{-n} \\ (\text{Corollary 1.52}) &\simeq Q_{p/q} + Q_n + Q_{-n} + Q_{-p/q} \\ (\text{II Reidemeister move}) &\simeq Q_{p/q} + Q_{-p/q} \\ &= T_{p/q}. \end{aligned}$$

Remark 5.3 (We can assume $0 < 2p < q$). We can now see that, without loss of generality, we can indeed consider only the tangles in the family $\mathcal{F}$. By Remarks 5.1 and 5.2 we can assume $0 < p < q$. Let p, q be such that $0 < q < 2p$ (i.e. $0 < 1/2 < p/q$); then, for same argument exploited in Remark 5.2, holds

$$T_{p/q} \simeq T_{(p/q)-1} = T_{(p-q)/q}.$$

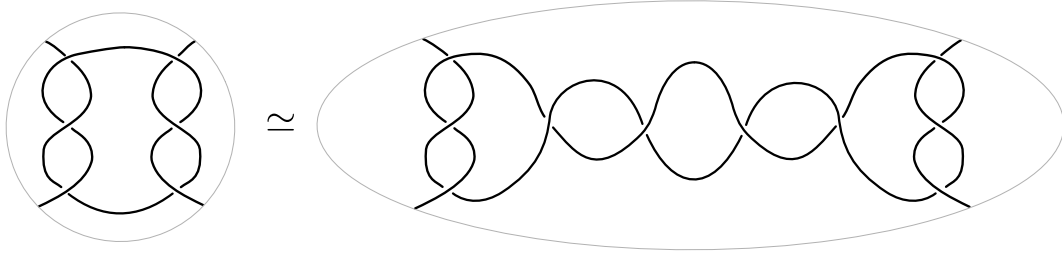


Figure 5.2: Adding n to p/q , we have $2n$ half-twists in the middle of $T_{(p/q)+n}$ that cancel out: example with $T_{1/3}$ and $T_{7/3}$.

Now, as we saw in Remark 5.1, we have

$$T_{(p-q)/q} \simeq m(T_{(q-p)/q})$$

and since

$$0 < \frac{q-p}{q} < \frac{1}{2},$$

we can obtain $\text{HFT}(T_{p/q})$ from $T_{(q-p)/q} \in \mathcal{F}$.

Therefore, by studying the result obtained applying HFT to tangles in the family $\mathcal{F}$, we are studying all $\text{HFT}(T_{p/q})$ for tangles of the form $T_{p/q} = Q_{p/q} + Q_{-p/q}$.

We obtained the computations showed in Table 5.1 for the tangle in $\mathcal{F}$. Here, there are three patterns that seem to be followed, which can be summed up in the following three conjectures.

Conjecture 5.4. *Let $T_{1/q}$ be $T_{1/q} = Q_{1/q} + Q_{-1/q}$ for q odd. Then,*

$$\text{HFT}(T_{1/q}) = \mathfrak{r}(0) + 2 \cdot \left(\sum_{i=1}^{(q-1)/2} \mathfrak{s}_{2i}(0) \right).$$

The idea for this conjecture comes by a direct observation of the first column of Table 5.1.

Conjecture 5.5. *Let $T_{p/q}$ be $T_{p/q} = Q_{p/q} + Q_{-p/q}$ for $p/q \in \mathbb{Q}P^1$, $0 < p$ and $q = 2p+1$. Then,*

$$\text{HFT}(T_{1/q}) = \mathfrak{r}(0) + (p^2 + p) \cdot \mathfrak{s}_4(0).$$

We can explain this conjecture by looking at the diagonal $q = 2p+1$ in Table 5.1. We can see that every time that we move to the next value of p , we are increasing the number of special components of length 4 and slope 0 by $2p$. Hence, the number of $\mathfrak{s}_4(0)$ for $T_{p/q}$ is $2(\sum_{i=1}^p i) = 2 \cdot (p(p+1))/2 = p^2 + p$.

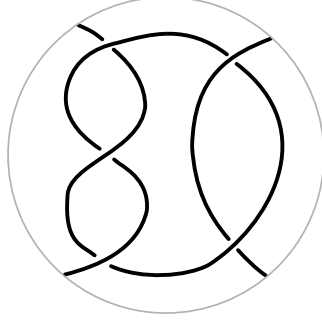
With a similar observation on the diagonal $q = 2p+3$, the following pattern seems to occur.

Conjecture 5.6. *Let $T_{p/q}$ be $T_{p/q} = Q_{p/q} + Q_{-p/q}$ for $p/q \in \mathbb{Q}P^1$, $0 < p$ and $q = 2p+3$. Then,*

$$\text{HFT}(T_{1/q}) = \mathfrak{r}(0) + (p^2 + 3p - 4) \cdot \mathfrak{s}_4(0) + 2 \cdot \mathfrak{s}_8(0).$$

$q \backslash p$	1	2	3	4	5	6
1	$\mathfrak{t}(0)$					
2						
3	$\mathfrak{t}(0) + 2 \cdot \mathfrak{s}_4(0)$					
4						
5	$\mathfrak{t}(0) + 2 \cdot \mathfrak{s}_4(0) + 2 \cdot \mathfrak{s}_8(0)$	$\mathfrak{t}(0) + 6 \cdot \mathfrak{s}_4(0)$				
6						
7	$\mathfrak{t}(0) + 2 \cdot \mathfrak{s}_4(0) + 2 \cdot \mathfrak{s}_8(0) + 2 \cdot \mathfrak{s}_{12}(0)$	$\mathfrak{t}(0) + 8 \cdot \mathfrak{s}_4(0) + 2 \cdot \mathfrak{s}_8(0)$	$\mathfrak{t}(0) + 12 \cdot \mathfrak{s}_4(0)$			
8						
9	$\mathfrak{t}(0) + 2 \cdot \mathfrak{s}_4(0) + 2 \cdot \mathfrak{s}_8(0) + 2 \cdot \mathfrak{s}_{12}(0) + 2 \cdot \mathfrak{s}_{16}(0)$	$\mathfrak{t}(0) + 8 \cdot \mathfrak{s}_4(0) + 6 \cdot \mathfrak{s}_8(0)$		$\mathfrak{t}(0) + 20 \cdot \mathfrak{s}_4(0)$		
10						
11			$\mathfrak{t}(0) + 18 \cdot \mathfrak{s}_4(0) + 6 \cdot \mathfrak{s}_8(0)$	$\mathfrak{t}(0) + 26 \cdot \mathfrak{s}_4(0) + 2 \cdot \mathfrak{s}_8(0)$	$\mathfrak{t}(0) + 30 \cdot \mathfrak{s}_4(0)$	
12						
13				$\mathfrak{t}(0) + 32 \cdot \mathfrak{s}_4(0) + 2 \cdot \mathfrak{s}_8(0) + 2 \cdot \mathfrak{s}_{12}(0)$	$\mathfrak{t}(0) + 38 \cdot \mathfrak{s}_4(0) + 5 \cdot \mathfrak{s}_8(0) + 2 \cdot \mathfrak{s}_{12}(0)$	$\mathfrak{t}(0) + 42 \cdot \mathfrak{s}_4(0)$
14						
15				$\mathfrak{t}(0) + 32 \cdot \mathfrak{s}_4(0) + 12 \cdot \mathfrak{s}_8(0)$		
16						
17				$\mathfrak{t}(0) + 32 \cdot \mathfrak{s}_4(0) + 20 \cdot \mathfrak{s}_8(0)$		$\mathfrak{t}(0) + 62 \cdot \mathfrak{s}_4(0) + 2 \cdot \mathfrak{s}_8(0) + 2 \cdot \mathfrak{s}_{12}(0)$

Table 5.1: HFT for tangles of the form $T_{p/q}$ with $p/q \in \mathbb{Q}P^1$, q odd and $0 < 2p < q$.

Figure 5.3: The tangle $S_{1/3}(2)$.

To motivate this conjecture, we look at the diagonal $q = 2p + 3$ in Table 5.1. We can see that, excluding the first column and taking it as ground level, every time that we move to the next value of p , we are increasing the number of special components of length 4 and slope 0 by $2(p + 1)$. This means that the number of $\mathfrak{s}_4(0)$ for $T_{p/q}$ is $2 + 2 \left(\sum_{i=3}^{p+1} i \right) = 2 + 2 \cdot ((p+1)(p+2))/2 - 6 = p^2 + 3p - 4$.

The author was not able to see any other promising patterns in Table 5.1.

5.2 HFT of $Q_{p/q} + Q_{1/(2n)}$

We compute now HFT on tangles of the form

$$S_{p/q}(2n) := Q_{p/q} + Q_{1/(2n)}$$

for $p/q \in \mathbb{Q}P^1$, q odd and $n \in \mathbb{Z}_{>0}$, i.e. rational tangles that we sum to a number of full vertical twists. See Figure 5.3 for the example $S_{1/3}(2)$.

As p, q and n grow, the computations require more time, therefore most of them were done for $n = 1, 2$. As we did in Section 5.1, we can find a symmetry among this class of tangles. This allows us to restrict ourselves to the investigation of a subset of $\{S_{p/q}(2n) \mid p/q \in \mathbb{Q}P^1, q \text{ odd and } n \in \mathbb{Z}_{>0}\}$: namely, we just need to compute HFT on the family $\mathcal{G}(2n) = \{S_{p/q}(2n) \mid 0 < p < q\}$. We prove this in the following two remarks.

Remark 5.7 (We can assume $p > 0$). Consider $S_{p/q}(2n)$, where $p < 0$ and let $h \in \mathbb{Z}_{>0}$ be such that $p/q + h > 0$. Then, it holds

$$\begin{aligned} S_{p/q}(2n) &= Q_{p/q} + Q_{1/(2n)} \\ \text{(II Reidemeister move)} &\simeq Q_{p/q} + Q_{1/(2n)} + Q_h + Q_{-h} \\ \text{(Corollary 1.52)} &\simeq Q_{p/q} + Q_h + Q_{1/(2n)} + Q_{-h} \\ &= Q_{(p/q)+h} + Q_{1/(2n)} + Q_{-h} \\ &= S_{(p/q)+h}(2n) + Q_{-h}. \end{aligned}$$

Therefore, we can deduce $\text{HFT}(S_{p/q}(2n))$ where $p/q < 0$ from $\text{HFT}(S_{(p/q)+h}(2n))$ where $0 < p/q + h$ via Theorem 3.63.

Remark 5.8 (We can assume $0 < p < q$). This follows again from Corollary 1.52: if we consider the tangle $S_{(p/q)+m}(2n)$ for $m \in \mathbb{Z}$, then it holds

$$\begin{aligned} S_{(p/q)+m}(2n) &= Q_{p/q} + Q_m + Q_{1/(2n)} \\ &= Q_{p/q} + Q_{1/(2n)} + Q_m \\ &= S_{p/q}(2n) + Q_m, \end{aligned}$$

and since we know by Theorem 3.63 how the addition of some horizontal twists affects HFT of a given tangle, we can recover $\text{HFT}(S_{(p/q)+m}(2n))$ from $\text{HFT}(S_{p/q}(2n))$. Therefore, without loss of generality, we can assume that $0 < p < q$.

5.2.1 Computations for $S_{p/q}(2)$

We begin by looking at the tangles obtained by adding a single full twist. Here, we can actually restrict ourselves to a smaller pool of tangles: we only need to compute the invariants of the family $\mathcal{G}'(2) = \{S_{p/q}(2) \mid 0 < 2p < q\} \subset \mathcal{G}(2)$. Once again, the key to prove this is to use Corollary 1.52. Let $0 < p < q$ and consider a tangle $S_{p/q}(2)$ where $0 < 1/2 < p/q$; then the following holds:

$$\begin{aligned} S_{p/q}(2) &= Q_{p/q} + Q_{1/2} \\ &\simeq m(Q_{-p/q} + Q_{-1/2}) \\ &\simeq m(Q_{-p/q} + Q_{-1} + Q_{1/2}) \\ &\simeq m(Q_{(-p/q)-1} + Q_{1/2}) \\ \text{(II Reidemeister move)} &\simeq m(Q_{(-p/q)+1} + Q_{-2} + Q_{1/2}) \\ \text{(Corollary 1.52)} &\simeq m(Q_{(-p/q)+1} + Q_{1/2} + Q_{-2}) \\ &= m(S_{(-p/q)+1}(2) + Q_{-2}) \\ &= m(S_{(-p/q)+1}(2)) + m(Q_{-2}) \\ &= m(S_{(-p/q)+1}(2)) + Q_2. \end{aligned}$$

Therefore, thanks to Proposition 3.60 and Theorem 3.63, we have proven that we can obtain $\text{HFT}(S_{p/q}(2))$ from $\text{HFT}(S_{(-p/q)+1}(2))$, where

$$0 < -\frac{p}{q} + 1 < \frac{1}{2}.$$

The computations obtained for $S_{p/q}(2)$ with $0 < 2p < q$ are collected in Table 5.2.

There seems to be a very regular pattern in the computations, which is presented in the following conjecture.

Conjecture 5.9. *Let $S_{p/q}(2)$ be $S_{p/q}(2) = Q_{p/q} + Q_{1/2}$ with $p/q \in \mathbb{Q}P^1$, q odd and $0 < 2p < q$. Then,*

$$\text{HFT}(S_{p/q}(2)) = (q - 2p) \cdot \mathfrak{r}(1/2) + p \cdot \mathfrak{s}_4(1).$$

$q \backslash p$	1	2	3	4	5	6	7	8
1								
2								
3	$1 \cdot \mathbf{r}^{(1/2)} + 1 \cdot \mathbf{s}_4(1)$							
4								
5	$3 \cdot \mathbf{r}^{(1/2)} + 1 \cdot \mathbf{s}_4(1)$	$1 \cdot \mathbf{r}^{(1/2)} + 2 \cdot \mathbf{s}_4(1)$						
6								
7	$5 \cdot \mathbf{r}^{(1/2)} + 1 \cdot \mathbf{s}_4(1)$	$3 \cdot \mathbf{r}^{(1/2)} + 2 \cdot \mathbf{s}_4(1)$	$1 \cdot \mathbf{r}^{(1/2)} + 3 \cdot \mathbf{s}_4(1)$					
8								
9	$7 \cdot \mathbf{r}^{(1/2)} + 1 \cdot \mathbf{s}_4(1)$	$5 \cdot \mathbf{r}^{(1/2)} + 2 \cdot \mathbf{s}_4(1)$		$1 \cdot \mathbf{r}^{(1/2)} + 4 \cdot \mathbf{s}_4(1)$				
10								
11	$9 \cdot \mathbf{r}^{(1/2)} + 1 \cdot \mathbf{s}_4(1)$	$7 \cdot \mathbf{r}^{(1/2)} + 2 \cdot \mathbf{s}_4(1)$	$5 \cdot \mathbf{r}^{(1/2)} + 3 \cdot \mathbf{s}_4(1)$	$3 \cdot \mathbf{r}^{(1/2)} + 4 \cdot \mathbf{s}_4(1)$	$1 \cdot \mathbf{r}^{(1/2)} + 5 \cdot \mathbf{s}_4(1)$			
12								
13	$11 \cdot \mathbf{r}^{(1/2)} + 1 \cdot \mathbf{s}_4(1)$	$9 \cdot \mathbf{r}^{(1/2)} + 2 \cdot \mathbf{s}_4(1)$	$7 \cdot \mathbf{r}^{(1/2)} + 3 \cdot \mathbf{s}_4(1)$	$5 \cdot \mathbf{r}^{(1/2)} + 4 \cdot \mathbf{s}_4(1)$	$3 \cdot \mathbf{r}^{(1/2)} + 5 \cdot \mathbf{s}_4(1)$			
14								
15								
16								
17						$5 \cdot \mathbf{r}^{(1/2)} + 6 \cdot \mathbf{s}_4(1)$		
18								
19		$15 \cdot \mathbf{r}^{(1/2)} + 2 \cdot \mathbf{s}_4(1)$					$5 \cdot \mathbf{r}^{(1/2)} + 7 \cdot \mathbf{s}_4(1)$	
20								
21								$5 \cdot \mathbf{r}^{(1/2)} + 8 \cdot \mathbf{s}_4(1)$

Table 5.2: HFT for tangles of the form $S_{p/q}(2)$ with $p/q \in \mathbb{QP}^1$, q odd and $0 < 2p < q$.

5.2.2 Computations for $S_{p/q}(4)$

We consider now tangles obtained from rational tangles by a sum with 2 complete vertical twists, i.e. tangles of the form $S_{p/q}(4)$. Here, it was not possible to find symmetries that could help us to reduce the pool of tangles that we need to compute HFT on; therefore the computations were done for tangles of the form $S_{p/q}(4)$ with $p/q \in \mathbb{Q}P^1$, $0 < p < q$ and q odd.

The computations obtained for these tangles are collected in in Table 5.3.

By looking at the computations for $S_{p/q}(4)$ with $p/q \in \mathbb{Q}P^1$, q odd and $0 < 4p < q$, the pattern presented in Conjecture 5.9 seems to generalise to the following conjecture.

Conjecture 5.10. *Let $S_{p/q}(4)$ be $S_{p/q}(4) = Q_{p/q} + Q_{1/4}$ with $p/q \in \mathbb{Q}P^1$, q odd and $0 < 4p < q$. Then,*

$$\text{HFT}(S_{p/q}(4)) = (q - 4p) \cdot \mathfrak{r}(1/4) + p \cdot (\mathfrak{s}_4(1) + \mathfrak{s}_4(1/3)).$$

The author could not see a particular pattern for the computations with $p < q < 4p$. However, a behaviour like this one seems reasonable: also in other cases (e.g. Pretzel tangles) the tangles split in two sub-families within whose there are substantially different patterns followed.

5.2.3 Conjecture for $\text{HFT}(S_{p/q}(2n))$

With the hints given by $S_{p/q}(2)$ and $S_{p/q}(4)$, also computations for some basic cases of $S_{p/q}(6)$ and $S_{p/q}(8)$ where $p/q \in \mathbb{Q}P^1$, q odd and $0 < 6p < q$ were done. These are collected in Tables 5.4 and 5.5.

$q \setminus p$	1	2
7	$1 \cdot \mathfrak{r}(1/6) + 1 \cdot \mathfrak{s}_4(1) + 1 \cdot \mathfrak{s}_4(1/3) + 1 \cdot \mathfrak{s}_4(1/5)$	
8		
9	$3 \cdot \mathfrak{r}(1/6) + 1 \cdot \mathfrak{s}_4(1) + 1 \cdot \mathfrak{s}_4(1/3) + 1 \cdot \mathfrak{s}_4(1/5)$	
10		
11		
12		
13		$1 \cdot \mathfrak{r}(1/6) + 2 \cdot \mathfrak{s}_4(1) + 2 \cdot \mathfrak{s}_4(1/3) + 2 \cdot \mathfrak{s}_4(1/5)$

Table 5.4: HFT for tangles of the form $S_{p/q}(6)$ with $p/q \in \mathbb{Q}P^1$, q odd and $0 < 6p < q$.

$q \backslash p$	1	2	3	4	5	6
1						
2						
3	$1 \cdot \mathfrak{r}(3/8) + 1 \cdot \mathfrak{s}_4(1)$	$1 \cdot \mathfrak{r}(3/4) + 2 \cdot \mathfrak{s}_4(1)$				
4						
5	$1 \cdot \mathfrak{r}(1/4) + 1 \cdot \mathfrak{s}_4(1) + 1 \cdot \mathfrak{s}_4(1/3)$	$1 \cdot \mathfrak{r}(5/12) + 2 \cdot \mathfrak{s}_4(1)$	$1 \cdot \mathfrak{r}(5/8) + 3 \cdot \mathfrak{s}_4(1)$	$1 \cdot \mathfrak{r}(5/4) + 2 \cdot \mathfrak{s}_4(1) + 1 \cdot \mathfrak{s}_8(1)$		
6						
7	$3 \cdot \mathfrak{r}(1/4) + 1 \cdot \mathfrak{s}_4(1) + 1 \cdot \mathfrak{s}_4(1/3)$	$1 \cdot \mathfrak{r}(3/8) + 2 \cdot \mathfrak{s}_4(1) + 1 \cdot \mathfrak{s}_4(1/3)$	$1 \cdot \mathfrak{r}(7/16) + 3 \cdot \mathfrak{s}_4(1)$	$1 \cdot \mathfrak{r}(7/12) + 4 \cdot \mathfrak{s}_4(1)$	$1 \cdot \mathfrak{r}(3/4) + 4 \cdot \mathfrak{s}_4(1) + 1 \cdot \mathfrak{s}_8(1)$	$3 \cdot \mathfrak{r}(5/4) + 2 \cdot \mathfrak{s}_4(1) + 1 \cdot \mathfrak{s}_8(1)$
8						
9	$5 \cdot \mathfrak{r}(1/4) + 1 \cdot \mathfrak{s}_4(1) + 1 \cdot \mathfrak{s}_4(1/3)$	$1 \cdot \mathfrak{r}(1/4) + 2 \cdot \mathfrak{s}_4(1) + 2 \cdot \mathfrak{s}_4(1/3)$		$1 \cdot \mathfrak{r}(9/20) + 4 \cdot \mathfrak{s}_4(1)$	$1 \cdot \mathfrak{r}(9/16) + 5 \cdot \mathfrak{s}_4(1)$	
10						
11	$7 \cdot \mathfrak{r}(1/4) + 1 \cdot \mathfrak{s}_4(1) + 1 \cdot \mathfrak{s}_4(1/3)$	$3 \cdot \mathfrak{r}(1/4) + 2 \cdot \mathfrak{s}_4(1) + 2 \cdot \mathfrak{s}_4(1/3)$	$1 \cdot \mathfrak{r}(3/8) + 3 \cdot \mathfrak{s}_4(1) + 2 \cdot \mathfrak{s}_4(1/3)$	$1 \cdot \mathfrak{r}(3/8) + 4 \cdot \mathfrak{s}_4(1) + 1 \cdot \mathfrak{s}_4(2/5)$	$1 \cdot \mathfrak{r}(11/24) + 5 \cdot \mathfrak{s}_4(1)$	$1 \cdot \mathfrak{r}(11/20) + 6 \cdot \mathfrak{s}_4(1)$
12						
13	$9 \cdot \mathfrak{r}(1/4) + 1 \cdot \mathfrak{s}_4(1) + 1 \cdot \mathfrak{s}_4(1/3)$		$1 \cdot \mathfrak{r}(3/8) + 3 \cdot \mathfrak{s}_4(1) + 3 \cdot \mathfrak{s}_4(1/3)$	$3 \cdot \mathfrak{r}(3/8) + 4 \cdot \mathfrak{s}_4(1) + 1 \cdot \mathfrak{s}_4(1/3)$	$1 \cdot \mathfrak{r}(5/12) + 5 \cdot \mathfrak{s}_4(1) + 1 \cdot \mathfrak{s}_4(2/5)$	$1 \cdot \mathfrak{r}(13/28) + 6 \cdot \mathfrak{s}_4(1)$
14						
15				$1 \cdot \mathfrak{r}(3/8) + 4 \cdot \mathfrak{s}_4(1) + 3 \cdot \mathfrak{s}_4(1/3)$		
16						
17						$1 \cdot \mathfrak{r}(3/8) + 6 \cdot \mathfrak{s}_4(1) + 1 \cdot \mathfrak{s}_4(2/5)$

Table 5.3: HFT for tangles of the form $S_{p/q}(4)$ with $p/q \in \mathbb{QP}^1$, q odd and $0 < p < q$.

$\mathbf{q} \backslash \mathbf{p}$	1
9	$1 \cdot \mathfrak{r}(1/8) + 1 \cdot \mathfrak{s}_4(1) + 1 \cdot \mathfrak{s}_4(1/3) +$ $1 \cdot \mathfrak{s}_4(1/5) + 1 \cdot \mathfrak{s}_4(1/7)$
10	
11	$3 \cdot \mathfrak{r}(1/8) + 1 \cdot \mathfrak{s}_4(1) + 1 \cdot \mathfrak{s}_4(1/3) +$ $1 \cdot \mathfrak{s}_4(1/5) + 1 \cdot \mathfrak{s}_4(1/7)$

Table 5.5: HFT for tangles of the form $S_{p/q}(8)$ with $p/q \in \mathbb{Q}\mathbb{P}^1$, q odd and $0 < 8p < q$.

All these computations, seem to point to a more general pattern followed by HFT for tangles of the form $S_{p/q}(2n)$ with $p/q \in \mathbb{Q}\mathbb{P}^1$, q odd and $0 < 2n \cdot p < q$, which is presented in the next conjecture.

Conjecture 5.11. *Let $S_{p/q}(2n)$ be the tangle $S_{p/q}(2n) = Q_{p/q} + Q_{1/2n}$ with $p/q \in \mathbb{Q}\mathbb{P}^1$, q odd and $0 < 2n \cdot p < q$. Then,*

$$\text{HFT}(S_{p/q}(2n)) = (q - 2n \cdot p) \cdot \mathfrak{r}(1/(2n)) + p \cdot \left(\sum_{i=0}^{n-1} \mathfrak{s}_4(1/(2i+1)) \right).$$

Remark 5.12 (A confirmation from the Pretzel tangles). As anticipated in the Introduction, Conjecture 5.11 seems to be in part confirmed by the know multicurves of the Pretzel tangles. In fact, the $(2r, -2s-1)$ -Pretzel tangle is defined as $Q_{1/(2r)} + Q_{-1/(2s+1)}$, which is clearly a subfamily of the tangles of the form $S_{p/q}(2n)$. For these tangles, the tangle Floer homology is known (see Theorem 6.9 of [Zib20]) and they agree with what is stated in Conjecture 5.11.

Appendix A

Manual for nicepy, version 1.0

This appendix is a manual for version 1.0 of the [nicepy] Python Package, an implementation of the Nicefication algorithm for Python 3.10.2.

A.1 Input preparation

The input is a text file, call it `input.txt`, with a certain number of lines. The first line, saved under `type_of_diagram`, states the type of input; it can be one of the following:

- a. "normal", which means that we are giving a (bordered sutured) Heegaard diagram as input;
- b. "tangle", which means that we are giving a 4-ended tangle diagram as input;
- c. "rational", which means that we are giving a (list of) rational tangle(s) of the form $Q_{p/q}$.

Depending on the type of `input.txt`, the successive lines change.

Input of type "normal"

Let $\mathcal{H} = (\Sigma, \alpha = \alpha^c \cup \alpha^a, \beta)$ be the Heegaard diagram that we want to give as input. This diagram can be of the following types: a closed Heegaard diagram (in which $\alpha^a = \emptyset$ and we have a basepoint z), a bordered Heegaard diagram (in which we have one basepoint z in one of the regions on the border), or, more generally, a bordered sutured Heegaard diagram with α -arcs (β -arcs are not allowed in this version of the program) with some basepoints on it such that every connected component of $\Sigma \setminus \alpha$ contains at least one multiplicity zero regions.

We require the following three properties on $\mathcal{H}$:

- every α -curve intersects at least one β -circle and that every β -circle intersects at least one α -curves,

- every region of $\mathcal{H}$ is a disc,
- every α - and β -circle has at least three intersection points on it.

Notice that the first two properties are the ones obtained by applying Step 1 of the Nicefication algorithm; see Subsection 4.2.1 for details on how is it possible to modify a diagram in order for it to have these properties. The third one is crucial for the program, as the edges obtained from a circle with only one or two intersection points are not uniquely identified by their ends; it is required to perform one or more finger moves in order to reach this property.

We first draw the diagram of interest and we label the regions from 1 to m ; we will refer to them as $e_1, \dots, e_m$. Then we label the intersection points, i.e. the points obtained by $\alpha \cap \beta$ and $\alpha^a \cap \partial\Sigma$ starting from the ones on the border (if there are any), and we denote them by $1, \dots, k, \dots, n$, where the points $1, \dots, k$ are the points in $\alpha^a \cap \partial\Sigma$. Moreover, for every border component, the labels for the border points have to be placed in an increasing order going counter-clock wise. Thus, if we have h_1 points on the first border component, the labels for them have to be $1, \dots, h_1$, with an increasing order going counter-clock wise; then if we have h_2 points on the second border component, the labels for them have to be $h_1 + 1, \dots, h_1 + h_2$, going increasingly counter-clock wise; and so on.

We need the following lines in `input.txt` in the presented order.

1. `number_border_points`. This is the number of border points, i.e. the number of intersection points $\alpha^a \cap \partial\Sigma$ (it can be 0). In the notation adopted above, this is k .
2. `number_intersection_points`. This is the total number of intersection points of the diagram (including the border points), i.e. all the points $\alpha \cap \beta$ and $\alpha^a \cap \partial\Sigma$. In the notation above it is n .
3. `regions_input`. This is the essential data of the diagram, constructed as a list `L`. Given any region e_i , we construct a list `L_i` with the labels of the corners such that
 - the labels are given reading them from the region going counter-clock wise,
 - the first two labels are the endpoints of an α -edge of e_i .

Then `L` is the list containing the lists `L_i` in order. Notice in particular that if a region has a corner whose label is less or equal to k , then it is a border region of the diagram.

4. `multiplicity_zero_regions`. This is the list of the labels of multiplicity zero regions, i.e. regions that have a basepoint. This is a list of labels, i.e. only numbers.

Input of type "tangle"

Let T be a 4-ended tangle without closed components. Let $\mathcal{H} = (\Sigma, \alpha = \alpha^c \cup \alpha^a, \beta)$ be the tangle Heegaard diagram that we want to give as input (not peculiar).

We require the following three properties on $\mathcal{H}$:

- every α -curve intersects at least one β -circle and that every β -circle intersects at least one α -curves,
- every region of $\mathcal{H}$ is a disc,
- every α - and β -circle has at least three intersection points on it.

Notice that the first two properties are the ones obtained by applying Step 1 of the Nicefication algorithm; see Subsection 4.2.1 for details on how is it possible to modify a diagram in order for it to have these properties. The third one is crucial for the program, as the edges obtained from a circle with only one or two intersection points are not uniquely identified by their ends; it is required to perform one or more finger moves in order to reach this property.

We first draw the diagram of interest and we label the regions from 1 to m ; we will refer to them as $e_1, \dots, e_m$. Then we label the intersection points, i.e. the points obtained by $\alpha \cap \beta$ and $\alpha^a \cap \partial\Sigma$ starting from the ones on the punctures, and we denote them by $1, \dots, n$, where the points $1, \dots, 8$ are the endpoints of the α -arcs.

We need the following lines in `input.txt` in the presented order.

1. `number_border_points`. This is the number of border points, i.e. it is 8.
2. `number_intersection_points`. This is the total number of intersection points of the diagram (including the border points), i.e. all the points $\alpha \cap \beta$ and $\alpha^a \cap \partial\Sigma$. In the notation above it is n .
3. `regions_input`. This is the essential data of the diagram, constructed as a list `L`. Given any region e_i , we construct a list `L_i` with the labels of the corners such that
 - the labels are given reading them from the region going counter-clock wise,
 - the first two labels are the endpoints of an α -edge of e_i .

Then `L` is the list containing the lists `L_i` in order. Notice in particular that if a region has a corner whose label is less or equal to k , then it is a border region of the diagram.

4. `multiplicity_zero_regions`. Recall that, once we make the tangle Heegaard diagram a peculiar Heegaard diagram, we have to choose two special basepoints: one among the p_i 's and one among the q_j 's. Therefore, we give here a list of the form `[[i, 0], [j, 1]]` as input, where e_i is the multiplicity

zero region on the front and e_j is a region on the back. The program is able to try to nicefy all the 16 possible combinations of multiplicity zero regions (see Section A.3), however it needs an initial input to understand which regions stay on the back of $\mathcal{S}_4^2$ and which stay on the front. Thus this line is required in any case.

5. **alpha_arcs_sites**. This list is needed in order to understand which α -arc corresponds to which site. We give a list of four labels, each one corresponding to a border point of the diagram, ordered with respect to the α -arc sites. For instance, the list `[i,j,k,l]` is interpreted to say that the border point i is one endpoint of the α -arc corresponding to the site a , the border point j is one endpoint of the α -arc corresponding to the site b , k is one endpoint of c , and l is one endpoint of d .
6. **alexander_grading**. This list encode the data of the Alexander gradings of the border regions. It is a list of 4 elements, where each element is an endpoint of a vertical α -arc and such points are in the following order with respect to Alexander grading: `[(1,0), (-1,0), (0,1), (0,-1)]`.

Input of type "rational"

Let T be a 4-ended tangle without closed components and cut it in order to obtain T as the gluing of some rational tangles $R_1, \dots, R_h$; in particular notice that each R_i 's is a 4-ended tangle. It is not possible to decompose a tangle in a way that to retrieve it we need to do $Q_2 = Q_1 + Q_1$; more than one $Q_{\pm 1}$ may be present, but it is not possible to glue them via sum or multiplication of 4-ended tangles (see Section 1.6 for the definition of these operations).

Given any R_i , we label the endpoints of its strings as follows: 1 corresponds to the end in the north-west puncture, 2 to the end in the south-west puncture, 3 to the end in the south-east puncture and 4 to the end in the north-east puncture; we label the endpoints of the strings of T in the same way. Moreover, we also label the $2h - 2$ gluing points and we refer to them as $g_1, \dots, g_{2h-2}$.

We need the following lines in `input.txt` in the presented order.

1. **number_of_tangles**. This is the number of tangles; in the setting above it is h .
2. **tangles_input**. This is a list `L` with the data of the single rational tangles. For any $R_i = Q_{p_i/q_i}$, we construct a list `L_i`, where the first element is `0` if $p_i/q_i < 0$ and `1` if $p_i/q_i > 0$, the second is `p` and the third is `q`. Then, `L` is the list where the i -th element is `L_i`.
3. **gluing_instructions**. This is a list `G` with the instruction for the gluing of the tangles R_i . Let g_j the gluing point between the tangles R_{i_1} and R_{i_2} , along `end_1` (one string's end for R_{i_1}) and `end_2` (one string's end for R_{i_2}). We construct the list `G_j` corresponding to g_j as `[i_1, end_1, i_2, end_2]`.

Then `G` is the list where the j -th element is `G_j`. If there is only one tangle in `tangles_input`, `G` is an empty list.

4. `alexander_grading_tangle`. This list encode the data of the Alexander gradings of the border regions. It is a list of the numbers from 1 to 4, indicating the endpoint of the string of T , where the order of them is given with respect to Alexander grading: `[(1,0), (-1,0), (0,1), (0,-1)]`.

In this case, we do not need the information about the α -arc sites for the diagram of T , as it is automatically interpreted as follows:

- the site a is from the string's endpoint 1 to the string's endpoint 2,
- the site b is from the string's endpoint 2 to the string's endpoint 3,
- the site c is from the string's endpoint 3 to the string's endpoint 4,
- the site d is from the string's endpoint 4 to the string's endpoint 1.

Placing the input

After the file with the input is ready, we have to give it to the program. We save the path to `input.txt` in the variable `input_path` which can be found in the top part of `main_run.py`. `input_path` is a tuple, where the first entry is a string containing the (possibly relative to the path of `main_run.py`) path to the folder in which `input.txt` is and the second entry is a string containing the name of the input file itself.

```
1 input_path = ("C:/Users/user1/Documents/", "input.txt")
```

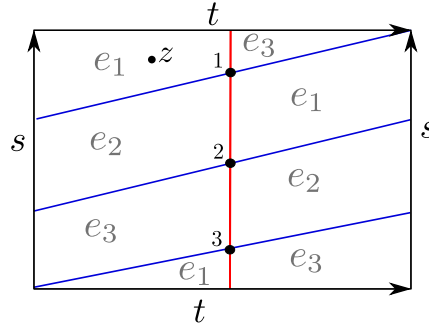
Listing A.1: Example of `input_path` if the path to the input file is `C:/Users/user1/Documents/input.txt`.

```
1 input_path = ("./inputs/", "input.txt")
```

Listing A.2: Example of `input_path` if the input file is in a subfolder called `inputs` of the folder in which `main_run.py` is.

A.2 Examples of input

We see some examples.

Figure A.1: Closed Heegaard diagram for $L(3, 1)$.

Closed Heegaard diagram

This is an example of an input where the first line is "normal".

Suppose to have the closed Heegaard diagram for the lens space $L(1, 3)$ as represented in Figure A.1.

Then, `input.txt` is the following.

```
1 normal
2 0
3 3
4 [[1, 2, 1, 3], [2, 3, 2, 1], [3, 1, 3, 2]]
5 [1]
```

Bordered sutured Heegaard diagram with α -arcs

This is an example of an input where the first line is "normal".

Suppose to have the bordered sutured Heegaard diagram in Figure A.2.

Then, `input.txt` is the following.

```
1 normal
2 10
3 25
4 [[15, 16, 17, 20, 23, 14], [20, 21, 22, 23], [23, 22], [21, 20], [18, 21, 20, 17],
   [14, 23, 22, 13], [2, 12, 15, 3], [12, 2, 5, 13], [13, 5, 4, 14], [3, 15, 14, 4],
   [19, 7, 6, 16], [16, 6, 9, 17], [8, 18, 17, 9], [18, 8, 7, 19], [1, 11, 24, 25, 11, 1],
   [16, 15, 12, 25, 24, 11, 25, 12, 13, 22, 21, 18, 19, 10, 10, 19], [25, 24],
   [11, 24]]
5 [15, 9, 13, 4, 3, 16]
```

Tangle Heegaard diagram

This is an example of an input where the first line is "tangle".

Consider the $(2, -3)$ -Pretzel tangle with the orientation on the strings as in Figure A.3 and the tangle Heegaard diagram for it represented in Figure A.4.

Then, `input.txt` is the following.

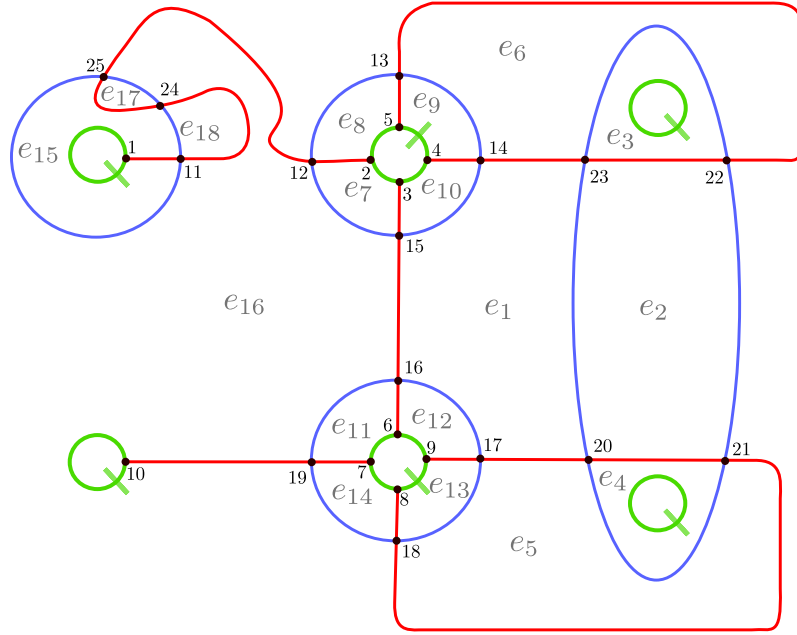
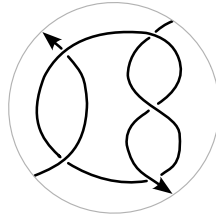
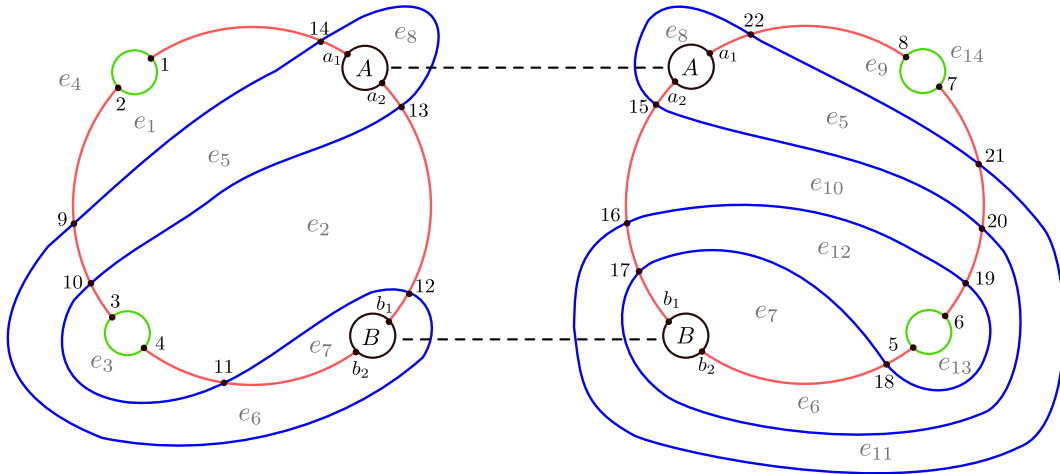
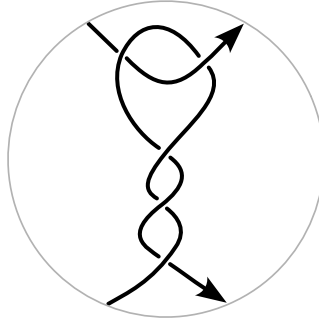


Figure A.2: A bordered sutured Heegaard diagram

Figure A.3: $(2, -3)$ -Pretzel tangle with oriented strings.Figure A.4: Tangle Heegaard diagram for the $(2, -3)$ -Pretzel tangle.

Figure A.5: Oriented tangle diagram for $Q_{3/7}$.

```

1 tangle
2 8
3 22
4 [[1, 11, 9, 2], [4, 13, 14, 10, 12, 3], [3, 12, 13, 4], [11, 1, 2, 9, 10, 14],
   [11, 12, 10, 16, 18, 17, 15, 9], [12, 11, 14, 19, 18, 21, 22, 13], [13, 22, 19, 14],
   [9, 15, 16, 10], [5, 15, 17, 6], [16, 20, 21, 18], [17, 18, 19, 20],
   [20, 19, 22, 8, 7, 21], [8, 22, 21, 7], [20, 16, 15, 5, 6, 17]]
5 [[1, 0], [3, 1]]
6 [2, 4, 6, 8]
7 [2, 3, 6, 7]

```

Single rational tangle

This is an example of an input where the first line is "rational".

Consider the rational tangle $Q_{3/7}$, represented in Figure A.5.

Then, `input.txt` is the following.

```

1 rational
2 1
3 [[1, 3, 7]]
4 []
5 [4, 1, 3, 2]

```

Non-rational tangle

This is an example of an input where the first line is "rational".

Consider the tangle T , represented in Figure A.6, where we label the rational pieces and the gluing points according to the image.

Then, `input.txt` is the following.

```

1 rational
2 3
3 [[1, 1, 2], [0, 1, 3], [1, 1, 1]]
4 [[1, 4, 2, 1], [1, 3, 2, 2], [1, 2, 3, 1], [2, 3, 3, 4]]
5 [3, 1, 4, 2]

```

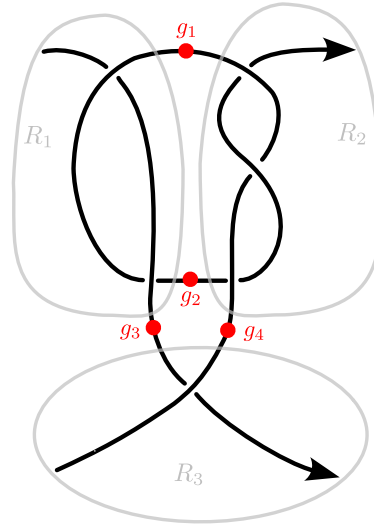


Figure A.6: Oriented tangle diagram for T , divided in rational components.

A.3 Additional settings

There are some additional parameters that can be set manually inside `main.py`. They are the following.

- `user_experience`. This flag, if set to `True`, stops the run in different parts and asks for an manual confirmation in order to proceed. If set to `False`, the run does not stop until the end or until it finds an error in the input data.
- `input_check`. This flag, if set to `True`, stops the run after the pre-process of the input to give the possibility to the user to check that the input given is correct. This is printed on the screen and the program asks for an manual confirmation in order to proceed. If set to `False`, nothing of the above happens and the run does not print the initial diagram.
- `try_multiplicity_zero_regions_choices`. This flag, if set to `True`, allows the program to generate others diagrams where the only difference from the one given in the input is the choice of the multiplicity zero regions (this is only implemented for the cases "tangle" and "rational"). If set to `False`, only the given input will be nicefied. In the case of a diagram of type "rational", this flag is automatically set to `True`.
- `print_distance_complexities`. This flag, if set to `True`, prints after every iteration of the algorithm the distance complexities, the total complexity and the number of generators of the diagram obtained with the last cycle. If set to `False`, the above data is not printed between the cycles.
- `print_intermediate_steps`. This flag, if set to `True`, prints after every iteration of the algorithm the new diagram obtained with the last modifica-

tion. One of the information displayed, is the last move done on the previous diagram; it is presented as showed below.

```

1 Regions of the last diagram that we modified:
2     Type of move: Handle slide
3     Starting region: Region 4
4     Edge modified: [12, 9]
5     Ending region: Region 4
6     Neighbor region: Region 6
7     Middle regions:
8         Region 27
9         Region 28
10    Edges crossed:
11        [5, 12]
12        [2, 33]
13        [4, 7]
```

If set to `False`, the diagrams generated are not printed between the cycles.

- `print_final_diagram`. This flag, if set to `True`, prints the final diagram obtained at the end of the run. If set to `False`, the final diagram is not printed at the end of the execution of the program.
- `print_details_nicefied_diagram`. This flag, if set to `True`, print the following details of the run: number of generators of the final diagram, number of regions of the final diagram, number of cycles of the algorithm needed to obtain the final diagram. If set to `False`, these information are not printed.
- `print_output_for_PQM`. This flag is effective only for diagrams of type "tangle" or "rational". If set to `True`, at the end of the run it formats in a string the final diagram as it is needed for the input of `[PQM.m]` and it prints this string on screen. If set to `False`, this string is not printed.
- `save_on_file`. This flag, if set to `True`, saves on a file various data relative to the run; these are specified by the parameters listed below here. In order to do so, in the same the folder in which there is `input.txt`, there has to be a folder named `nicefied_diagrams`. After the run, the file is saved in such folder under the name `input_nicefied_diagram.txt`. If the flag is set to `False`, the program will not save anything (hence, even if the next flags are set on `True`, they will not have any effect).
- `save_intermediate_steps`. This flag, if set to `True`, saves all the intermediate diagrams obtained in `input_nicefied_diagram.txt` (if we previously set `try_multiplicity_zero_regions_choices = True`, it saves the intermediate diagrams only for the final diagram). If set to `False`, the intermediate diagrams are not saved.

- `save_final_diagram`. This flag, if set to `True`, saves the final diagram obtained at the end of the run in `input_nicefied_diagram.txt`. If set to `False`, the final diagram is not saved.
- `save_details_nicefied_diagram`. This flag, if set to `True`, saves some details of the run in `input_nicefied_diagram.txt`. These details are: number of generators of the final diagram, number of regions of the final diagram, number of cycles of the algorithm needed to obtain the final diagram. If set to `False`, these information are not saved.
- `save_output_for_PQM`. This flag is effective only for diagrams of type "tangle" or "rational". If set to `True`, at the end of the run it formats in a string the final diagram as it is needed for the input of [PQM.m] and it saves this string in `input_nicefied_diagram.txt`. If set to `False`, these string is not saved.

A.4 Pre-processing of the input

After the input is ready, it is possible to run the program. This is done by executing the python script `main_run.py`.

Before the application of the algorithm, the program processes the input given. The first step of the run is the input reading and some sanity checks.

Pre-processing of type "normal"

In the case the diagram is of type "normal", the program first computes the curves of the diagram and it checks if all the circles (hence, arcs excluded) have at least three intersection points on them. If the check fails, the run terminate and the user is asked to perform some finger moves on the diagram to increase the number of intersection points on the circles that are missing some. If this check is passed, then a second sanity check is done: the program scans over the edges of the diagram (identified by the endpoints), trying to find all the edges twice in opposite orientation. If also this second sanity check is passed, the Heegaard diagram is generated.

Then the program checks if our diagram is bordered and, in such case, it checks the border regions to see if they are already squares or if an initial finger move is needed. If such move is required, the method `finger_move_beginning_bordered()` performs it.

Pre-processing of type "tangle"

In the case the diagram is of type "tangle", the program first computes the curves of the diagram and it checks if all the circles (hence, arcs excluded) have at least three intersection points on them. If the check fails, the run terminate and the user is asked to perform some finger moves on the diagram to increase the number of intersection points on the circles that are missing some. If this check is passed, then a second

sanity check is done: the program scans over the edges of the diagram (identified by the endpoints), trying to find all the edges twice in opposite orientation. If also this second sanity check is passed, the Heegaard diagram is generated.

Then the program transforms it in a peculiar Heegaard diagram, removing the sutures around the tangle's endpoints, concatenating the α -arcs and placing the basepoints p_i 's and q_j 's and the two special ones in the indicated regions. If the flag `try_multiplicity_zero_regions_choices` is activated, then the programs generates all the 16 Heegaard diagrams carrying the different combinations for the special basepoints p_i and q_j .

Pre-processing of type "rational"

If the diagram is of type "rational", the program constructs itself the tangle Heegaard diagram for T : it builds the tangle Heegaard diagrams of $R_1, \dots, R_h$ as showed in Section 2.6.1 and it glues them together as indicated in `gluing_instructions`. While doing so, it does two sanity checks. Firstly, for any rational tangle $Q_{p/q}$ it checks that p and q are coprime; if they are not, the run is terminated. As second check, when it glues two tangles Q_{p_1/q_1} and Q_{p_2/q_2} it checks that not both p_1 and p_2 are odd and that not both q_1 and q_2 are even; this is done to avoid the creation of a closed component in T . If such a case occurs, the program alerts the user but it does not terminate: the user can decide to manually override this alert and continue the execution of the program.

Once the tangle Heegaard diagram is built, the program transforms it in a peculiar Heegaard diagram. It removes the sutures around the tangle's endpoints, concatenates the α -arcs, places the basepoints p_i 's and q_j 's and it generates all the 16 Heegaard diagrams carrying the different combinations for the special basepoints p_i and q_j .

A.5 Application algorithm

After the pre-processing, the Nicefication algorithm is applied to the Heegaard diagram given as input. The algorithm is applied as presented in Section 4.2 with the modifications described in Section 4.5.

If the flag `try_multiplicity_zero_regions_choices` is activated, the program cycles the algorithm over the different possible diagrams. It starts with the diagrams that seem more promising (i.e. with a smaller total complexity), moving afterwards to the others. The first diagram is fully nicefied, let n be the number of generators of the final diagram. Then the others are nicefied until the number of generators become greater than n : if this happens, the algorithm stops and the program passes to the next diagram of the list. If one diagram is nicefied and the final number of generators is $n' < n$, then the next ones are compared with this one.

A.6 Examples of output

The output of the program is built with respect to the flags presented in Section A.3. We see some examples of output obtained with these flags.

If `print_final_diagram` or `save_final_diagram`

If one of the above flags is activated, the respective string generated is of the following form.

```

1 The algorithm worked!
2
3 We were able to nicefy the input given in ./inputs/ for the diagram
  input.txt
4
5 Details of the Heegaard diagram:
6
7     Is it nice: True
8
9     Number of regions: 22
10    Number of intersection points: 29
11    Number of border points: 10
12
13    Regions of the diagram:
14        Region 1:
15            Input: [27, 16, 17, 20]
16            Badness: 0
17            Distance: 2
18            Is a border region: False
19
20        Region 2:
21            Input: [29, 27, 20, 21]
22            Badness: 0
23            Distance: 1
24            Is a border region: False
25
26        ...
27
28        Region 22:
29            Input: [29, 28]
30            Badness: 0
31            Distance: 2
32            Is a border region: False
33

```

If `print_details_nicefyed_diagram` or `save_details_nicefyed_diagram`

If one of the above flags is activated, the respective string generated is of the following form.

```

1 Number of generators of the diagram: 45
2 Number of cycles of the algorithm: 2

```

3 Number of regions of the diagram: 22

If `print_output_for_PQM` or `save_output_for_PQM`

If one of the above flags is activated, the respective string generated is of the following form.

```

1 regionsInput = ( {{1,9,36,35,38},{2,12,13,14,29},{3,13,12,11,21},...
...,{39,42,41,39,40},{40,41,42,\[Placeholder],\[Placeholder]}} );
2
3 alphasInput = ( {{1,{1,24,19,20,25,8,29,14,13,12,11,21,4,33,42,41,...
...28,10,9,36,7,26,18,17,23,6,31,39,40,34}}}} );
4
5 betasInput = ( {{1,{1,8,7,2,5,4,3,6}},{2,{9,38,40,42,41,39,37,14,...
...15,11,12,29,30,31,32,33,34,35,36}}}} );
6
7 cancellationSortListInput = ( { } );
8
9 alphaArcs = ( {"a
",{42,41,32,3,22,15,16,27,2,35,38,37,30,1,24,19,20}}, {"b",{11,21,
4,33}}, {"c",{13,12}}, {"d",{25,8,29,14}} } );
10
11 basepointRegions = ( {{7,p,{1,0},-2},{8,q,{1,0},-2},{9,p,{-1,0},-2},
{10,q,{-1,0},-2},{11,p,{0,-1},-2},{12,q,{0,1},-2}} );
12
13 multiplicity0Regions = ( {{41,{{4,5},{28,10},{9,38},{37,14},{13,21}},
{0,1},-4},{42,{{8,7},{36,9},{10,16},{15,11},{12,29}},{0,-1},-4}} );

```

This string is already formatted to be used as input of `[PQM.m]`. For details on this output, we refer to the documentation of `[PQM.m]`.

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